Online Supplement

A Sufficient Statistic for Algorithmic Feedback

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The supplement contains the proofs of the results stated in the main text (Section S1), the further equilibrium results with their proofs (Section S2), execution cost and capital (Section S3), the two-group case used by Theorem 5.4(ii) (Section S4), the supervisory decision in detail, with the targeted halt for every K (Section S5), the proofs for saturating traders (Section S6), which questions prices answer cheaply (Section S7); surveillance over time and against strategic traders (Section S8); then further results in four groups: identification and cost (general memory, Section S11; near-colliding rates, Section S12; the Cramér–Rao benchmark; finite-sample surveillance, Section S14; real market noise, Section S15), what prices hide in economic terms (Sections S16, S18 and S19), supervision and policy (Section S20), and robustness beyond the linear model (Section S21); finally machine checks in detail (Section S22), the machine-checked declarations (Section S23) and the notation (Section S24). Proofs of results that also appear in the full working paper of 5 November 2026 are taken from it. Two things are changed in them: cross-references to material not included here, and the word for a trader group, which the full paper calls a "cohort". New in this version, with their proofs: Section 7, Section S5.1, Propositions 5.5 and S5.6, Theorems 7.7, 6.4, 6.5, S7.1 and S8.1, and Propositions S7.2, S8.2 and S8.3. Every numerical check cited in the text is reproduced by one notebook (`research_extension/theorem_checks/Paper1_Theorem_Checks_Colab.ipynb`).

S1 Proofs of the main results

A lemma used in the proofs.

Lemma S1.1 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_\ell \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_\ell \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, even though J is non-normal. In price/memory coordinates the price-only gain is $\sup_t \|e_x^\top J^t\|_2 = 1$.

Lemma S1.1. Omit zero-weight groups. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots), \qquad u = (\sqrt{a}, \sqrt{b_1\eta_1}, \ldots)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to κu and $J'e_x = \sqrt{a}\,u$, so $h_j = (\kappa/a)e_x^\top (J')^{j+1} e_x$. The spectral decomposition of J' gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0, 1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since J is nonnegative with row sums at most one, every price row of J^t has ℓ_1, hence ℓ_2, norm at most one, attained at $t = 0$.

Lemma 2.1. (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance P solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = (\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2})/2$.

The gain is $\eta = P/(P+\sigma_\nu^2) \in (0,1)$, which simplifies to (7), and the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ has the stated form. (ii) $\nabla_\theta \frac{1}{2} H(x_t - \theta)^2 = -H(x_t - \theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

Theorem 3.3. With zero initial states, $y_{i,t} = \sum_{s \le t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s \le t} k_{t-s} x_s$ and every response is determined by (a, κ, k), hence by M. Conversely, equal responses for all probes imply equal H, hence equal G, and partial fractions of G identify the distinct poles with nonzero residues w_i, i.e. M. After cancellation, $H = \kappa z D(z)/[(z-a)D(z) - \kappa N(z)]$ with $D = \prod(z - r_i)$ and $N(z) = \sum_i w_i z \prod_{j \ne i}(z - r_j)$, as in (4); at a retained r_i the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i \prod_{j \ne i}(r_i - r_j) \ne 0$ (memory rates are positive because $\eta_i < 1$), so no root of D cancels. The numerator $\kappa z D(z)$ has no roots other than 0 and the r_i, so these are the only possible cancellations. The remaining numerator factor z cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1 + |\mathrm{supp}(M) \cap (0,1)|$ for $a' \ne 0$ and $|\mathrm{supp}(M) \cap (0,1)|$ for $a' = 0$ (an atom at rate 0, a group with $\eta = 1$, contributes the constant $M(\{0\})$ to G and no pole, so it shifts a to $a' = a + \kappa M(\{0\})$ and the argument above applies with a') (numerically, the rank of the Hankel matrix of $(h_j)_{j \ge 1}$ is $K+1$ and K respectively for $K = 1, 2, 3$). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z - 1 + \bar{\eta})\big((z - 1 + \bar{\eta})^2 - \delta^2\big)}.$$

The denominator is the product of the three linear factors $z - r$ with $r \in \{1 - \bar{\eta},\ 1 - \bar{\eta} \pm \delta\}$, all in $[\epsilon, 1 - \epsilon]$ by assumption; on $|z| = 1$ each satisfies $|z - r| \ge 1 - r \ge \epsilon$, so $|G_\delta - G_0| \le 2q\delta^2/\epsilon^3$ there. Uniform stability gives $M_H = \sup |H_\theta| < \infty$, and the identity $H_\delta - H_0 = z^{-1} H_\delta H_0 (G_\delta - G_0)$, whose factor z^{-1} has modulus one on the circle, gives an ℓ_2 gain at most $C_{\mathcal{F}} \delta^2$. The mean difference of a power-constrained probe has squared norm at most $C_{\mathcal{F}}^2 \delta^4 PT$; the equal-noise KL is at most $C_{\mathcal{F}}^2 \delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (9).

Theorem 5.1. Let (s^A, ω^A) and (s^B, ω^B) be the K-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K - 2$, and their moments of degree $2K - 1$ differ by $g_K \ne 0$. To see that $g_K \ne 0$, let the Radau nodes be $-1, s_2, \ldots, s_K$ and put $P(s) = (s+1)\prod_{j \ge 2}(s - s_j)^2$. This is monic of degree $2K - 1$ and vanishes at every Radau node, so the Radau rule gives it 0. Gauss is exact to degree $2K - 1$, so it gives $\int P \, d\mu > 0$, since $P \ge 0$ on $[-1, 1]$ and vanishes only at the nodes. The two rules agree on every lower-degree term, so they differ on s^{2K-1}. Put $r = r_c + \Delta\eta \, s$ and weights $W\omega$, and let $\mu_j^X = W \sum_i \omega_i^X (s_i^X)^j$ for $X \in \{A, B\}$. Then $|\mu_j^X| \le W$, and $\mu_j^A = \mu_j^B$ for $j \le 2K - 2$.

Kernel gap. By the binomial theorem, $k_m^X = \sum_j \binom{m}{j} r_c^{m-j} (\Delta\eta)^j \mu_j^X$. So, with $n = 2K - 1$, $\Delta k_m = k_m^A - k_m^B = \sum_{j \ge n} \binom{m}{j} r_c^{m-j} (\Delta\eta)^j (\mu_j^A - \mu_j^B)$. Since $\binom{m}{j} \le \binom{m}{n}\binom{m-n}{j-n}$ for $j \ge n$, this gives $|\Delta k_m| \le 2W \binom{m}{n} (\Delta\eta)^n (r_c + \Delta\eta)^{m-n}$. Summing with $\sum_m \binom{m}{n} x^{m-n} = (1-x)^{-(n+1)}$ gives $\|\Delta k\|_1 \le 2W(\Delta\eta)^{2K-1}/(1 - r_c - \Delta\eta)^{2K}$ over all lags.

Response gap. Let $\varsigma = a + \kappa W/(1 - r_c - \Delta\eta) < 1$, the theorem's hypothesis. Each kernel has $\|k^X\|_1 \le W/(1 - r_c - \Delta\eta)$. From $h_0 = \kappa$ and $h_{t+1} = a h_t + \kappa \sum_j k_j h_{t-j}$, every partial sum of $|h^B|$ is at most $\kappa/(1 - \varsigma)$. The difference $\Delta h = h^A - h^B$ has $\Delta h_0 = 0$ and $\Delta h_{t+1} = a \Delta h_t + \kappa \sum_j (k_j^A \Delta h_{t-j} +$

$\Delta k_j h^B_{t-j}$), so every partial sum of $|\Delta h|$ is at most $\kappa\|\Delta k\|_1\|h^B\|_1/(1-\varsigma)$. Hence, for every horizon,

$$\|h^A - h^B\|_2 \le \|h^A - h^B\|_1 \le C_U(\Delta\eta)^{2K-1}, \qquad C_U = \frac{2\kappa^2 W}{(1-r_c-\Delta\eta)^{2K}(1-\varsigma)^2}.$$

In the other direction, $\Delta k_j = 0$ for $j < 2K-1$, so by induction $\Delta h_t = 0$ for $t \le 2K-1$, and $\Delta h_{2K} = \kappa\,\Delta k_{2K-1} h_0^B = \kappa^2 W g_K(\Delta\eta)^{2K-1}$. When the taps $h_0, \dots, h_{T-1}$ are observed with $T \ge 2K+1$, this gives $\|h^A - h^B\|_2 \ge \kappa^2 W|g_K|(\Delta\eta)^{2K-1}$. With $T \le 2K$ the observed responses coincide. The impossibility bound holds for every T, since it uses only the upper order.

Testing. N reset experiments have divergence $N\|h^A - h^B\|_2^2/(2\sigma^2)$. By Pinsker's inequality, a test whose two error probabilities are at most α needs divergence at least $2(1-2\alpha)^2$. So $N \ge 4(1-2\alpha)^2\sigma^2(\Delta\eta)^{-(4K-2)}/C_U^2$, and the constant does not depend on T. For the W_1 version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass W, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

Theorem 5.4. For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi}\int_{-\pi}^{\pi} (\varrho - 1 - \log\varrho)\, d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = z^{-1}H_1\Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(z^{-1}H_1\Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k\phi(z,\eta_k)$, with w_k the group weights (any factor of η_k is absorbed in ϕ), where ϕ is analytic in η on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1-\eta$ stays inside the disk). Expanding ϕ around $\bar\eta$, $\Delta G = \sum_{j\ge 0} \frac{\partial_\eta^j \phi(z,\bar\eta)}{j!}\,\Delta\tilde M_j$, where $\tilde M_j = \sum_k w_k(\eta_k - \bar\eta)^j$. For the pair, $\Delta\tilde M_0 = \Delta\tilde M_1 = 0$ and $\Delta\tilde M_2 = q\delta^2$; for the moment-matched economies, $\Delta\tilde M_j = 0$ for $j \le 2$ and $\Delta\tilde M_3 = 4\tau p_- p_+(\Delta\eta)^3$ by (20); for the K-group Gauss and Gauss–Radau economies with rates $\bar\eta + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K-2$ exactly, so $\Delta\tilde M_j = W(\Delta\eta)^j \sum_k(\omega_k^A(s_k^A)^j - \omega_k^B(s_k^B)^j)$ vanishes for $j \le 2K-2$ and equals $\pm W g_K(\Delta\eta)^{2K-1} \ne 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta\tilde M_j| \le 2W(\Delta\eta)^j$, and analyticity of ϕ in η on a disk of radius $\varrho > 0$ around $\bar\eta$, uniformly on the circle, gives $|\partial_\eta^j\phi|/j! \le C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}\big(z^{-1}H_0\,\partial_\eta^{j_0}\phi(\cdot,\bar\eta)\big)/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function c does not vanish identically: $f = z^{-1}H_0\partial_\eta^j\phi$ is analytic outside the disk including at infinity, where it vanishes, and is not identically zero; an analytic function on the exterior whose real part vanishes on the circle is an imaginary constant, which here would have to be zero. For $|u| \le \frac12$, $u^2/4 \le u - \log(1+u) \le u^2$, so $D \asymp \int c(\omega)^2 d\omega\,\epsilon^2$, which gives (i)–(iii). The sample requirements need a finite-T bound that is uniform in the separation, which we obtain directly. The first T observations are $N(0,\Sigma_j)$ with Σ_j the $T\times T$ Toeplitz matrices of the spectral densities $f_j = \sigma^2|H_j(e^{i\omega})|^2$. On the compact, uniformly stable family the transfer functions have no zeros or poles on the circle (Theorem 5.3), so $0 < \underline f \le f_j \le \bar f < \infty$ with constants that do not depend on the separation. A Toeplitz matrix has eigenvalues within the range of its symbol, so $\lambda_{\min}(\Sigma_0) \ge \underline f$ and $\|\Sigma_1 - \Sigma_0\| \le \sup_\omega |f_1 - f_0| = \sup_\omega f_0|u| \le \bar f \sup_\omega |u|$. With $A = \Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}$,

whose eigenvalues lie in $[\underline{f}/\bar{f}, \bar{f}/\underline{f}]$, the inequality $\lambda - 1 - \log\lambda \le (\lambda-1)^2/\lambda$ gives

$$D_{\mathrm{KL}}(P_1^T \| P_0^T) = \tfrac{1}{2}\sum_k (\lambda_k - 1 - \log\lambda_k) \le \frac{\bar{f}}{2\underline{f}}\,\|A - I\|_F^2 \le \frac{\bar{f}}{2\underline{f}}\,T\Big(\frac{\bar{f}}{\underline{f}}\Big)^2 \sup_\omega |u|^2 \le C\,T\epsilon^2,$$

with C independent of T and of the separation, because $\sup_\omega |u| = O(\epsilon)$ uniformly. The Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6) then gives the stated lower bounds on T for every T, not only asymptotically. For a fixed pair, $T^{-1}D_{\mathrm{KL}}(P_1^T\|P_0^T) \to D$ and Stein's lemma (one error fixed) and the Chernoff bound (both errors vanishing) give the error exponents of the likelihood-ratio test, of the same order in the separation; that statement is pointwise in the pair.

Theorem 5.3. In the delay variable $\lambda = z^{-1}$, $H = \kappa\prod_i(1 - r_i\lambda)/[(1-a\lambda)\prod_i(1-r_i\lambda) - \kappa\lambda\sum_i w_i\prod_{j\neq i}(1-r_j\lambda)]$, which is (4) divided through by z^{K+1}; its nonstructural zeros are the memory rates, which do not cancel under the nondegenerate assumptions of Theorem 5.3. In the z-plane there is also the structural timing zero at zero: the numerator is proportional to $z\prod_i(z - r_i)$. Stability puts the poles inside the disk. A rational spectral density $\sigma^2|H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is H; Theorem 3.3 then gives the merged measure.

Proposition 6.2. With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only group i, so $\tilde{u}_{i,t} = r_i\tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and, by the ergodic theorem for the stationary linear price process, the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde{u}_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i\sum_{s\ge0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1}u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives Σ_i. The price is a nonnegative mixture of geometric responses to its innovation (Lemma S1.1), so $\gamma_x(s) = \sum_\ell c_\ell\lambda_\ell^s$ with $c_\ell = \sigma_\xi^2\kappa^{-2}\omega_\ell\sum_m\omega_m/(1-\lambda_\ell\lambda_m) \ge 0$; substituting, $\det\Sigma_i = w_i(\gamma_x(0)A_0 - \gamma_x(1)A_1) = w_i\sum_{\ell,m} c_\ell c_m(1-\lambda_\ell\lambda_m)/(1-r_i\lambda_m)$, each term positive. By summation by parts, $\sum_{t=1}^T Z_t e_{i,t} = \sum_{t=1}^T(Z_t - r_i Z_{t+1})\varepsilon_{i,t} + O_P(1)$. Since $(\varepsilon_{i,t})$ is i.i.d. with mean zero and variance σ_f^2 and independent of the whole stationary price process, the summands $m_t = (Z_t - r_i Z_{t+1})\varepsilon_{i,t}$ form a stationary ergodic martingale difference sequence with respect to $\mathcal{F}_t = \sigma((x_s)_{s\in\mathbb{Z}}, \varepsilon_{i,s} : s \le t)$, with $\mathbb{E}[m_t m_t^\top] = \sigma_f^2[(1+r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)] = \Omega_i$, so the Billingsley–Ibragimov martingale central limit theorem gives (11). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \le \bar{\lambda} < 1$, so with $w_i \ge w_{\min} > 0$ we get $\det\Sigma_i \ge w_{\min}(1-\bar{\lambda}^2)\gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

Theorem 5.2.

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = a h_{t-1} + \kappa\sum_m k_m h_{t-1-m}$ (plus κ at $t = 0$) can be solved for $k_0, \dots, k_{2K-1}$ from $h_1, \dots, h_{2K}$. Because $h_0 = \kappa$ is a known primitive, each k_m is a polynomial in $h_1, \dots, h_{m+1}$ with coefficients depending on (a, κ) (the only division is by $\kappa h_0 = \kappa^2$), so on the bounded set of responses within sup-distance 1 of the class this map is L_h-Lipschitz, with L_h depending on the class but not on $\Delta\eta$. Let E be the event $\max_{1\le t\le 2K}|\bar{h}_t - h_t| \le \sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$, where $\bar{h}_t$ averages the N responses; a Gaussian tail bound and a union bound over the $2K$ estimated taps give $P(E) \ge 1 - \beta$. On E, and provided this radius is at most 1, the estimated kernel values satisfy $\max_m|\hat{k}_m - k_m| \le L_h\sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$. Centring at r_c is a fixed triangular

linear map, $\mu_m = \sum_{j \le m} \binom{m}{j} (-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \dots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j} (s_i - s_j)^4$, bounded away from zero on the compact class. On ordered, γ-separated nodes F is injective, because a measure with K distinct atoms is determined by its first $2K$ moments. A C^1 map that is injective on a compact set Q and has a nonsingular Jacobian there is injective on an open neighbourhood U of Q: otherwise there are $x_n \neq y_n$ with $F(x_n) = F(y_n)$ and distance to Q tending to zero, whose limits $x, y \in Q$ satisfy $F(x) = F(y)$, so $x = y$, contradicting local injectivity at x. Applying the quantitative inverse function theorem uniformly on Q, there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F(U)$ contains the ball of radius ρ_0 around every point of $F(Q)$ and the single inverse $F^{-1} : F(U) \to U$ is L-Lipschitz on each such ball. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K-1$. Since $\Delta\eta < 1$ and the centring map is fixed, on E we get $\|\hat{t} - t\|_\infty \le C_1 \sqrt{\log(4K/\beta)}\, \epsilon_N$ with C_1 equal to $\sqrt{2} L_h$ times the norm of the centring map; and because $\sigma N^{-1/2} \sqrt{2 \log(4K/\beta)} \le \sqrt{2}\, \epsilon_N \sqrt{\log(4K/\beta)}$, the radius condition holds whenever $c_0 \le 1/\sqrt{2}$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1} is L-Lipschitz. For an input in such a ball, $x = F^{-1}(\hat{t})$ has K distinct nodes and nonzero weights, so the $K \times K$ Hankel matrix of $\hat{t}$ is nonsingular and Prony's method returns exactly x: the Prony polynomial is determined uniquely by the Hankel system, and since the K real nodes of x reproduce $\hat{t}$ it equals $\prod_i (z - \hat{s}_i)$, so its roots are real and distinct and no complex pair can appear; Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C\sqrt{\log(4K/\beta)}\, \epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \min\{\min(\rho_0, \gamma/(2L))/C_1,\, 1/\sqrt{2}\}$, which also keeps $C\sqrt{\log(4K/\beta)}\, \epsilon_N < \gamma/2$. No separate root-perturbation bound is needed: on the ball where F^{-1} is L-Lipschitz, Prony's roots and weights are the coordinates of $F^{-1}(\hat{t})$, so the conditioning of the root step is part of L (for an explicit step-by-step analysis of Prony's method with clustered exponents see Katz et al., 2023). The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem 5.1 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight W); for any such class the orders match, with constants that depend on the class. □

Theorem 6.3.

Proof. $N_p \asymp \sigma^2 (\Delta\eta)^{-(4K-2)}$ by Theorems 5.1 and 5.2. *Upper bounds.* Average the N price responses to estimate h with error of order $\sigma N^{-1/2}$ per tap. Given h, class i's mean flow is a two-parameter family $F(w_i, r_i; h)$ whose Jacobian columns $\sum_s r_i^{t-s} h_s$ and $w_i \sum_s (t-s) r_i^{t-s-1} h_s$ are linearly independent, with a Gram determinant bounded below uniformly on the class because $w_i \ge \omega_{\min}$, the rates lie in a compact subset of $(0, 1)$ and $h_0 = \kappa > 0$; the bound does not involve the other classes or $\Delta\eta$. Least squares on class i's flows with the plug-in h therefore estimates (w_i, r_i) with error $O_P(N^{-1/2})$, uniformly on the class, which gives the upper bounds in (i) and in (ii) for accuracy $\epsilon\Delta\eta$. For (i) there is also a finite-sample version: in the first period the tagged flow of class i is $w_i\kappa$, so the mean of the N first-period records gives $P(|\hat{w}_i - w_i| \ge \epsilon) \le \sigma_f^2/(N\kappa^2\epsilon^2)$ by Chebyshev's inequality, for every configuration (this step is machine-checked). *Lower bounds.* At a configuration in the interior of the class, perturb one class's weight by ϵ, or its rate by $\epsilon\Delta\eta$; for small ϵ both perturbed configurations remain in the class because the weight interval is nondegenerate and the rate set has nonempty interior. The mean differences of both the price and the flow records are of order ϵ, respectively $\epsilon\Delta\eta$, so the Kullback–Leibler divergence of N experiments is at most $CN\epsilon^2$, respectively $CN\epsilon^2(\Delta\eta)^2$, and Le Cam's two-point lemma gives the lower bounds. □

Unknown primitives.

Corollary S1.2 (Unknown primitives cost nothing in the rate)**.** *In the setting of Theorem 5.2, suppose in addition that every memory rate is at least $\varepsilon > 0$ (that is, $r_c - \Delta\eta \geq \varepsilon$), and that the fundamental value p^*, the impact $\kappa \in [\underline{\kappa}, \overline{\kappa}] \subset (0, \infty)$ and the persistence $a \in [0, \bar{a}] \subset [0, 1)$ are unknown. Put $\tilde{k}_0 = k_0 + a/\kappa$ and $\tilde{k}_m = k_m$ for $m \geq 1$.*

(i) The response depends on (a, κ, M) only through κ and $\tilde{k}$: $h_0 = \kappa$ and $h_{t+1} = \kappa \sum_{m=0}^{t} \tilde{k}_m h_{t-m}$. The persistence a acts exactly like an extra group with memory rate zero and merged weight a/κ.

(ii) The taps $h_0, \ldots, h_{2K+1}$ identify (κ, a, M): $\kappa = h_0$; the recursion gives $\tilde{k}_0, \ldots, \tilde{k}_{2K}$; the values $\tilde{k}_1, \ldots, \tilde{k}_{2K}$ are the moments of orders $0, \ldots, 2K-1$ of $\mu' = \sum_i w_i r_i \, \delta_{r_i}$; and $a = \kappa(\tilde{k}_0 - \sum_i w_i)$.

(iii) Subtract the pre-probe price to remove p^* (doubling the noise variance), estimate $\hat{\kappa} = \hat{h}_0$, deconvolve with $\hat{\kappa}$ alone, apply Prony's method to the estimated moments of μ', and set $\hat{w}_i = \widehat{w_i r_i}/\hat{r}_i$ and $\hat{a} = \hat{\kappa}(\hat{\tilde{k}}_0 - \sum_i \hat{w}_i)$. There are constants $c_0, C > 0$ depending on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, \varepsilon, \underline{\kappa}, \overline{\kappa}, \bar{a}, r_c)$ but not on $\Delta\eta$ such that, whenever $\epsilon_N \leq c_0$,

$$\max_i |\hat{w}_i - w_i| + |\hat{a} - a| + |\hat{\kappa} - \kappa| \;\leq\; C\sqrt{\log(8K/\beta)}\,\epsilon_N$$

with probability at least $1 - \beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice, the same order as with known primitives.

In words. Not knowing the market's own persistence or its price impact costs a constant factor, not a worse rate. Persistence enters the price exactly like one more group of traders with no memory at all, and that group sits far from the clustered groups the supervisor is trying to separate. Measured in the merged weights times the rates, the clustered groups' problem is the same Prony problem as before, with weights bounded below because rates are bounded away from zero; one more tap pins down the extra group.

Proof of Corollary S1.2. (i) Substitute $k_0 = \tilde{k}_0 - a/\kappa$ in $h_{t+1} = a h_t + \kappa \sum_{m \leq t} k_m h_{t-m}$: the term $a h_t$ cancels against $\kappa(a/\kappa)h_t$. (ii) Solving the recursion of (i) for $\tilde{k}_t$ gives $\tilde{k}_t = (h_{t+1} - \kappa \sum_{m<t} \tilde{k}_m h_{t-m})/\kappa^2$, a polynomial in $h_1, \ldots, h_{t+1}$ divided by a power of $h_0 = \kappa$. For $m \geq 1$, $\tilde{k}_m = \sum_i w_i r_i^m = \sum_i (w_i r_i) r_i^{m-1}$, so $\tilde{k}_1, \ldots, \tilde{k}_{2K}$ are the moments of orders $0, \ldots, 2K-1$ of μ', which has K distinct atoms with positive weights; these moments determine μ' (Theorem 3.3). Since $r_i \geq \varepsilon > 0$, $w_i = (w_i r_i)/r_i$, and then $a = \kappa(\tilde{k}_0 - \sum_i w_i)$. (iii) On the event that every estimated tap is within $\sigma N^{-1/2}\sqrt{2\log(8K/\beta)}$ of its mean (probability at least $1 - \beta/2$ by the Gaussian tail and a union bound over $2K + 2$ taps, with the doubled variance absorbed into the constant), $\hat{\kappa} \geq \underline{\kappa}/2$ once $\epsilon_N \leq c_0$, so the map from taps to $\hat{\tilde{k}}$ is Lipschitz with a constant that depends on $(K, \underline{\kappa}, \overline{\kappa}, \bar{a}, W)$ and not on $\Delta\eta$. The weights of μ' lie in $[\varepsilon\omega_{\min}, \omega_{\max}]$ and its rates are those of the class of Theorem 5.2; the proof of that theorem uses only a $\Delta\eta$-free Lipschitz deconvolution, the rescaling by $(\Delta\eta)^{-m}$ and weights bounded below, so it applies verbatim to μ' and gives $\max_i |\widehat{w_i r_i} - w_i r_i| \leq C_1\sqrt{\log(8K/\beta)}\,\epsilon_N$ and rate errors smaller by a factor $\Delta\eta$, with probability at least $1 - \beta/2$. The maps $(v, r) \mapsto v/r$ on $r \geq \varepsilon/2$ and $(\kappa, \tilde{k}_0, w) \mapsto \kappa(\tilde{k}_0 - \sum_i w_i)$ are Lipschitz on the bounded set reached, and $|\hat{\kappa} - \kappa|$ and $|\hat{\tilde{k}}_0 - \tilde{k}_0|$ are of order $\sigma N^{-1/2} \leq \epsilon_N$. Combining gives the bound. □

Proposition 3.2. The proof is given after the statement in the main text.

S2 Equilibrium: further results and proofs

Corollary S2.1 (Disclosure in equilibrium)**.** *Take a positive-impact equilibrium of Theorem 4.1 with positive weights and $D > \max_i r_i$.*

(i) The groups' flows obey $u_{i,t+1} = r_i u_{i,t} + w_i x_t$ with their true *weights. The price is the stable closed loop of Theorem 4.2, driven by ξ, with positive effective weights. So Proposition 6.2 applies: tagged flow identifies each group's (r_i, w_i), with a limiting covariance that has no singularity as rates collide.*

(ii) On classes in which both the true and the effective weights lie in compact positive intervals, Theorem 6.3 holds in equilibrium. Tagging cuts the experiments needed by a factor of order $(\Delta\eta)^{-(4K-2)}$ for weights and $(\Delta\eta)^{-(4K-4)}$ for rates.

In words. What makes disclosure valuable survives the dealers' optimization. Prices still hide the groups at the same polynomial cost, and each group's own flow still reveals that group's memory and weight directly.

Theorem S2.2 (Fragility: where liquidity disappears)**.** *Let there be one feedback group with weight $w > 0$ and memory rate $r \in (0,1)$, with $\bar{a} \in [0,1)$ and $\sigma_F, \sigma_\xi > 0$. Call an equilibrium of Theorem 4.1 with $\kappa > 0$ and $D > 0$ a positive-impact equilibrium. For $D \in (m,1)$ with $m = \max(\bar{a}, r)$, let*

$$\kappa_w(D) = \frac{(1-D)(D-r)}{w}, \qquad \Psi_w(D) = \frac{\kappa_w(D)\,(D-\bar{a})\,D}{\sigma_F^2 D^2 + \kappa_w(D)^2 \sigma_\xi^2}, \qquad A^*(w) = \max_{D\in(m,1)} \Psi_w(D).$$

(i) Threshold.

- *The positive-impact equilibria are exactly the pairs $(\kappa_w(D), D)$ with $D \in (m,1)$ and $\Psi_w(D) = A$, and every one of them is stable.*
- *A positive-impact equilibrium exists if and only if $A \le A^*(w)$.*
- *For $A < A^*(w)$ there are at least two; for $A > A^*(w)$ there are none.*

(ii) Monotonicity. *$A^*(w)$ is strictly decreasing in w. As $w \downarrow 0$ it tends to $A_0^* = (1-\bar{a})/(2\sigma_F\sigma_\xi)$, the threshold of Theorem 4.1(ii).*

(iii) Bounds and rate. *For every $w > 0$,*

$$A^*(w) < \min\Big\{A_0^*,\ \frac{\Phi^*}{\sigma_F^2\, w}\Big\},$$

where

$$\Phi^* = \max_{D\in[m,1]} \frac{(1-D)(D-r)(D-\bar{a})}{D} \le \frac{(1-\bar{a})(1-r)^2}{4},$$

and $w\,A^*(w) \to \Phi^*/\sigma_F^2$ as $w \to \infty$. In terms of the feedback intensity $G(1) = w/(1-r)$, strong feedback gives $A^* \approx \Phi^*/(\sigma_F^2(1-r)G(1))$, and always $A^* < (1-\bar{a})(1-r)/(4\sigma_F^2 G(1))$.

(iv) Several groups. *Let all weights be positive.*

- *The positive-impact equilibria with $D > \max_i r_i$ are exactly the pairs $((1-D)/G(D), D)$ with $D \in (\max(\bar{a}, \max_i r_i), 1)$ at which the analogue of Ψ_w, with $\kappa = (1-D)/G(D)$, equals A.*

- *Every such equilibrium has*

$$A < \min\Big\{A_0^*,\ \frac{(1-\sqrt{\bar{a}})^2}{\sigma_F^2\, G(1)}\Big\},$$

  whatever the number of groups and their rates.
- *Every such equilibrium is stable.*

In words. With positive feedback, dealers can make a market only if their dislike of risk is below a threshold, and the threshold falls as feedback grows: for strong feedback it is inversely proportional to the feedback intensity. Past the threshold, no price impact is consistent with dealers' optimization. Below it, every equilibrium has stable prices. Liquidity does not erode through growing instability; it disappears when the two equilibria merge, at a point that stable prices give no warning of. At the calibration of Supplement S2 ($\bar{a} = 0.6$, $r = 0.7$, $\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$), A^* is 0.40 without feedback, 0.25 at $G(1) = \frac{1}{15}$ and 0.034 at $G(1) = \frac{2}{3}$, and $w\,A^*$ approaches $\Phi^* = 0.00688$. The necessary conditions behind parts (i) and (iii) are machine-checked (`Paper1EquilibriumReducedForm.lean`). With several groups, part (iv) gives a threshold that depends on the population only through its total feedback intensity $G(1)$. At the same calibration, numerically, the exact thresholds for two and three groups lie between 28% and 86% of that bound, and the equilibria on the threshold curve are stable. Every such equilibrium is stable (part (iv)), as the scan also shows.

Theorem S2.3 (A strategic insider in the inventory equilibrium)**.** *Add three things to the model of Theorem 4.1:*

- News. *The fundamental moves by $F_{t+1} = F_t + \theta_t + \varepsilon_{t+1}$, with $\theta_t \sim N(0, \sigma_\theta^2)$ and ε_t i.i.d. of variance σ_ε^2.*
- An insider. *At date t a short-lived insider privately observes θ_t, and nothing else, submits d_t, and values the position at F_{t+1}.*
- What groups and dealers see. *The groups trade on the observed deviation $s_t = p_t - F_t$. The dealers price from total flow.*

Let $\nu_t = \xi_t + d_t$ be the unexpected flow.

(i) Characterization. *The strategy $d_t = b\theta_t$ and the price $p_t = F_t + \lambda\nu_t + x_t$, with x_t the price law of Theorem 4.1, form an equilibrium exactly when the induced dynamics are stable and*

$$D = 1 - \kappa G(D), \quad \kappa(D - \bar{a}) = AVD, \quad \lambda = \frac{b\sigma_\theta^2}{\sigma_\nu^2}, \quad b = \frac{D}{2(\kappa + \lambda)},$$

$$\sigma_\nu^2 = \sigma_\xi^2 + b^2\sigma_\theta^2, \qquad V = \sigma_\varepsilon^2 + \frac{\sigma_\theta^2\sigma_\xi^2}{\sigma_\nu^2} + \Big(\frac{\kappa + \lambda}{D}\Big)^2\sigma_\nu^2.$$

One unit of unexpected flow moves the price by $(\kappa + \lambda)/D$, and the insider's expected profit per date is $b\sigma_\theta^2/2$.

(ii) Kyle limit. *Without inventory risk ($A = 0$), $\kappa = 0$, $D = 1$, $\lambda = \sigma_\theta/(2\sigma_\xi)$ and $b = \sigma_\xi/\sigma_\theta$. These are Kyle's one-period values at every date, and the feedback groups do not move prices.*

(iii) Invariance. *The coefficients depend on the groups only through G, as Theorem 3.1 requires.*

(iv) Observer. *The observed deviation follows*

$$s_t = a^* s_{t-1} + \kappa^* \sum_i \tilde{u}_{i,t} + c_0 \nu_t - c_1 \nu_{t-1}, \qquad c_0 = \frac{\kappa + \lambda}{D}, \quad c_1 = \frac{\bar{a}\lambda}{D},$$

with $(a^*, \kappa^*, \tilde{u})$ as in Theorem 4.2. This is the closed loop of Section 2 driven through a known moving-average filter $c_0 - c_1 z^{-1}$, which is invertible because $c_1 < c_0$.

- *The memory rates remain the transmission zeros of the feedback part.*
- *The lower-bound pair of Theorem 5.1 is realized by two such equilibria with identical primitives.*
- *So the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold with an insider present.*

In words. A short-lived insider who knows the next piece of news trades against noise traders and against the feedback traders' predictable flow. Dealers net the predictable flow out. So the insider's trades move prices through two channels: an information part λ and an inventory part, both amplified by feedback through D. Positive feedback raises the total impact. The insider then trades less and earns less, and prices reveal less of her news. Negative feedback does the opposite.

- *An example from Supplement S2.* One positive-feedback group lowers the share of the insider's information impounded in prices from 0.372 to 0.366. One negative-feedback group raises it to 0.415.
- *What prices cannot identify.* It is still worth nothing to anyone trading on public information.
- *Machine check.* `Paper1EquilibriumInsider.lean` checks the observer's pathwise reduced form, the insider's best response, the Kyle limit and the posterior variance.
- *Scope.* The insider's information is short-lived, and she conditions only on her news. Two cases are not solved: an insider who also saw the public state, who would trade against the predictable mispricing; and a long-lived Kyle insider, who would also time her trades against the groups' memory.

A misjudged dealer and a targeted halt. The next two results use the simplest linear market inside Theorem 3.1. They are summarized in Section 4.

- Dates are $t = 0, \dots, T-1$ and the price change is $s_t = p_t - p_{t-1}$, with $p_{-1} = 0$.
- The dealer moves the price by $s_t = \lambda(y_t - \hat{f}_t)$, with $\lambda > 0$, where $\hat{f}_t = \sum_{m<t} \hat{k}_m s_{t-1-m}$ is the feedback flow it predicts from its kernel $\hat{k}$.
- The insider, with value v of mean zero and variance Σ_0 independent of the noise, commits to a schedule $x = bv$, $b \in \mathbb{R}^T$. Schedules may change sign, so pump-and-dump patterns are allowed.
- The kernel error is $\Delta k = k - \hat{k}$, with $\varepsilon = \sum_{m \le T-2} |\Delta k_m|$.

Let C be the lower-triangular matrix of ones and L the strictly lower Toeplitz matrix with $L_{t,t-1-m} = \Delta k_m$.

Theorem S2.4 (Rents from a misjudged aggregate response)**.** *(i) The insider's expected profit is $J(b) = \Sigma_0(\mathbf{1}^\top b - b^\top A b)$, with $A = \lambda C(I - \lambda L)^{-1}$. If $Q = \frac{1}{2}(A + A^\top) \succ 0$, the optimal schedule is $b^* = \frac{1}{2}Q^{-1}\mathbf{1}$ and the value is $V(\Delta k) = \frac{\Sigma_0}{4}\mathbf{1}^\top Q^{-1}\mathbf{1}$.*

(ii) If $\lambda T\varepsilon \le \frac{1}{4}(1-\lambda\varepsilon)$, then $Q \succ 0$ and

$$|V(\Delta k) - V(0)| \le \frac{2\Sigma_0 T^2 \varepsilon}{1-\lambda\varepsilon}.$$

(iii) To first order, $V(\Delta k) = V(0) - \Sigma_0 \sum_m \omega_m \Delta k_m + O(\varepsilon^2)$, with $\omega_m = \frac{(T-m)(T-m-1)}{2(T+1)^2} > 0$ and $V(0) = \frac{\Sigma_0 T}{2\lambda(T+1)}$. The rent is therefore of exact order ε. The insider gains when the dealer overstates the feedback ($\Delta k < 0$) and loses when it understates it.

(iv) If the dealer estimates the aggregate kernel from n observations with error $\varepsilon_n = O_p(n^{-1/2})$, the rent is $O_p(n^{-1/2})$. Separating the groups behind the same kernel still needs order $(\Delta\eta)^{-(4K-2)}$ experiments (Theorem 5.1).

In words. A dealer that misjudges how strongly feedback traders respond hands the insider a rent, or charges it one, in proportion to the error. That error concerns only the aggregate response, which public data reveal at the ordinary rate. So the information that protects price formation is cheap to learn, while the information about who the groups are stays expensive. The dealer's coefficient λ is held fixed: this is the insider's best response to a misspecified dealer, not a re-solved equilibrium.

Theorem S2.5 (Targeted intervention in equilibrium)**.** *Take the Gauss and Radau pairs of Theorem 3.1(iv), with total weight W, on rates $c+ds$ with $\rho = |c|+|d| < 1$. Place them in the market above, with the same dealer rule $(\lambda, \hat{k})$, the same insider schedule and the same draws.*

(i) The kernels agree for $m \le 2$, and $|k^G_m - k^R_m| \le 2W\binom{m}{3}|d|^3\rho^{m-3}$ for $m \ge 3$. Hence $\|k^G - k^R\|_1 \le 2W|d|^3/(1-\rho)^4$.

(ii) Before any intervention, the price paths differ by at most $\lambda^2 T\|k^G - k^R\|_1\|x+z\|$ divided by $(1-\lambda\|k^G-\hat{k}\|_1)(1-\lambda\|k^R-\hat{k}\|_1)$, which is $O(|d|^3)$. The insider's values differ by $O(|d|^3)$.

(iii) Halting the largest group leaves kernels $\frac{W}{2}(c-d\sigma)^m$ (the Gauss weights tie; we remove the group at $c+d\sigma$) and $\frac{W}{4}(c-d)^m$, which differ by $\frac{W}{4}$ at $m=0$ for every spacing. The post-halt responses therefore differ at order one, and so do the post-halt feedback budgets $\sum_m k_m$.

In words. Two markets can look the same in equilibrium to within the cube of the memory spacing, yet halting their largest group leaves feedback that differs at order one. Equilibrium does not remove the case for targeted information; it sharpens it. In the simulation below, the pre-halt price gap is about $3d^3$ at all three spacings tried, while the post-halt gap stays near 0.19.

Proof of Theorem 3.1. Part (i). By induction, $m_{i,t} = \sum_{m<t} r_i^m s_{t-1-m}$. Multiplying by w_i and summing over i gives $f_t = \sum_{m<t} k_m s_{t-1-m}$.

Part (ii). If $k_m = k'_m$ for $m < T$, then by (i) the two aggregate flows agree on $t \le T$ for every signal. Conversely, the impulse signal sequence $s_j = \mathbf{1}\{j=0\}$ gives $f_{m+1} = k_m$, so equal flows on $t \le T$ for every signal sequence force $k_m = k'_m$ for $m < T$. This direction quantifies over signal sequences, not over price paths under a fixed rule S: a rule such as $S \equiv 0$ never produces the impulse.

On an unbounded horizon, equality of all $k_m = \int r^m \, dM(r)$ is equality of all moments of two finite signed measures on $[0,1]$. Polynomials are dense in $C[0,1]$, so the measures coincide. In terms of η this is the condition (8) of Theorem 3.3.

Part (iii). Fix a strategy profile and a realization. We argue by strong induction on $t \le T$. If prices and flows agree before t:

- the signals agree before t, because S_j depends on prices up to j;
- the aggregate flows at t agree, by part (i) and the equal kernels, since f_t uses signals before t only;
- the insider's orders at t agree, by causality;
- hence the total flows y_t agree, and then the prices p_t agree, by causality of the dealer's rule.

The insider's profit $\sum_{t<T}(v - p_t)x_t$ and the dealer's inventory $-\sum_{t<T} y_t$ are functions of these paths.

Strategy sets are the same in the two games, because the rules are defined on the same histories. Every payoff is an expectation of the same function of (v, z), so best responses coincide and so do the equilibria.

Part (iv). The Gauss rule has weights $\frac{1}{2}, \frac{1}{2}$ and nodes $\pm\sigma$ with $\sigma^2 = \frac{1}{3}$. The Radau rule has weights $\frac{1}{4}, \frac{3}{4}$ and nodes $-1, \frac{1}{3}$. Both integrate $1, s, s^2$ exactly, so on rates $c + ds$ the kernels agree for $m \le 2$. The largest weights are $\frac{1}{2}$ and $\frac{3}{4}$. □

Machine check. `Paper1EquilibriumInvariance.lean` proves the following:

- part (i): `p1_eqAggFlow_kernel`;
- both directions of part (ii) on a finite horizon: `p1_eqAggFlow_of_kernel` and `p1_kernel_of_eqAggFlow`;
- the pathwise statement of part (iii) for arbitrary causal insider, dealer and signal rules: `p1_eq_market_invariance` and `p1_eq_insider_profit_invariance`;
- part (iv): `p1_eq_group_level_differs`.

The unbounded-horizon moment statement and the passage from pathwise equality to equal equilibria are on paper.

Group-specific demand shocks. Suppose group i also trades an idiosyncratic shock $\epsilon_{i,t}$ through its memory, so that f_t gains the term $\sum_i w_i \sum_{m<t} r_i^m \epsilon_{i,t-1-m}$. The same induction shows that the equilibrium depends on the groups only through the kernel and the law of this aggregate shock process. For independent Gaussian shocks with variances σ_i^2, that law is determined by the spectral density $\sum_i w_i^2 \sigma_i^2 |1 - r_i e^{i\omega}|^{-2}$. This is a second merged object, and the invariance class shrinks accordingly. The paper's groups have no idiosyncratic shocks.

The textbook benchmark. Take a risk-neutral, competitive dealer that knows the kernel. It can compute f_t from public prices and price only $y_t - f_t$, so predictable feedback flow then has no price impact, and the closed loop of Section 2 is absent in that equilibrium. The theorem holds in that benchmark as well. Feedback reaches prices in equilibrium when the dealer bears inventory risk, or when it does not know the kernel and must learn it. These are the cases studied in the equilibrium extension.

Numerical illustration. `equilibrium/e1_demo.py` runs one market. The insider submits $x_t = 0.3(v - p_{t-1})$. The dealer sets $p_t = p_{t-1} + 0.3\, y_t - 0.05\,(I_t - I_{t-1})$. The signal is $s_t = p_t - p_{t-1}$. Two group configurations are compared, the Gauss and Radau pairs of part (iv) on rates $0.6 + 0.2s$, with total weight 0.5, under the same (v, z) draws. Over 200 draws:

- price paths coincide to machine precision on the first four dates ($t \le 3$), with largest gap 2×10^{-16};

- from date five, where k_3 first differs (by $\frac{2}{9}d^3$ times the total weight), they separate, with largest gap 1.8×10^{-4} at date five and 1.9×10^{-2} by date twenty;
- the largest group's share of feedback positions stays at medians of 0.59 and 0.79.

The result file is `equilibrium/e1_demo_results.json`.

Proof of Theorem S2.4. *Part (i).* Total flow is $y = x + z + f$. The dealer subtracts $\hat{f}$, so $s = \lambda(x + z + Ls)$. The matrix L is strictly lower triangular, so $I - \lambda L$ is invertible and $s = \lambda(I-\lambda L)^{-1}(x+z)$. Prices are $p = Cs$. With $x = bv$, realized profit is $\sum_t (v-p_t)x_t = v^2\mathbf{1}^\top b - v\, b^\top Cs$. Taking expectations with $E[v] = 0$ and v independent of z gives $J(b)$. When $Q \succ 0$, J is strictly concave, and its maximizer and value are as stated.

Part (ii). A Toeplitz operator is bounded by the ℓ_1 norm of its symbol, so $\|L\| \le \varepsilon$. For $\lambda\varepsilon < 1$ this gives $\|(I - \lambda L)^{-1}\| \le 1/(1 - \lambda\varepsilon)$ and $(I - \lambda L)^{-1} - I = \lambda L(I - \lambda L)^{-1}$. Since $\|C\| \le T$, we get $\|Q - Q_0\| \le \|A - A_0\| \le \lambda^2 T\varepsilon/(1 - \lambda\varepsilon)$. Moreover $Q_0 = \frac{\lambda}{2}(I + \mathbf{1}\mathbf{1}^\top) \succeq \frac{\lambda}{2} I$. Under the stated condition $\|Q - Q_0\| \le \lambda/4$, so $Q \succeq \frac{\lambda}{4} I$. Then

$$|V - V_0| = \tfrac{\Sigma_0}{4}|\mathbf{1}^\top Q^{-1}(Q_0 - Q)Q_0^{-1}\mathbf{1}| \le \tfrac{\Sigma_0}{4}\, T \cdot \tfrac{4}{\lambda} \cdot \tfrac{2}{\lambda} \cdot \tfrac{\lambda^2 T\varepsilon}{1-\lambda\varepsilon}.$$

Part (iii). $Q_0^{-1}\mathbf{1} = \frac{2}{\lambda(T+1)}\mathbf{1}$, which gives $V(0)$. Since $A = A_0 + \lambda^2 CD + O(\varepsilon^2)$, the derivative of $\mathbf{1}^\top Q^{-1}\mathbf{1}$ is $-u^\top(\lambda^2 CD)u$ with $u = Q_0^{-1}\mathbf{1}$. Using $\mathbf{1}^\top C = (T, T-1, \ldots, 1)$ and $(L\mathbf{1})_t = \sum_{m<t} \Delta k_m$, we get $\mathbf{1}^\top CL\mathbf{1} = \sum_m \Delta k_m \frac{(T-m)(T-m-1)}{2}$. Multiplying by $\frac{\Sigma_0}{4} \cdot \frac{4}{\lambda^2(T+1)^2} \cdot \lambda^2$ gives ω_m.

Part (iv). This is part (ii) with $\varepsilon = \varepsilon_n$. The rate for ε_n is an assumption on the dealer's estimator, for example least squares of order flow on lagged price changes in a stationary market. It is not proved here.

Proof of Theorem S2.5. *Part (i).* Write μ_j^G, μ_j^R for the jth moments of the two rules, which are at most one in absolute value. Expanding $(c + ds)^m$ gives

$$k_m^G - k_m^R = W \sum_{j \ge 3} \binom{m}{j} c^{m-j} d^j (\mu_j^G - \mu_j^R).$$

The terms with $j \le 2$ vanish because both rules are exact for those degrees. Since $\binom{m}{j} \le \binom{m}{3}\binom{m-3}{j-3}$, the sum is at most $2W\binom{m}{3}|d|^3\rho^{m-3}$. Finally, $\sum_{m\ge3}\binom{m}{3}\rho^{m-3} = (1-\rho)^{-4}$.

Part (ii). With residual matrices L_G, L_R built from $k^G - \hat{k}$ and $k^R - \hat{k}$,

$$\begin{aligned} p^G - p^R &= \lambda C\big[(I - \lambda L_G)^{-1} - (I - \lambda L_R)^{-1}\big](x + z) \\ &= \lambda^2 C(I - \lambda L_G)^{-1}(L_G - L_R)(I - \lambda L_R)^{-1}(x + z). \end{aligned}$$

Bound each factor as in the proof of Theorem S2.4. The value comparison follows from the same perturbation argument, with base point L_R instead of 0.

Part (iii). Removing one of the two Gauss groups leaves weight $W/2$. Removing the Radau group with weight $3/4$ leaves $W/4$. At $m = 0$ the kernel is the remaining weight.

Proof of Theorem 4.1. *Part (i).* Conjecture $x_t = -\kappa H_t + \sum_i \beta_i u_{i,t+1}$ and set $D = 1 - \sum_i \beta_i w_i$. The next-period expectation follows from three facts: $E_t\xi_{t+1} = 0$, $E_t H_{t+1} = \bar{a} H_t - \sum_i u_{i,t+1}$, and

$E_t u_{i,t+2} = r_i u_{i,t+1} + w_i E_t x_{t+1}$. Together they give

$$D\, E_t x_{t+1} = -\kappa \bar{a} H_t + \sum_i (\kappa + \beta_i r_i) u_{i,t+1}.$$

The dealers' condition $x_t = E_t x_{t+1} - AVH_t$ must hold for all states. Matching the coefficient on $u_{i,t+1}$ gives $\beta_i = (\kappa + \beta_i r_i)/D$, that is, $\beta_i = \kappa/(D - r_i)$. The definition of D then becomes $D = 1 - \kappa G(D)$, and matching on H_t gives $\kappa(D - \bar{a}) = AVD$.

For the variance, x_{t+1} depends on ξ_{t+1} through H_{t+1} and through its own effect on $u_{i,t+2}$. Its innovation δ therefore satisfies $\delta = \kappa \xi_{t+1} + \sum_i \beta_i w_i \delta$, so $\delta = \kappa \xi_{t+1}/D$ and $V = \sigma_F^2 + \kappa^2 \sigma_\xi^2 / D^2$.

Conversely, take a solution of (10). Then the conjectured law satisfies the dealers' condition in every state (`p1_inventory_foc`), and V is consistent with it (`p1_inventory_innovation`). If the induced dynamics are stable, the process is stationary and the solution is an equilibrium.

Part (ii). With no groups, β is empty and $D = 1$. Then $x_t = -\kappa H_t$ and $H_t = \bar{a} H_{t-1} - y_t$ give $x_t = \bar{a} x_{t-1} + \kappa y_t$. The quadratic $A\sigma_\xi^2 \kappa^2 - (1 - \bar{a})\kappa + A\sigma_F^2 = 0$ has discriminant Δ (`p1_inventory_benchmark`).

Part (iii). The system (10) involves the groups only through G. The price process is a function of the history of flows, and by Theorem 3.1(i) it depends on the groups only through the kernel $k_m = \int r^m \, dM$. A finite measure on $[0, 1]$ is determined by G off its support, and also by its moments.

Part (iv). Write $\Phi(\kappa, D; M) = \big(D + \kappa G(D) - 1,\ \kappa(D - \bar{a}) - A(\sigma_F^2 D + \kappa^2 \sigma_\xi^2 / D)\big)$. At $M = 0$ the zero is $(\kappa_-, 1)$. There the Jacobian with respect to (κ, D) is triangular with determinant $\sqrt{\Delta} \neq 0$: $\partial_D \Phi_1 = 1$, $\partial_\kappa \Phi_1 = G = 0$, and $\partial_\kappa \Phi_2 = (1 - \bar{a}) - 2A\kappa_- \sigma_\xi^2 = \sqrt{\Delta}$. The implicit function theorem, applied in the norm of the statement, gives a unique zero near $(\kappa_-, 1)$ that is continuous in M.

Linearizing, $dD = -\kappa_- G(1)$. Differentiating the second equation and using $A(\sigma_F^2 + \kappa_-^2 \sigma_\xi^2) = \kappa_-(1 - \bar{a})$ gives $d\kappa = -\kappa_-(\bar{a} + 2A\kappa_- \sigma_\xi^2)\, dD / \sqrt{\Delta}$, and so the stated formula. Finally, $G(1) = \sum_i w_i/(1 - r_i) = \sum_i q_i$.

At $M = 0$ the transition matrix has eigenvalues $\bar{a}$ and the r_i, all inside the unit disk, so the dynamics remain stable for small M.

Proof of Theorem 4.2. *Part (i).* Write the implicit form of the price at t, $Dx_t = -\kappa H_t + \sum_i \beta_i r_i u_{i,t}$, and the price law at $t - 1$, $x_{t-1} = -\kappa H_{t-1} + \sum_i \beta_i u_{i,t}$. Use $H_t = \bar{a} H_{t-1} - \sum_i u_{i,t} - \xi_t$. Then

$$Dx_t - \bar{a} x_{t-1} = \kappa \xi_t + \sum_i (\kappa + \beta_i (r_i - \bar{a})) u_{i,t}.$$

Since $\kappa = \beta_i (D - r_i)$, the coefficient is $\beta_i (D - \bar{a}) = \kappa (D - \bar{a})/(D - r_i)$. Divide by D.

The scaled flows inherit the group recursion, because the scaling is constant (`p1_eq_scaled_cohort`). The states $(H_t, u_{\cdot,t+1})$ and $(x_t, \tilde{u}_{\cdot,t+1})$ are related by an invertible linear map, since $\kappa \neq 0$ and every scale factor is nonzero. So the two dynamics are conjugate and have the same eigenvalues.

Part (ii). The effective weights are nonzero because $D \neq \bar{a}$. Apply Theorem 5.3(i) to the closed loop of part (i).

Part (iii). Since $G(D) = \sum_i w_i/(D - r_i) = \sum_i \tilde{w}_i/(D - \bar{a})$, the equation $D = 1 - \kappa G(D)$ reads $D = 1 - \kappa^* D\, \tilde{M}([0, 1])/(D(1 - a^*))$. Solve for D; then $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. Conversely, $(\bar{a}, \kappa, D, M)$ determine the effective objects. So two equilibria give the same price response if and only if their effective objects agree, and then their merged measures agree. This is consistent with Theorem 3.1.

Part (iv).

- *D exceeds every rate.* The hypothesis of Theorem 5.1 is $a^* + \kappa^* W/(1 - r_c - \Delta\eta) < 1$. It gives $\kappa^* W/(1 - a^*) < 1 - r_c - \Delta\eta$, so $D > r_c + \Delta\eta$, which is at least every rate.
- *Each economy is an equilibrium.* With the stated choices, $D + \kappa G(D) = 1$ holds (`p1_eq_realize_D`), and so does $\kappa(D - \bar{a}) = AVD$ with $V = \sigma_F^2 + \kappa^{*2}\sigma_\xi^2$ (`p1_eq_realize_dealer`). By part (i) the equilibrium price response is the given closed loop, which is stable. So each economy is an equilibrium by Theorem 4.1(i).
- *The two economies share everything but the groups.* They have the same total effective weight W, so they share D, κ and the primitives, and their price responses are those of Theorem 5.1.
- *Upper bound.* On a class where the effective parameters are known or estimated at the parametric rate (Corollary S1.2), Theorem 5.2 recovers $\tilde{w}$. Then D follows from part (iii), and $w_i = \tilde{w}_i(D - r_i)/(D - \bar{a})$ is Lipschitz on the class.

Proof of Theorem S2.2. *Part (i).*

- *D lies in $(m, 1)$.* Let $\kappa > 0$ and $D > 0$. The dealer equation $\kappa(D - \bar{a}) = AVD > 0$ gives $D > \bar{a}$. The group equation is $(D - 1)(D - r) + \kappa w = 0$. If $D < r$, it forces $D > 1$, which is impossible; and $D = r$ is excluded. So $D \in (r, 1)$, and $\kappa = \kappa_w(D)$.
- *The equilibria are the solutions of $\Psi_w(D) = A$.* Multiplying the dealer equation by D gives $\kappa(D - \bar{a})D = A(\sigma_F^2 D^2 + \kappa^2\sigma_\xi^2)$, which is $\Psi_w(D) = A$. Conversely, every such D solves the system.
- *Stability.* By Theorem 4.2(i) the dynamics are those of the closed loop with $c = \kappa^*\tilde{w} = (1 - D)(D - \bar{a})/D \in (0, 1)$. Its characteristic polynomial is $p(\lambda) = \lambda^2 - (a^* + r + c)\lambda + a^* r$. The roots are real and nonnegative, because the discriminant is at least $(a^* - r)^2$ and the coefficients alternate in sign. Next, $p(1) = (1 - a^*)(1 - r) - c = (D - \bar{a})(D - r)/D > 0$, so 1 lies outside the interval between the roots. Their product $a^* r$ is below 1, so both roots are below 1.
- *Existence and count.* Ψ_w is continuous and positive on $(m, 1)$ and tends to 0 at both ends: as $D \to 1$, $\kappa_w \to 0$; as $D \to m$, either $D - \bar{a} \to 0$ or $\kappa_w \to 0$. So its maximum A^* is attained. Every $A \in (0, A^*)$ is taken at least twice, once on each side of a maximizer, and no $A > A^*$ is taken.

Part (ii).

- *A parametrization by κ.* For fixed $\kappa \in (0, (1 - r)^2/(4w)]$, the solutions of $\kappa_w(D) = \kappa$ are $D_\pm(\kappa, w) = \big((1 + r) \pm \sqrt{(1 - r)^2 - 4\kappa w}\big)/2$.
- *The larger root gives the maximum.* $\tilde{\Psi}(\kappa, D) = \kappa(D - \bar{a})D/(\sigma_F^2 D^2 + \kappa^2\sigma_\xi^2)$ is strictly increasing in D for $D > \bar{a}$. The numerator of its D-derivative is $\bar{a}\sigma_F^2 D^2 + (2D - \bar{a})\kappa^2\sigma_\xi^2 > 0$. So $A^*(w) = \max_\kappa \tilde{\Psi}(\kappa, D_+(\kappa, w))$.
- *Strict decrease.* Let $w_1 < w_2$, and let κ° attain $A^*(w_2)$. Then κ° is feasible at w_1, and $D_+(\kappa^\circ, w_1) > D_+(\kappa^\circ, w_2)$. So $A^*(w_1) \geq \tilde{\Psi}(\kappa^\circ, D_+(\kappa^\circ, w_1)) > A^*(w_2)$.
- *The limit as $w \downarrow 0$.* The bound $A^* < A_0^*$ of part (iii) is one half. For the other, fix $\kappa_0 = \sigma_F/\sigma_\xi$. As $w \downarrow 0$, $D_+(\kappa_0, w) \to 1$, and $\tilde{\Psi}(\kappa_0, D_+) \to (1 - \bar{a})/(2\sigma_F\sigma_\xi)$.

Part (iii).

- *The bound A_0^*.* By the arithmetic–geometric mean inequality, $\sigma_F^2 D^2 + \kappa^2\sigma_\xi^2 \geq 2\sigma_F\sigma_\xi\kappa D$. So $\Psi_w(D) \leq (D - \bar{a})/(2\sigma_F\sigma_\xi) < A_0^*$. Machine-checked in squared form as $4A^2\sigma_F^2\sigma_\xi^2 \leq (D - \bar{a})^2$ (`p1_fragility_benchmark_bound`).

- *The bound $\Phi^*/(\sigma_F^2 w)$.* Dropping $\kappa^2\sigma_\xi^2 > 0$ from the denominator gives $\Psi_w(D) < (1-D)(D-r)(D-\bar{a})/(\sigma_F^2 wD) \le \Phi^*/(\sigma_F^2 w)$. Machine-checked as $A\sigma_F^2 wD \le (1-D)(D-r)(D-\bar{a})$ (`p1_fragility_feedback_bound`).
- *The bound on Φ^*.* It uses $(1-D)(D-r) \le (1-r)^2/4$ and $(D-\bar{a})/D \le 1-\bar{a}$.
- *The limit as $w \to \infty$.* We have $\kappa_w = O(1/w)$ uniformly on $[m,1]$. So $w\Psi_w(D) \to (1-D)(D-r)(D-\bar{a})/(\sigma_F^2 D)$ uniformly, and the maxima converge.

Part (iv).

- *κ is determined by D.* Let $D > \max_i r_i$. All weights are positive, so $G(D) > 0$. Then $D = 1-\kappa G(D)$ with $\kappa > 0$ gives $D < 1$ and $\kappa = (1-D)/G(D)$. The dealer equation then reads as in part (i), and $D > \bar{a}$ as there.
- *Comparison with $G(1)$.* For $D \in (\max_i r_i, 1)$, each $D - r_i < 1 - r_i$, so $G(D) > G(1)$.
- *The bound.* The dealer equation gives $A\sigma_F^2 D < \kappa(D-\bar{a}) < (1-D)(D-\bar{a})/G(1)$. Moreover

$$(1-D)(D-\bar{a}) = D(1-\sqrt{\bar{a}})^2 - (D-\sqrt{\bar{a}})^2 \le D(1-\sqrt{\bar{a}})^2.$$

  Machine-checked as `p1_fragility_K_bound`. The bound A_0^* follows as in part (iii).
- *Stability.* By Theorem 4.2(i), the dynamics are those of the closed loop with $(a^*, \kappa^*, \tilde{w})$. Here $a^* \ge 0$, $\kappa^* > 0$, $\tilde{w}_i > 0$ and $r_i \ge 0$, so its transition matrix in the state $(x_t, y_{1,t}, \ldots, y_{K,t})$ is nonnegative.
  - *The spectral radius is the root above the rates.* By Perron–Frobenius, the spectral radius ρ is an eigenvalue. Every eigenvalue $z \notin \{0, r_i\}$ solves $F(z) = 1$, where $F(z) = a^*/z + \kappa^* \sum_i \tilde{w}_i/(z - r_i)$. No r_i is an eigenvalue, because $\tilde{w}_i \ne 0$ (Theorem 5.3(i)). F decreases strictly from $+\infty$ to 0 on $(\max_i r_i, \infty)$, so it has exactly one root z^* there, and $\rho \ge z^*$. Being real and at least z^*, ρ equals z^*.
  - *The criterion.* Hence $\rho < 1$ if and only if $F(1) < 1$, that is, $a^* + \kappa^* \sum_i \tilde{w}_i/(1-r_i) < 1$. With $g_i = w_i/(D-r_i)$ this reads $\kappa \sum_i g_i/(1-r_i) < 1$ (`p1_eq_effective_stability_iff`).
  - *It holds.* Each $1 - r_i > 1 - D > 0$ and $\kappa \sum_i g_i = 1 - D$, so $\kappa \sum_i g_i/(1-r_i) < 1$ (`p1_eq_stability_K`).

Proof of Corollary S2.1. *Part (i).*

- The group recursion is part of the model.
- By Theorem 4.2(i), the price is the closed loop with $a^* \ge 0$, $\kappa^* > 0$ and $\tilde{w}_i > 0$, driven by the i.i.d. flow ξ.
- It is stable by Theorem S2.2(iv).
- These are the hypotheses of Proposition 6.2. Its regression involves only group i's own flow and the price, so it delivers the true (r_i, w_i).

Part (ii).

- *Price-only side.* By Theorem 4.2, the price-only experiment is the reduced form at the effective parameters. The realization of Theorem 4.2(iv) puts the Gauss and Gauss–Radau pair inside the equilibrium class, so the price-only side of Theorem 6.3 holds with the effective weights.

- *Tagged side.* Each group's flow response to a unit impulse is $f_{i,t} = w_i \sum_{s \le t} r_i^{t-s} h_s$, where h is the equilibrium response and $h_0 = \kappa^* > 0$. That is the tagged record of Theorem 6.3, and its argument uses only that h is bounded with $h_0 > 0$.
- On the stated classes the true weights are a Lipschitz function of the effective ones, and conversely, so the orders are unchanged.

Numerical checks of Theorems 4.2 and S2.2. The script `equilibrium/e4_reduced_form.py` runs three checks.

- *Reduced form.* For four solved equilibria, the equilibrium price path and the reduced-form path with $(a^*, \kappa^*, \tilde{w})$ agree to 4.4×10^{-16} over 5,000 simulated periods. The four are: one group with positive feedback, one with negative feedback, two groups, and the Gauss pair. Their impulse responses agree to 10^{-16}. For one group of weight 0.02 at $r = 0.7$, for example, $a^* = 0.629$, $\kappa^* = 0.621$ and $\tilde{w} = 0.0279$.
- *Realization.* Take the Gauss and Gauss–Radau pair with $K = 2$, $a^* = 0.5$, $\kappa^* = 0.5$, $W = 0.2$, $r_c = 0.6$ and $\Delta\eta = 0.1$, so that the stability index of Theorem 5.1 is 0.833. Both economies are equilibria with $D = 0.8$, $\bar{a} = 0.4$, $\kappa = 0.4$ and $A = 0.188$, and both equilibrium equations hold to machine precision.
  - Their true group weights are $(0.064, 0.036)$ and $(0.0375, 0.0625)$.
  - Their equilibrium impulse responses differ in ℓ_2 by $1.22\,d^3$, $1.25\,d^3$ and $1.29\,d^3$ at $d = 0.1$, 0.05 and 0.025. That is the order $(\Delta\eta)^{2K-1}$.
- *Fragility.* The maximum of Ψ_w reproduces the bisection thresholds of the solver to 10^{-12} ($A^* = 0.2547$, 0.1297, 0.0677, 0.0342 at $w = 0.02$, 0.05, 0.1, 0.2).
  - $A^* = 0.3967$ at $w = 5 \times 10^{-4}$, against the limit 0.4.
  - $w\,A^* = 0.006875$ at $w = 1$ and 0.0068764 at $w = 10$, against $\Phi^* = 0.0068764$.

Several groups. `equilibrium/e4_fragility_K.py` scans D for three populations at weights 0.005 to 1 per group. The populations are two groups at rates 0.5 and 0.85, three at 0.3, 0.6 and 0.8, and the Gauss pair.

- The exact threshold A^* lies below the part (iv) bound in all 15 cases, at 28% to 86% of it. For example, for two groups with $G(1) = 1.73$, $A^* = 0.0084$ against a bound of 0.0293.
- Every equilibrium on the threshold curve is stable. At the threshold the spectral radius is between 0.79 and 0.93; it approaches 1 only at the end of the curve where $D \to \max_i r_i$ and $A \to 0$.

The result files are `equilibrium/e4_reduced_form_results.json` and `equilibrium/e4_fragility_K_results.json`.

Proof of Theorem S2.3. *Part (i).*

- *Dealers' information.* The groups' flow is predictable. So the dealers' information about θ_t is ν_t, and $E[\theta_t \mid \nu_t] = \lambda \nu_t$ with $\lambda = \mathrm{Cov}(\theta, \nu)/\mathrm{Var}(\nu) = b\sigma_\theta^2/\sigma_\nu^2$.
- *The price decomposes.* $E_t F_{t+1} = F_t + \lambda \nu_t$ and $E_t \nu_{t+1} = 0$, so $E_t p_{t+1} - p_t = E_t x_{t+1} - x_t$. The dealers' condition is therefore the one of Theorem 4.1. Its state is $(H_t, u_{\cdot,t+1})$, and $E_t u_{i,t+2} = r_i u_{i,t+1} + w_i E_t x_{t+1}$. Coefficient matching gives the same equations for β, D and κ (`p1_inventory_necessity`).

- *The price's response to ν_t.* Since the groups trade on $s_t = x_t + \lambda\nu_t$ and $\sum_i \beta_i w_i = 1 - D$, the implicit form is $Dx_t = -\kappa H_t + \sum_i \beta_i r_i u_{i,t} + (1-D)\lambda\nu_t$. The response of s_t to ν_t is therefore $(\kappa + (1-D)\lambda)/D + \lambda = (\kappa + \lambda)/D$.
- *The insider.* $F_{t+1} - p_t = \theta_t + \varepsilon_{t+1} - m_t - c\,\xi_t - c\,d_t$, where $c = (\kappa + \lambda)/D$ and m_t is the predictable part of s_t. The insider does not observe the public state, and m_t has mean zero and is independent of θ_t. So she maximizes $(\theta - c\,d)\,d$. So $d = \theta/(2c)$ and the profit is $\theta^2/(4c)$ (`p1_insider_best_response`). Taking expectations, $b\sigma_\theta^2/2$.
- *The variance.* The innovation of p_{t+1} given dealers' information is $(\theta_t - \lambda\nu_t) + \varepsilon_{t+1} + c\,\nu_{t+1}$, which gives V (`p1_insider_posterior_variance`).
- *Converse.* A solution of the system satisfies every condition just used.

Part (ii). With $A = 0$, $\kappa(D - \bar{a}) = 0$ forces $\kappa = 0$, and then $D = 1$. Then $b = 1/(2\lambda)$ and $\lambda\sigma_\nu^2 = b\sigma_\theta^2$ give $4\lambda^2\sigma_\xi^2 = \sigma_\theta^2$ (`p1_insider_kyle_limit`).

Part (iii). The groups enter only through $D = 1 - \kappa G(D)$.

Part (iv).

- *The reduced form.* Combine the implicit form at t, the price law at $t-1$ and $H_t = \bar{a}H_{t-1} - \sum_i u_{i,t} - \nu_t$, as in Theorem 4.2. This gives $Ds_t - \bar{a}s_{t-1} = \sum_i \beta_i(D - \bar{a})u_{i,t} + (\kappa + \lambda)\nu_t - \bar{a}\lambda\nu_{t-1}$ (`p1_insider_reduced_form`).
- *The filter is invertible.* $c_1 < c_0$ because $\bar{a} < 1$, $\lambda \geq 0$ and $\kappa \geq 0$. So the observer, knowing the primitives, inverts the filter, and is left with the closed loop of Theorem 4.2.
- *Realization.* Proceed as in Theorem 4.2(iv):
  - set $D = 1 - \kappa^* W/(1 - a^*)$, $\bar{a} = a^* D$ and $\kappa = \kappa^* D$;
  - solve the insider's pair (λ, b) given (κ, D);
  - set $A = \kappa(D - \bar{a})/(VD)$.

  The two economies then share every primitive.

Numerical check of Theorem S2.3. The script `equilibrium/e6_insider.py` solves the system at $A = 0.05$, $\sigma_\varepsilon^2 = 1$, $\sigma_\theta^2 = \sigma_\xi^2 = 0.25$ and $\bar{a} = 0.6$, and simulates 2×10^5 dates for each solution.

Groups	κ	D	λ	b	$(\kappa+\lambda)/D$	profit	inform.
none	0.1656	1	0.4835	0.7704	0.6490	0.0963	0.372
+0.02 at 0.7	0.1692	0.9883	0.4816	0.7592	0.6586	0.0949	0.366
−0.2 at 0.7	0.1463	1.0775	0.4928	0.8430	0.5931	0.1054	0.415
two groups, or split†	0.1679	0.9924	0.4823	0.7632	0.6552	0.0954	0.368

Informativeness (inform.) is $b^2\sigma_\theta^2/\sigma_\nu^2$, the share of $\mathrm{Var}(\theta)$ resolved by the price. †Weights $0.005, 0.005$ at rates $0.5, 0.85$, or the same merged measure split into three groups; both give identical coefficients.

- In every case, the dealers' first-order residual is below 5×10^{-17}, and the observer's reduced form holds to 4.4×10^{-16}.
- The regression of θ on ν reproduces λ to the third decimal.
- The simulated profit at $b' = 0.6b, \ldots, 1.4b$ peaks at b, as does the analytic profit.
- As $A \downarrow 0$ (from 0.05 to 0.001), (λ, b) moves from $(0.482, 0.759)$ to $(0.49999, 0.994)$, toward Kyle's $(0.5, 1)$.

Numerical solution of Theorem 4.1. The script `equilibrium/e4_inventory.py` solves the system (10) at $A = 0.2$, $\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$ and $\bar{a} = 0.6$.

How the solver finds equilibria.

- For one group with $w > 0$, every equilibrium with $\kappa > 0$ and $D > 0$ has $D \in (\max(\bar{a}, r), 1)$ and $\kappa = (1 - D)(D - r)/w$. A scan over D therefore finds all of them.
- For several groups, the solver follows the branch of $D(\kappa)$ that is continuous with $D = 1$.
- Each solution is checked by simulating 2×10^5 periods of the closed loop. The check computes the dealers' first-order residual and the innovation variance of the price.

Benchmark. Here $\Delta = 0.12$, and the two equilibria are $\kappa_- = 0.536$ and $\kappa_+ = 7.46$, as in part (ii).

Results.

Groups (weights at memories)	κ	D	κ/D	spectral radius
none	0.5359	1	0.5359	0.600
0.02 at 0.7	0.5916	0.9533	0.6206	0.787
-0.2 at 0.7	0.4212	1.1767	0.3579	0.597
$0.005, 0.005$ at $0.5, 0.85$	0.569147	0.970292	0.586573	0.881
$0.0025, 0.0025, 0.005$ at $0.5, 0.5, 0.85$	0.569147	0.970292	0.586573	0.881
Gauss pair, $0.01, 0.01$	0.5759	0.9649	0.5969	0.766
Radau pair, $0.005, 0.015$	0.5743	0.9662	0.5945	0.738
0.2 at 0.7; or $0.1, 0.1$ at $0.5, 0.85$	no equilibrium			

- *The simulation checks hold in every case.* The relative first-order residual is below 4×10^{-16}. The simulated innovation variance matches $\kappa^2 \sigma_\xi^2 / D^2$ to within 0.3%.
- *Invariance (part (iii)).* The split configuration has the same merged measure as the two-group one, and its coefficients agree to all printed digits.
- *Gauss and Radau pairs.* They agree on the first three kernel moments, and their impacts differ by 0.0016.
- *Negative feedback* lowers impact.

First-order formula of part (iv), for one group at $r = 0.7$:

- at $w = 0.005$, $\kappa = 0.5457$ against the first-order value 0.5449, and $D = 0.9906$ against 0.9911;
- at $w = 0.01$, $\kappa = 0.5574$ against 0.5540.

Doubling w multiplies the error by about 4.5, as a second-order remainder should.

Existence region. For one group at $r = 0.7$, an equilibrium exists for dealer risk aversion below a threshold A^*. This is computed by bisection, with the full scan over D at each step:

w	0	0.02	0.05	0.1	0.2
$G(1) = w/(1 - r)$	0	0.067	0.167	0.333	0.667
A^*	0.400	0.255	0.130	0.068	0.034

- Without feedback, $A^* = (1 - \bar{a})/(2\sigma_F \sigma_\xi) = 0.4$, which is the closed form of part (ii).
- For larger $G(1)$, A^* falls roughly in proportion to $1/G(1)$.
- At the threshold the two equilibria merge and disappear.

The results file is `equilibrium/e4_inventory_results.json`.

Numerical checks.

- `equilibrium/e2_rents.py`, with $T = 10$, $\lambda = 0.5$ and $\Delta k_m = \pm 0.4\,\epsilon\,(0.7)^m$:
  - the exact rent is -0.322ϵ when the dealer understates feedback and $+0.322\epsilon$ when it overstates it, as $\epsilon \to 0$;
  - this matches the first-order formula $\mp\Sigma_0 \sum_m \omega_m |\Delta k_m|$ to three digits at $\epsilon = 0.0125$, and to within 6% at $\epsilon = 0.4$.
- `equilibrium/e3_halt.py`, with the market of `e1_demo.py`, $c = 0.6$, $W = 0.5$, 200 draws and 40 dates:
  - the largest pre-halt price gap is 3.8×10^{-4}, 2.9×10^{-3} and 2.3×10^{-2} for $d = 0.05$, 0.1 and 0.2, a ratio to d^3 of 3.0, 2.9 and 2.8;
  - the largest post-halt gap is 0.196, 0.190 and 0.177.

The result files are `e2_rents_results.json` and `e3_halt_results.json`. Parts of both theorems are machine-checked.

- *Theorem S2.4(i):* the symmetrization, the optimal schedule and its value (`Paper1EquilibriumExtras.lean`).
- *Theorem S2.5(i):* the kernel bound $|k^G_m - k^R_m| \le 2W\binom{m}{3}|d|^3(|c| + |d|)^{m-3}$ for every $m \ge 3$ (`Paper1HaltKernelBound.lean`).
- *Theorem S2.5(iii):* the post-halt gap $W/4$.

The remaining parts are proved on paper.

S3 Execution cost and capital

Theorem S3.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0,1)$. Compare a single group of weight q and rate η with two groups of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1-\epsilon]$. All parameters are known. The price receives executed quantity through κv_t, starts from zero, has no process noise, and is observed with independent $N(0,\sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h^\delta_j - h^0_j)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.*

(i) For every probe with $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{13}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e^*_T(V) = \Phi\Big(-\frac{V\|d_T\|_2}{2\sigma}\Big), \tag{14}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed V. Error at most $\alpha < 1/2$ requires $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ independently of T.

(iii) The first two taps coincide and $d_2 = -\kappa^2 q\delta^2$. A single pulse of size

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{15}$$

and the third observation achieve error exactly α. The necessary and sufficient quantity order is δ^{-2}.

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$, the necessary horizon order is δ^{-4}, attained by the pulse whenever $T \ge \max\{3, \lceil A^2/P \rceil\}$.

(v) For a fixed monetary price exposure $\lambda \neq 0$, additive reserve L and cost envelope $c\sum|v_t|$, the sharp robust quantity allowance under budget $B > L$ is $V_B = (B-L)/(|\lambda|\kappa + c)$.

In words. To tell one desk from two desks whose memories differ slightly, the observer must trade against them, and the quantity it must trade grows like $1/\delta^2$: halving the difference quadruples the trading needed. Waiting longer after a fixed trade does not help, because the trade's echo in the price dies out.

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and κ currency per squared quantity.

Lemma S3.2 (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{16}$$

In the positive stable economy T_h is positive definite, so no nonzero round trip produces a deterministic gain.

Positivity here means that the response is a positive mixture of decaying exponentials (Lemma S1.1), whose spectral density is positive. Positive, decreasing taps alone do not suffice: for $h = (1, 0.9, 0.1)$ the 3×3 matrix T_h has determinant -0.468, so some round trip earns money. The conclusion concerns modelled execution cost only.

We verify the ledger identity (16) against an explicit cash account to 5.7×10^{-13} on 1,000 random probes, and the smallest eigenvalue of T_h over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

S3.1 Capital supplied by a counterparty

Theorem S3.3 (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \ge 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \le E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \le V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{17}$$

equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \le B$. The cap holds across all positive group representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least δ^{-4}, and a single positive pulse attains that order.

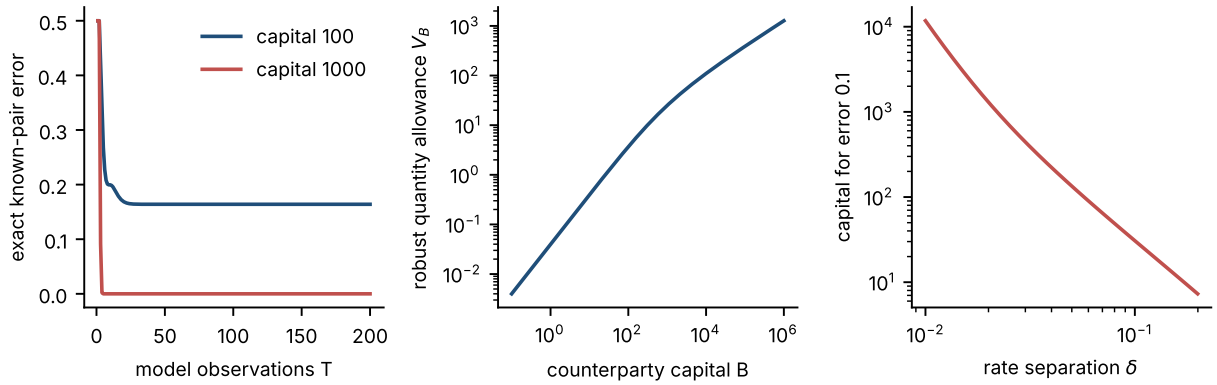


Figure S1: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (17). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

In words. A probe moves the price against the account that makes it, so the margin it needs grows with the square of the probe. Because the probe grows like $1/\delta^2$, the capital grows like $1/\delta^4$: halving the difference multiplies the capital needed by 16.

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure S1 illustrates the experiment.

Proposition S3.4 (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \le V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

In words. If the observer can see an account's exact equity, a single test trade settles the question, however close the two candidate markets are. The hard part of the problem is therefore not the market itself but what the observer is allowed to see.

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (see the disclosure contracts of the full working paper).

S3.2 Capital paid by the probe trader

Lemma S3.5 (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence v changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to P_*. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa\frac{1-r}{1+r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3(1-S)^2}.$$

Every solvent strategy satisfies $D_{\mathrm{KL}}(P_0^v \| P_\delta^v) \le K_*^2 B\delta^4/(\sigma^2 m)$*, so error at most* $\alpha < 1/2$ *requires*

$$B \;\ge\; \frac{2\sigma^2 m(1-2\alpha)^2}{K_*^2\delta^4}, \tag{18}$$

without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$*, the pulse* A *in* (15) *achieves error* α *and remains margin-solvent at every later mark with capital* $B_A = \frac{\kappa}{2}A^2 + (c+\mu P_*)A$*. Necessary and sufficient capital orders are* δ^{-4}*.*

Lemma S3.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the δ^{-4} rate before the exact frequency-domain optimization of the account-level analysis of the full working paper. Its explicit constant is very conservative, because it scales as $m/K_*^2 \propto \epsilon^7(1-S)^5$: at the calibrated parameters it is 4.9×10^{-4} par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any feasible design. We therefore state it as a lemma; the account-level analysis of the full working paper computes the sharp constant, within 5% of a feasible design at every separation.

Theorem S3.1. In either positive economy $J \ge 0$ and $\|J\|_\infty \le 1-\epsilon(1-S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1-S)$. From the display above, $|G_\delta - G_0| \le 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \le K_*\delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \le K_*\delta^2\|v\|_1$, and Gaussian KL with Pinsker gives (13). For (ii), the mean difference is a lower-triangular convolution matrix times v; each column is a shifted copy of a prefix of d_T, so $\|\mu_\delta - \mu_0\|_2 \le V\|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters h_2 with factor κ^2 while h_0, h_1 coincide; a pulse A separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the ℓ_2 gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t |\lambda x_t| \le |\lambda|\kappa V$, attained by a single pulse.

Lemma S3.2. After the uncontrolled update, the pre-probe displacement at step j is $\sum_{i<j} h_{j-i}v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2}v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \le \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

Theorem S3.3. Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \le x_t \le \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x_{j+1}^0 - \frac{\kappa}{2}\sum_{j<t} v_j^2 + cU_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of h bounds every entry by $\kappa/2$. Hence $B - E_t \le \kappa U_t^2/2 + cU_t \le \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (17). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse A funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

Proposition S3.4. The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

Lemma S3.5. Marking residual inventory at P_* gives $E_\infty = B - \frac{1}{2}v^\top T_h v - c\sum|v_t| \geq 0$. For each mode $\lambda \leq r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1-2\lambda\cos\omega+\lambda^2) \geq (1-r)/(1+r)$, so $v^\top T_h v \geq m\|v\|_2^2$ and $\|v\|_2^2 \leq 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (18). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory A; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \leq 1$.

Account-level capital. The full working paper refines these orders at the level of a single trading account (a three-region partition of probe designs, adaptive probing, heteroskedastic impact) and gives a hypothetical illustration in dollars. No result here uses them.

S4 Two unknown groups: pole fitting and the global two-point bound

Because the memory rates are zeros of H (Theorem 3.3), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A three-step inversion repairs this: (1) compute a minimal realization of H itself from the Hankel matrix of $h_1, h_2, \ldots$ (with h_0 as the direct term) by Kung's eigensystem realization; (2) compute its transmission zeros, which are the memory rates and the timing zero at 0 (transforming the sequence first, for example to $h_{t+1} - ah_t$, cancels these zeros); (3) project reconstructed flow onto the recovered memory filters to obtain the weights (full working paper, internet appendix). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-group systems exactly; with noise, per-group weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

S4.1 A global two-point bound for any estimator

A Cramér–Rao benchmark (full working paper) applies only to estimators that are unbiased near the true parameter. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

Theorem S4.1 (Global two-point bound and matching upper bound, $K = 2$)**.** *In this theorem w_k denotes group k's trading intensity q_k, not the merged weight $q_k\eta_k$ of Section 2, and $\Delta\eta$ is the actual spacing of the two rates. Fix total intensity $q = 0.20$, mean rate $\bar\eta = 0.40$, target separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$ (so that $\bar\eta - p_+\Delta\eta > 0$). Let $p_\mp = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar\eta - p_+\Delta\eta, \bar\eta + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar\eta - p_-\Delta\eta, \bar\eta + p_+\Delta\eta). \tag{19}$$

(i) Label groups in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k\eta_k^m$ for $m \in \{0,1,2\}$, while

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{20}$$

Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.

(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-group economies

Table S1: Two-point sample complexity for two-group weight recovery

$\Delta\eta$	ΔM_3	$\|\Delta h\|_2$	N^*_{LeCam} (95%)	N_{CR}
0.30	2.67×10^{-4}	1.98×10^{-3}	7.1	47.8
0.20	7.92×10^{-5}	5.60×10^{-4}	88.2	597.0
0.14	2.72×10^{-5}	1.89×10^{-4}	775.3	5,267.9
0.10	9.90×10^{-6}	6.84×10^{-5}	5,927.6	40,374.2
0.08	5.07×10^{-6}	3.49×10^{-5}	22,740.5	155,048.9
0.06	2.14×10^{-6}	1.47×10^{-5}	128,333.0	875,732.3
0.05	1.24×10^{-6}	8.50×10^{-6}	383,859.1	2,620,321.2
0.04	6.34×10^{-7}	4.35×10^{-6}	1,466,361.5	10,012,662.4

Worst-case failure at $\Delta\eta = 0.10$, $N = 50$, *1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Estimators: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least τ.

with total weight q, *mean rate* $\bar\eta$ *and spacing* $\Delta\eta$,

$$\sup_\theta P_\theta(\|\hat w - w(\theta)\|_\infty \ge \tau) \;\ge\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{21}$$

so failure probability at most 0.05 *requires* $N \ge 4\sigma_h^2 z^2_{0.95}/\|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. *For* $\Delta\eta \ge 2\tau q/(q + 2\tau)$ *(*≈ 0.018 *at the stated values) the same bound holds without any labelling convention, for the permutation-invariant loss*

$$d_\infty(\hat\theta, \theta) = \min_{\pi \in S_2} \max_k \big(|\hat w_{\pi(k)} - w_k| + |\hat\eta_{\pi(k)} - \eta_k|\big),$$

because then $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1 + \Delta\eta/q),\, (1 + 2\tau/q)\Delta\eta\} \ge 2\tau$. *For smaller spacings the relevant permutation-invariant statement is the Wasserstein bound of Theorem 5.1.*

(iii) Conversely, the matching upper bound $N = O((\Delta\eta)^{-6})$ *on rich separated two-group classes is Theorem 5.2 with* $K = 2$. *On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant* $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ *for the deconvolved kernel* $g_m = \sum_i w_i \eta_i r_i^m$, *and perturbation analysis gives the same order.*

In words. For two unknown groups, no method can tell the two candidate markets apart reliably with fewer than a number of experiments of order $(\Delta\eta)^{-6}$, and a practical method (Prony's method, Theorem 5.2) achieves this order. Halving the spacing multiplies the data needed by 64.

We recomputed the two-point quantities exactly (Table S1): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 re-sets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table S1).

Remark S4.2 (The hard directions carry no risk). The two economies of Theorem S4.1 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1 - a - \kappa \sum_i q_i)$ is identical, 2.353 at the calibration of Table S1: the directions in which prices are least informative about the

groups are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; the companion paper turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Niroula et al., 2026). Identification and safety ask different questions of the same data.

S5 The supervisory decision in detail

This section completes Section 6.

Proof of Proposition 6.1.

Proof of Proposition 6.1. Part (i) follows by scaling the intensities. For (ii), expand $(1-r_c-\Delta\eta s)^{-1}$: the first three moments agree, so aggregate budgets differ by order $(\Delta\eta)^3$. Each intensity converges to $W\omega_i/(1-r_c)$, giving the stated dominant-weight gap. This exceeds the aggregate discrepancy at sufficiently small spacing. A ceiling between the two post-halt budgets reduces the action problem to testing two Gaussian mean vectors. Their common-covariance testing error is $\Phi(-\sqrt{N}\|h^G - h^R\|_2/(2\sigma))$; the response bound in Theorem 5.1 gives the lower bound. Part (iii) follows from continuity of $w/(1-r)$ on the declared compact parameter set and the fixed margin. □

General-K scope. The same reduction gives exponent $4K-2$ for a pair whose price responses have that moment-matching order *and* whose dominant intensities have a fixed gap. Moment matching alone does not supply the gap: the symmetric odd/even binomial pair has equal maximal normalized weights. Lemma S5.3 and Proposition S5.4 below supply a pair with a gap for every K. The $K=3$ rows below are numerical examples for the Gauss and Gauss–Radau pair.

Explicit robustness condition. Smooth best responses alone do not preserve separation: the constant smooth map $\Phi(M)=0$ sends both post-halt budgets to a. Instead assume that, over the stated action and holding horizon, $|\widetilde{R}_j - R_j| \le \varepsilon$ for $j \in \{G,R\}$, with $R_R < R_G$. Then

$$R_R + \varepsilon \le \bar{s} < R_G - \varepsilon \quad \Longrightarrow \quad \widetilde{R}_R \le \bar{s} < \widetilde{R}_G. \tag{22}$$

Indeed, $\widetilde{R}_R \le R_R + \varepsilon$ and $\widetilde{R}_G \ge R_G - \varepsilon$. A nonempty robust window exists if $2\varepsilon < R_G - R_R$. This is a maintained response bound, not an equilibrium result. The same allowance must cover the alternative uniform action. At the $K=2$, $\Delta\eta = 0.10$ calibration, $R_R = 0.532$, $R_G = 0.569909$, and $\bar{s} = 0.55$ has margin 0.018; an allowance 0.005 fits within it.

A complete conditional cost-and-error model. Let j denote the economy, π_j its declared prior, H the holding horizon, b the surplus per active class per period, C_h the implementation cost of halting and L the penalty per period for violating the ceiling. These are assumed accounting primitives, not estimated social welfare. An envelope $S_{\max}$ supports the robust uniform action

$$\theta_\varepsilon = \min\{1, \max\{0, (\bar{s} - \varepsilon - a)/(S_{\max} - a)\}\}, \qquad L_T = HbK(1-\theta_\varepsilon).$$

This action is feasible only if its worst-case budget meets the ceiling (in particular $a + \varepsilon \le \bar{s}$). For a halt,

$$L_{h,j} = Hb + C_h + HL\,\mathbf{1}\{R_j + d_j > \bar{s}\}, \quad \mathcal{L} = F + Nc + \sum_j \pi_j\{p_j L_{h,j} + (1-p_j)L_T\}.$$

Here $F + Nc$ is the information bill, p_j the probability of halting, and $|d_j| \leq \varepsilon$ the action response. Require $\Pr_j(\text{halt and violate}) \leq \alpha$ separately in each world. Compare this loss with the best feasible no-purchase lottery over the *same two actions*, and allow refusing to buy information. Neither arbitrary alternative policies nor an optimum over sample size are claimed. Active probing additionally incurs its execution loss and capital requirement; these are not automatically the per-record fee c and must not be counted twice.

Existing runs and the remaining synthetic check. The archived `R_P1_targeted_intervention.json` supplies six response/decision rows. Recomputing them independently from the stated state recursion agrees to 2.1×10^{-17} in the response discrepancy. Those rows supply S_j, R_j and Gaussian testing distances; they contain no measured disclosure fees, intervention payoffs or strategic response bounds. The existing GPU and historical diagnostics cannot fill these quantities by relabelling their outputs. The new CPU study `studies/supervisory_release_20261105/run_study.py` therefore evaluates only this missing accounting layer, with exact Gaussian probabilities rather than a new large simulation.

For this check alone, assume independent scalar disclosures $Z_n \sim N(R_j, \sigma_t^2)$ and halt when $\bar{Z} + z_{1-\alpha}\sigma_t/\sqrt{N} + \varepsilon \leq \bar{s}$. This is an illustrative measurement contract, *not* a claim that class positions or tagged flow directly observe post-halt budgets. If a world is unsafe under the response allowance, this gate falsely halts with probability at most α under the stated unbiased Gaussian law. The price comparator observes the full 400-tap Gaussian reset record of the same known pair, with a likelihood-ratio rule constrained by the same per-world error budget. Both information channels have the same assumed monetary fee here; neither represents free passive observation.

With $K = 2$, $N = 32$, $\sigma_t = 0.02$, $\varepsilon = 0.005$, $\alpha = 0.05$, $\pi_R = 1/2$, $H = b = 1$, $C_h = 0.02$, $L = 10$, $F = 0.05$, and $c = 0.0005$, the uniform loss is 1.449219, the information bill is 0.066, and disclosure loss is 1.305132. Its conditional net advantage is 0.144087 surplus units. Halt probabilities are approximately zero in G and 0.978928 in R. Buying the price record yields loss 1.515219 at this sample size; its optimal acquisition decision is to decline and retain the no-data action. An informed oracle before information fees has loss 1.234609.

Across 2,304 deterministic configurations varying fees, noise, horizon, prior, sample size and allowance, paid disclosure improves on the no-data benchmark in 1,207, worsens it in 1,079, and ties in 18. These counts have no population or statistical-significance interpretation. Negative cases remain in the output. A downward measurement bias of 0.03 produces false-halt probability 0.418776; a future response of 0.025 despite an allowance of 0.005 makes halting unsafe in R with probability 0.978928. The latter also invalidates the uniform-action guarantee, so payoff comparisons under that misspecification are not presented as a fair safety comparison. The experiment establishes conditional accounting and failure modes, not real-market gains, a validated equilibrium response, or the general-K theorem.

Proposition S5.1 (A robust scalar disclosure gate)**.** *Let $Z_1, \ldots, Z_N$ be independent $N(R, v)$ readings of the fixed post-halt feedback budget, $N > 0$, and let the realized action response d satisfy $d \leq \varepsilon$. For $t > 0$, halt only if $\bar{Z} + t + \varepsilon \leq \bar{s}$. Then the probability of halting while the realized budget exceeds the ceiling is at most*

$$\Pr\{\bar{Z} + t + \varepsilon \leq \bar{s},\ R + d > \bar{s}\} \leq \frac{v}{Nt^2}.$$

Consequently $v \leq \alpha N t^2$ gives a per-world error bound α.

Proof. On an unsafe-halt event, $\bar{Z} + t + \varepsilon \leq \bar{s} < R + d \leq R + \varepsilon$, so $t \leq |\bar{Z} - R|$. Chebyshev's inequality for the actual Gaussian sample mean gives the bound. Lean checks the product Gaussian

law, the sample statistic, this event inclusion and the probability inequality in `p1_supervisory_gaussian_unsafe_halt`; it is not merely a check of a scalar named "probability bound." The response allowance and the interpretation of Z remain explicit assumptions. □

This conservative gate uses $t = \sigma_t/\sqrt{N\alpha}$. It differs from the sharper Gaussian-quantile gate above, whose exact normal probabilities are numerically evaluated but whose quantile identity is not formalized here. Under the same calibration with $N = 128$, the conservative gate has radius 0.007906, information bill 0.114, loss 1.349034 and conditional net advantage 0.100185. Its halt probabilities are approximately zero in G and 0.998023 in R. Across the same 2,304 sensitivity configurations it improves on the fixed-menu no-data benchmark in 781, worsens it in 1,499 and ties in 24. Both gates' rows remain in the result file: 4,608 configurations in total. The welfare and error interpretation remains conditional on the declared observation and response contracts.

Operational bridge from tagged flows to a halt decision. The scalar-reading contract above can be replaced, within the reset model, by five observed channels: two first class-flow responses, two second responses, and the first price response. For a unit pulse write $f_{i,0} = \kappa w_i$, $f_{i,1} = w_i(\kappa r_i + h_1)$, and $h_1 = \kappa a + \kappa^2 \sum_i w_i$. Thus

$$w_i = f_{i,0}/\kappa, \qquad r_i = f_{i,1}/f_{i,0} - h_1/\kappa, \qquad q_i = w_i/(1 - r_i).$$

Assume known impact, verified stable class membership, $0 \le r_i \le 0.95$, independent unbiased Gaussian reset readings with per-channel variance σ_f^2, and N independent records. A simultaneous radius $e = \sigma_f\sqrt{5/(N\alpha)}$ covers the five sample means with probability at least $1 - \alpha$ by Chebyshev and a union bound. Interval division, only when the first-flow lower endpoint is positive, gives bounds on w_i, r_i, q_i; intersect rate bounds with the declared class. Halt only if the remaining upper budget plus the response allowance meets the ceiling. Otherwise retain the same robust uniform fallback. On simultaneous coverage, every selected halt is safe. This supplies a conditional observation-to-action argument without feeding the unknown R, w or r to the rule.

The separate Lean module `Paper1FlowDecision.lean` covers exact flow inversion, ratio and clipped intervals, the intensity interval, selected-halt inequality, actual product-Gaussian joint mean bound and a probability-composition theorem with an explicit rule-soundness premise. Their deterministic composition, clipping implementation and executable simulation still require separate correspondence checks; no claim of a fully formalized executable decision algorithm is made. The pinned build receipt records the checked declarations.

The exploratory script `studies/paper1_uplift_20261105/flow_decision.py` runs 20,000 independent replications per world in each of 60 configurations: $N \in \{32, 128, 512, 2048, 8192\}$, $\sigma_f \in \{0.0005, 0.002, 0.005\}$, $H \in \{1, 10\}$ and $F \in \{0.05, 0.25\}$. Other quantities use the two-group calibration above, a per-channel-vector record fee 0.0005, and the same class-halt or throttle action menu. Fourteen configurations have positive estimated net information value; the others are retained. The largest observed gain is exploratory, selected from the grid: at $N = 128$, $\sigma_f = 0.0005$, $H = 10$ and $F = 0.05$, no-data loss is 14.492188, information-inclusive loss 12.370094 and net gain 2.122094 surplus units. A simultaneous grid-adjusted Hoeffding interval is reported in the result file; zero observed unsafe halts is not a proof of zero probability. Costs, horizons and response bounds are synthetic, and no real-market gain or calibrated welfare conclusion follows. Six additional fixed stress cases retain both smaller biases with no observed unsafe halt and failures: a downward second-flow bias of 0.010 produces unsafe halts in G in all 20,000 replications, while a response 0.030 exceeding the allowance 0.005 produces unsafe halts in R in all replications. Swapped

second-flow labels lead to abstention in both worlds. These are synthetic contract falsifications, not guarantees under misspecification.

Proposition S5.2 (No-data action and information purchase)**.** *For the fixed halt/throttle menu, let* $\overline{L}_h = \sum_j \pi_j L_{h,j}$ *and let* $p_{\max} = \alpha$ *if a halt is unsafe in at least one world, otherwise* $p_{\max} = 1$*. Assuming the fallback meets the ceiling, the best constant no-data halt probability is* $p = 0$ *when* $\overline{L}_h \geq L_T$*, and* $p = p_{\max}$ *when* $\overline{L}_h < L_T$*. A specified information rule with expected action loss* L_I *is strictly preferred if and only if*

$$F + Nc < \mathcal{L}_0 - L_I.$$

Allowing refusal to purchase gives loss $\min\{\mathcal{L}_0, F + Nc + L_I\} \leq \mathcal{L}_0$*.*

Proof. The no-data loss is affine in its feasible halt probability: $L_T + p(\overline{L}_h - L_T)$. Its slope determines the minimizing endpoint. Subtracting the information rule's total loss gives the break-even inequality, and the minimum inequality proves the optional-purchase statement. These algebraic decisions are machine checked in `Paper1SupervisoryDecision.lean`. They do not establish an optimum over other actions, sample sizes, or information contracts. □

S5.1 The targeted halt for every K

Proposition 6.1(ii) is stated for two groups. Its argument needs two populations whose kernels agree to order $(\Delta\eta)^{2K-2}$ and whose largest groups differ at order one. This subsection supplies such a pair for every K.

Let μ be the uniform probability measure on $[-1, 1]$, let $p_0, p_1, \ldots$ be its orthonormal polynomials, and put $\lambda_K(s) = 1/\sum_{j<K} p_j(s)^2$. Call a rule *admissible* if it has K distinct nodes in $[-1, 1]$ and integrates every polynomial of degree at most $2K - 2$ exactly against μ. The Gauss rule and the Gauss–Radau rule of Theorem 5.1 are admissible.

Lemma S5.3 (Largest weights in the family)**.** *Fix* $K \geq 2$*.*

(i) An admissible rule with nodes $t_1, \ldots, t_K$ *has weights* $\omega_i = \lambda_K(t_i)$*. In particular its weights are positive.*

(ii) There is $c_0 > 0$ *such that, for every* $|c| \leq c_0$*, the zeros of* $p_K - c\,p_{K-1}$ *are the nodes of an admissible rule* R_c*. A point is a node of* R_c *for at most one* c*.*

(iii) There are $c_1 \neq c_2$ *in* $[-c_0, c_0]$ *for which* R_{c_1} *and* R_{c_2} *have different largest weights. Their moments of degree* $2K - 1$ *differ.*

Proof. (i) Let ℓ_i be the Lagrange polynomials of the nodes. Each product $\ell_i\ell_j$ has degree $2K - 2$, so the rule integrates it exactly: $\int \ell_i\ell_j\, d\mu = \sum_m \omega_m \ell_i(t_m)\ell_j(t_m) = \omega_i \mathbf{1}\{i = j\}$. Taking $i = j$ gives $\omega_i = \int \ell_i^2 d\mu > 0$. So $\ell_i/\sqrt{\omega_i}$, $i = 1, \ldots, K$, is an orthonormal basis of the polynomials of degree at most $K - 1$, as is $p_0, \ldots, p_{K-1}$. The sum of squares of an orthonormal basis at a point does not depend on the basis, so $\sum_{j<K} p_j(s)^2 = \sum_i \ell_i(s)^2/\omega_i$. At $s = t_i$ the right side is $1/\omega_i$.

(ii) The zeros of p_K are real, simple and inside $(-1, 1)$. Simple real zeros move continuously with the coefficients, so for $|c| \leq c_0$, with c_0 small, $q_c = p_K - c\,p_{K-1}$ has K simple zeros in $(-1, 1)$. Give these nodes the interpolatory weights. Let f have degree at most $2K-2$ and write $f = q_c g + \rho$ with $\deg g \leq K - 2$ and $\deg \rho \leq K - 1$. Both p_K and p_{K-1} are orthogonal to polynomials of degree at most $K - 2$, so $\int q_c g\, d\mu = 0$, and the rule integrates ρ exactly and gives $q_c g$ the value zero. Hence the rule is exact for f. If s is a zero of q_c and of $q_{c'}$ with $c \neq c'$, then $p_{K-1}(s) = 0$ and so $p_K(s) = 0$. By the three-term recurrence a common zero of p_K and p_{K-1} is a zero of p_{K-2}, and so on down to $p_0 = 1$, which has none.

(iii) The reciprocal $1/\lambda_K$ is a polynomial of degree $2K-2 \geq 2$, so every level set of λ_K is finite. Suppose all the rules R_c, $|c| \leq c_0$, had the same largest weight v. By (i) each R_c would have a node in the finite set $\{\lambda_K = v\}$, and by (ii) different values of c would give different nodes. There are infinitely many c, so this is impossible. For the last statement, let ν be the difference of the two rules. It is supported on at most $2K$ points. If the two rules also agreed at degree $2K-1$, ν would vanish on every polynomial of degree at most $2K-1$, including the one that equals 1 at one support point and 0 at the others. So ν would be zero, which contradicts (ii) and the positivity of the weights. □

Proposition S5.4 (Supervisory separation for every K)**.** *Fix $K \geq 2$ and let R_{c_1} and R_{c_2} be the rules of Lemma S5.3(iii), with nodes s_i^j, weights ω_i^j and largest weights $v_1 < v_2$. Under the hypotheses of Theorem 5.1, let economy $j \in \{1,2\}$ have K groups with rates $r_c + \Delta\eta\, s_i^j$ and merged weights $W\omega_i^j$, and write $q_i^j = W\omega_i^j/(1 - r_c - \Delta\eta\, s_i^j)$.*

(i) The response bounds of Theorem 5.1 hold for this pair: $\|h^1 - h^2\|_2 \leq C_U(\Delta\eta)^{2K-1}$ for every horizon, and $\|h^1 - h^2\|_2 \geq \kappa^2 W|g|(\Delta\eta)^{2K-1}$ when at least $2K+1$ taps are observed, where $g \neq 0$ is the difference of the two rules at degree $2K-1$.

(ii) The feedback budgets satisfy $|S^1 - S^2| = O((\Delta\eta)^{2K-1})$, and the largest groups satisfy $\max_i q_i^2 - \max_i q_i^1 = W(v_2 - v_1)/(1 - r_c) + O(\Delta\eta)$.

(iii) For sufficiently small spacing there is a nonempty interval of ceilings at which halting the largest group meets the ceiling in economy 2 and not in economy 1. For N independent reset records with Gaussian noise $\sigma^2 I$, a rule that makes this decision with both errors at most $1/4$ needs

$$N \;\geq\; \left(2\Phi^{-1}(0.75)\,\sigma/\|h^1 - h^2\|_2\right)^2 \;\geq\; c\,\sigma^2(\Delta\eta)^{-(4K-2)}.$$

Proof. (i) The proof of Theorem 5.1 uses four properties of its pair: nodes in $[-1,1]$, positive weights of unit mass, equal moments up to degree $2K-2$, and a nonzero difference at degree $2K-1$. Lemma S5.3 gives all four. (ii) Expanding $(1 - r_c - \Delta\eta\, s)^{-1}$ in powers of $\Delta\eta$, the sum $\sum_i q_i^j$ is $W\sum_{m\geq 0}(\Delta\eta)^m \mu_m^j/(1-r_c)^{m+1}$, where μ_m^j is the moment of degree m of rule j. The moments agree for $m \leq 2K-2$, which gives the first statement. Each q_i^j equals $W\omega_i^j/(1-r_c) + O(\Delta\eta)$, and the maximum is continuous, which gives the second. (iii) The post-halt budgets are $R^j = S^j - \kappa \max_i q_i^j$, so $R^1 - R^2 = \kappa W(v_2 - v_1)/(1-r_c) + O(\Delta\eta) > 0$ for small spacing, and any ceiling strictly between R^2 and R^1 separates the two decisions. The decision is then a test between two Gaussian mean vectors, as in the proof of Proposition 6.1, and (i) gives the bound. □

What is and is not shown. The pair in Proposition S5.4 is two members of the family R_c, not the Gauss and Gauss–Radau pair itself. For that pair Supplement Proposition S10.13 proves that the largest weights differ for every $K \geq 2$, with difference $\pi/(4K^2) + O(K^{-3})$. Lemma S5.3(ii) proves admissibility only for c near zero. Numerically, for $K = 2, \ldots, 14$ the rules R_c are admissible on the whole range between the two Gauss–Radau rules, their weights equal λ_K at the nodes to 7×10^{-13}, and on a grid of 41 values of c the largest weight is smallest at the Gauss rule and largest at the Gauss–Radau rules (`equilibrium/every_k_halt_check.py`). The counting step in part (iii) is machine-checked (`p1_largest_weight_not_constant` in `Paper1EveryKHalt.lean`), with parts (i) and (ii) and the finiteness of the level sets as hypotheses. The rest is proved on paper.

S5.2 Proof of Proposition 5.5

Write $D_X(z) = z - a_X - \kappa G_X(z)$ for economy X, so that $H_X = \kappa z/D_X$ and the stationary price spectrum is $f_X(\omega) = \sigma^2\kappa^2/|D_X(e^{i\omega})|^2$. Because both economies have budget $1-\epsilon$, $D_A(1) = D_B(1) = \epsilon$.

Step 1: the difference vanishes at zero frequency. $D_B - D_A = -(a_B - a_A) - \kappa(G_B - G_A)$, and the choice of persistences gives $a_B - a_A = -\kappa(G_B(1) - G_A(1))$. With $E = G_B - G_A$ this is $\Delta(z) \equiv D_B(z) - D_A(z) = -\kappa\,(E(z) - E(1))$, which does not depend on ϵ. Write $E(z) = \sum_m e_m z^{-m}$ with $e_m = k_m^B - k_m^A$. Then $|E(e^{i\omega}) - E(1)| \le |\omega| \sum_m m|e_m|$. The rates are $r_c + \Delta\eta\, s$, and expanding $(r_c + \Delta\eta\, s)^m$ in powers of $\Delta\eta$, the moments of order up to $2K-2$ of the two normalized rules cancel, so $|e_m| \le 2W \sum_{j \ge 2K-1} \binom{m}{j} r_c^{m-j} (\Delta\eta)^j$. Summing, $\sum_m m|e_m| \le C_1(\Delta\eta)^{2K-1}$, with C_1 depending on K, W, r_c and a bound on $\Delta\eta$. Hence $|\Delta(e^{i\omega})| \le \kappa C_1(\Delta\eta)^{2K-1}|\omega|$.

Step 2: the denominator is bounded below by $\epsilon + |\omega|$. Let $a^* = \lim_{\epsilon\to 0} a_A(\epsilon)$ and $D^* = D_A - \epsilon$, the denominator of economy A at budget one; $D_A = D^* + \epsilon$. At budget one the closed-loop matrix J of the price and the groups is entrywise nonnegative with a positive diagonal and positive first row and column (since $a^* > 0$ and all weights are positive), so it is primitive. Its Perron root is 1, it is simple, and no other eigenvalue lies on the unit circle. The zeros of D^* in $|z| \ge 1$ are eigenvalues of J, so D^* has no zero on the unit circle except $z = 1$, and there $D^{*\prime}(1) = 1 + \kappa \sum_i w_i r_i/(1-r_i)^2 > 0$. Hence there are $\delta, c_1, C_2, m > 0$ with $|\mathrm{Im}\, D^*(e^{i\omega})| \ge c_1|\omega|$ and $\mathrm{Re}\, D^*(e^{i\omega}) \ge -C_2\omega^2$ for $|\omega| \le \delta$, and $|D^*(e^{i\omega})| \ge m$ for $\delta \le |\omega| \le \pi$. For $|\omega| \le \delta$: if $\epsilon \ge 2C_2\omega^2$ then $\mathrm{Re}\, D_A \ge \epsilon/2$ and $|D_A| \ge \max(\epsilon/2, c_1|\omega|)$; if $\epsilon < 2C_2\omega^2$ then $|D_A| \ge c_1|\omega|$ and $\epsilon < 2C_2\delta|\omega|$. In both cases $|D_A(e^{i\omega})| \ge c_2(\epsilon + |\omega|)$. For $|\omega| \ge \delta$ and $\epsilon \le m/2$, $|D_A| \ge m/2$. So $|D_A(e^{i\omega})| \ge c_3(\epsilon + |\omega|)$ on $[-\pi, \pi]$ for all $\epsilon \le \epsilon_1$. These constants can be taken uniform in $\Delta\eta \in (0, \Delta\eta_0]$. As $\Delta\eta \to 0$ the kernel of A converges to that of the one-group economy at rate r_c in the norm $\sum_m (1+m^2)|k_m|$, so D^* and its first two derivatives converge uniformly on the circle. The limit has the same properties: its closed-loop matrix is primitive with Perron root 1, and its derivative at 1 is at least 1. Hence c_1, C_2, δ and m can be chosen for the limit and kept, after halving c_1 and m and doubling C_2, for all small $\Delta\eta$.

Step 3: the bound. By Steps 1 and 2, $\rho(\omega) = |\Delta/D_A| \le \kappa C_1(\Delta\eta)^{2K-1}|\omega|/(c_3(\epsilon + |\omega|)) \le C_3(\Delta\eta)^{2K-1}$, uniformly in ϵ and ω. The ratio of spectra is $u = f_B/f_A = |D_A/D_B|^2 = |1 + \Delta/D_A|^{-2}$, so for $\rho \le 1/8$, $|u-1| \le 4\rho$. The divergence rate of stationary Gaussian processes is $D_\epsilon(B\|A) = \frac{1}{4\pi}\int_{-\pi}^{\pi} (u - 1 - \log u)\, d\omega$, and $u - 1 - \log u \le 2(u-1)^2$ for $|u-1| \le 1/2$. Hence $D_\epsilon(B\|A) \le 16 C_3^2 (\Delta\eta)^{4K-2}$ for all $\epsilon \le \min(\epsilon_0, \epsilon_1)$ and all $\Delta\eta$ small enough that $C_3(\Delta\eta)^{2K-1} \le 1/8$. For $\epsilon_1 < \epsilon \le \epsilon_0$ both economies are uniformly stable, so $|D_A|$ is bounded below by a constant on the circle, while $|\Delta| \le |a_B - a_A| + \kappa|E(e^{i\omega})| = O((\Delta\eta)^{2K-1})$; Step 3 then applies with this lower bound in place of Step 2. Steps 2 and 3 are machine-checked, with Whittle's formula, Step 1 and the local properties of D^* as hypotheses (`p1_nearcrit_lower`, `p1_ratio_bound`, `p1_nearcrit_kl_bound`, `Paper1NearCritical.lean`). □

Numerical check. For $K = 2, 3$, $\Delta\eta \in \{0.1, 0.05, 0.025\}$ and margins from 10^{-1} to 10^{-6}, the ratio $D_\epsilon(B\|A)/(\Delta\eta)^{4K-2}$ increases to a finite limit as $\epsilon \to 0$ (0.41 to 0.48 for $K = 2$, 0.65 to 0.71 for $K = 3$; `fragility_uniform_check.py`). With a common persistence instead of a common budget, the divergence rate grows like ϵ^{-1}, which is the budget difference becoming visible. This proposition is not machine-checked.

Finite samples. The proposition bounds the information per observation; the same bound holds for every record length.

Proposition S5.5 (Near instability, finite samples)**.** *Under the conditions of Proposition 5.5, with $C_3(\Delta\eta)^{2K-1} \le 1/8$, the divergence between the laws of n consecutive stationary observations of the two economies is at most $16nC_3^2(\Delta\eta)^{4K-2}$, for every n and uniformly in the margin ϵ.*

Proof. The covariance of n consecutive observations is the Toeplitz matrix $T_n(f)$ of the spectral density, and $x^\top T_n(f)x = \frac{1}{2\pi}\int |\hat{x}(\omega)|^2 f(\omega)\,d\omega$. Step 3 gives $|u-1| \le 4\rho$ for $u = f_B/f_A$ at every frequency, so $(1-4\rho)T_n(f_A) \preceq T_n(f_B) \preceq (1+4\rho)T_n(f_A)$, and every eigenvalue λ_k of $T_n(f_A)^{-1}T_n(f_B)$ lies within 4ρ of one. The divergence is $\frac{1}{2}\sum_k(\lambda_k - 1 - \log\lambda_k) \le \frac{1}{2}n \cdot 2(4\rho)^2 = 16n\rho^2$, using $u - 1 - \log u \le 2(u-1)^2$ for $|u-1| \le 1/2$. □

The eigenvalue step is machine-checked (`p1_nearcrit_finite_sample`, `Paper1OpenResolved.lean`); the Toeplitz sandwich is on paper. Numerically, the exact divergence of n consecutive stationary observations lies between 97.8% and 99.8% of n times the per-observation rate for $K = 2$, $\Delta\eta \in \{0.1, 0.2\}$, $\epsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and $n \in \{250, 1000, 4000\}$ (`research_extension/fragility_reveals/finite_sample_check.py`). In every such case at budgets $1-10^{-1}$, $1-10^{-2}$ and $1-10^{-3}$, all eigenvalues lie in the band of the proof, and the divergence is below the bound (`research_extension/open_problems/open_checks.py`).

S5.3 Stress identification with noisy observations

Theorem 7.4 assumes exactly observed responses. The next result shows that the identification survives observation noise: a minimum-distance estimator recovers the signed capacity measure consistently. Proposition S10.7 adds the rate for smooth shapes.

Proposition S5.6 (Noisy stress identification)**.** *Let ψ be continuous, odd and with independent dilations. Fix $n \ge 1$, $0 < \ell_{\min} \le \ell_{\max}$, $0 \le \bar{r} < 1$ and $\bar{c} > 0$, and let Θ be the set of populations of at most n traders, written $\theta = (c_i, r_i, \ell_i)_{i \le n}$ with $c_i \in [0, \bar{c}]$, $r_i \in [0, \bar{r}]$ and $\ell_{\min} \le |\ell_i| \le \ell_{\max}$ (a trader with $c_i = 0$ is absent). Let $(v^{(j)})_{j \ge 1}$ enumerate the probe sequences of finite length with rational entries, $v^{(j)}$ of length $T_j \le j$, and let $h_j(\theta) \in \mathbb{R}^{T_j}$ be the reset response of the price to $v^{(j)}$, without process noise. In round N, each probe $j \le N$ is run N times, every price is observed with independent $N(0, \sigma^2)$ noise, and $\hat{h}_{j,N}$ is the average of the N observed responses. Let $\hat{\theta}_N$ minimize*

$$Q_N(\theta) = \sum_{j \le N} 2^{-j} \min\bigl(1,\ \|\hat{h}_{j,N} - h_j(\theta)\|\bigr)$$

over Θ. If the true population $\theta_0 \in \Theta$, then $\tilde{\mu}(\hat{\theta}_N) \to \tilde{\mu}(\theta_0)$ in probability, in the weak topology on signed measures on $[0, \bar{r}] \times [\ell_{\min}, \ell_{\max}]$.

Proof. Continuity. For a fixed probe, the response at each date is a finite composition of the market recursion, sums, products and ψ, so h_j is continuous on Θ, which is compact. The map $\theta \mapsto \tilde{\mu}(\theta) = \sum_i c_i \operatorname{sgn}(\ell_i)\,\delta_{(r_i, |\ell_i|)}$ is continuous into the weak topology, because $\operatorname{sgn}(\ell_i)$ is constant on each of the two components $\ell_i \ge \ell_{\min}$ and $\ell_i \le -\ell_{\min}$; the weak topology on this bounded set of measures is metrized by the bounded-Lipschitz distance ρ.

Identification. Let $Q(\theta) = \sum_j 2^{-j}\min(1, \|h_j(\theta_0) - h_j(\theta)\|)$. If $Q(\theta) = 0$, the two populations give identical responses to every rational probe sequence. Responses are continuous in the probe, by the same composition argument, and rational sequences of each length are dense, so the responses agree for every probe sequence. Drop the traders with $c_i = 0$ from both populations; the remaining traders satisfy the hypotheses of Theorem 7.4, so $\tilde{\mu}(\theta) = \tilde{\mu}(\theta_0)$. Conversely $\tilde{\mu}(\theta) = \tilde{\mu}(\theta_0)$ gives $Q(\theta) = 0$ by the same theorem.

Uniform convergence. Since $\min(1, \|\cdot\|)$ satisfies the triangle inequality,

$$\sup_\theta |Q_N(\theta) - Q(\theta)| \le \sum_{j \le N} 2^{-j} \|\hat{h}_{j,N} - h_j(\theta_0)\| + 2^{-N},$$

and $\mathbb{E}\|\hat{h}_{j,N} - h_j(\theta_0)\| \le \sigma\sqrt{T_j/N} \le \sigma\sqrt{j/N}$. The expectation of the right side is at most $\sigma N^{-1/2} \sum_j 2^{-j}\sqrt{j} + 2^{-N} \to 0$, so $\sup_\theta |Q_N - Q| \to 0$ in probability.

Consistency. $Q_N(\hat{\theta}_N) \le Q_N(\theta_0) \le \sup |Q_N - Q|$, since $Q(\theta_0) = 0$; hence $Q(\hat{\theta}_N) \le 2 \sup |Q_N - Q| \to 0$ in probability. Fix $\delta > 0$. The set $F_\delta = \{\theta \in \Theta : \rho(\tilde{\mu}(\theta), \tilde{\mu}(\theta_0)) \ge \delta\}$ is closed in a compact set, hence compact, and Q is continuous and, by identification, strictly positive on it, so $q_\delta = \min_{F_\delta} Q > 0$ (if F_δ is empty there is nothing to prove). Then $\Pr(\rho(\tilde{\mu}(\hat{\theta}_N), \tilde{\mu}(\theta_0)) \ge \delta) \le \Pr(Q(\hat{\theta}_N) \ge q_\delta) \to 0$. □

The argument is the standard one for extremum estimators: identification, compactness, continuity and uniform convergence. Its abstract form is machine-checked on any compact parameter set, in probability, together with the uniform bound for the truncated criteria (`p1_extremum_consistency_prob`, `p1_criterion_uniform_close`, `Paper1NoisyStress.lean`); continuity of the responses and of $\tilde{\mu}$ and the expectation bound are on paper. It inherits the hard limit's requirement of probes of all sizes; for smooth shapes, Proposition S10.7 gives the parametric rate. A Monte Carlo run with a trend follower and a contrarian of the tanh shape, seven probes and observation noise, shows the consistency: the median parameter error falls from 0.24 with one run per probe to 0.050, 0.024, 0.0052 and 0.0009 with 16, 256, 4,096 and 65,536 runs, close to the $N^{-1/2}$ rate (`research_extension/theorem_checks/mc_stress_and_general.py`). For tanh, Proposition 7.3 shows the information per probe grows like the sixth power of its size for small probes, so rates for the full measure will depend on how far the design pushes the market.

S6 Saturating traders: proofs and checks

Theorem 7.1.

Proof. $f_t = \sum_i c_i g_t(r_i, \ell_i)$ with $g_t(r, \ell) = \psi(\ell \sum_{j<t} r^{t-1-j} s_j)$, so equal measures give equal flows for every signal. The rest is the induction of Theorem 3.1(iii), which uses only that the two flow maps agree and depend on signals before t. The last sentence follows from the example below Theorem 7.2. □

Theorem 7.2.

Proof. (i) By strong induction on $t < \tau$. If the paths agree before t, the signals agree before t, so every trader of either population has the same state in both markets. All are inside their bands, so both aggregate flows equal the linear flow $\sum_{j<t} k_{t-1-j} s_j$, which is the same for P and P' because the kernels agree. The insider's order and the dealer's price are causal, so they agree at t. The same argument with P' replaced by the linear market (the band-linear shape $\psi'(u) = u$) gives the second statement. (ii) Build both markets, and the record noise, on one probability space. By (i) the records coincide on $\{\tau \ge T\}$, so the total-variation distance is at most $\Pr(\tau < T)$, and τ is computed on the linear market. (iii) In the linear market from rest, $x = g * \varepsilon$ with $g = h/\kappa$, and $m_t(r) = (\varphi_r * x)_t$ with $\varphi_r(j) = r^j$ shifted by one date. So $m_t(r)$ is centred Gaussian with variance at most $s_r^2 \le \sigma_\varepsilon^2 \|\varphi_r\|_1^2 \|g\|_1^2$. Here $\|\varphi_r\|_1 = 1/(1-r)$. Also $|h_t|$ is dominated by the response of the positive economy with persistence a and kernel $|k_m|$, whose total mass is $\kappa/(1 - S_{\mathrm{abs}})$, so

$\|g\|_1 \le 1/(1 - S_{\text{abs}})$. For a standard normal Z and $z \ge 0$, $\Pr(|Z| > z) \le e^{-z^2/2}$. A union bound over the T dates and $n + n'$ traders gives the bound. A test with both errors at most α needs TV $\ge 1 - 2\alpha$. (iv) Without process noise, the linear path is deterministic, so $\tau \ge T$ surely and (ii) gives distance zero. For a pulse, the linear state is $m_t(r) = A \sum_{j<t} r^{t-1-j} h_j$. □

Proposition 7.3.

Proof. For $u \ge 0$, $\phi(u) = u - \tanh u$ has $\phi(0) = 0$ and $\phi'(u) = \tanh^2 u \in [0, u^2]$. So $0 \le u - \tanh u \le u^3/3$, and by oddness $|\tanh u| \le |u|$ and $|\tanh u - u| \le |u|^3/3$ for all u. (i) *Size of each path.* Since $|\tanh u| \le |u|$, $|x_t| \le a|x_{t-1}| + \kappa \sum_i c_i|\ell_i|\,(\varphi_{r_i} * |x|)_t$ for $t \ge 1$, with $|x_0| = \kappa|A|$. So $|x|$ is dominated by the pulse response of the positive economy with weights $c_i|\ell_i|$. That response has peak $\kappa|A|$ (Lemma S1.1) and total mass $\kappa|A|/(1 - \rho_P)$. Hence $\|x\|_\infty \le \kappa|A|$ and $\|x\|_2 \le \|x\|_1 \le \kappa|A|/(1 - \rho_P)$, and the same holds for the memory states, with an extra factor $1/(1 - r_i)$. *Difference.* Write $c_i \tanh(\ell_i m_{i,t}) = c_i\ell_i m_{i,t} + R_{i,t}$ with $|R_{i,t}| \le c_i|\ell_i|^3|m_{i,t}|^3/3$. The linear parts give $(k * x)_t$ with the common kernel. So $e = x^P - x^{P'}$ solves the linear closed loop driven by $u_t = \sum_i R_{i,t} - \sum_j R'_{j,t}$ in place of the probe, and $e = h * u$. By Young's inequality, $\|e\|_2 \le \|h\|_1\|u\|_2$, and $\|h\|_1 \le \kappa/(1 - S_{\text{abs}}) \le \kappa/(1 - \hat{S})$ as in Theorem 7.2(iii). Also $\| |m_i|^3 \|_2 \le \|m_i\|_\infty^2 \|m_i\|_2 \le \kappa^3|A|^3/((1 - r_i)^3(1 - \rho_P))$, so $\| \sum_i R_i \|_2 \le \Lambda_P \kappa^3 |A|^3/(1 - \rho_P)$. Combining gives the bound on $\|e\|_2$. For N experiments, the Kullback–Leibler divergence is $N\|e\|_2^2/(2\sigma^2)$. Pinsker's inequality and TV $\ge 1 - 2\alpha$ give the requirement. (ii) Both markets have $x_0 = \kappa A$ and $m_{i,1} = x_0$, so $x_1^P - x_1^{P'} = \kappa(\sum_i c_i \tanh(\ell_i \kappa A) - \sum_j c'_j \tanh(\ell'_j \kappa A))$. The linear terms cancel because k_0 is common, and $\tanh u = u - u^3/3 + O(u^5)$ gives the expansion. For $|A|$ small, $|x_1^P - x_1^{P'}| \ge \kappa^4 |\Delta M_3| |A|^3/6$. The test that compares the average of N first taps with the midpoint has error $\Phi(-\sqrt{N}\,|x_1^P - x_1^{P'}|/(2\sigma))$, which gives the stated N. □

Theorem 7.4.

Proof. Step 1: probes steer the price. Write F_t for aggregate feedback flow, a function of $x_0, \dots, x_{t-1}$. Since $x_t = a x_{t-1} + \kappa(F_t + v_t)$ with $\kappa \ne 0$, the probe $v_t = (x_t^* - a x_{t-1}^*)/\kappa - F_t(x_{<t}^*)$ realizes any target path x^*, and $F_t = (x_t - a x_{t-1})/\kappa - v_t$ is read from the response. So two populations give identical responses to all probes if and only if their flow maps agree on all paths.

Step 2: sufficiency of $\tilde{\mu}$. The shape is odd, so $c\,\psi(\ell m) = c\,\mathrm{sgn}(\ell)\,\psi(|\ell| m)$ and $F_t = \int \psi(\lambda\, m_t(r))\, \tilde{\mu}(dr, d\lambda)$. Equal $\tilde{\mu}$ give equal flow maps, hence identical responses.

Step 3: identification. Let $D = \tilde{\mu}_P - \tilde{\mu}_{P'} = \sum_k d_k \delta_{(r_k, \lambda_k)}$ with distinct atoms and $\lambda_k > 0$. Take the path with $x_0 = \theta u$ and $x_1 = u$. Then $m_2(r) = (r\theta + 1)u$, and equal flow maps at date 2 give $g(u) = \sum_k d_k \psi(s_k u) = 0$ for all u, with $s_k = \lambda_k(r_k\theta + 1)$. For two distinct atoms, $\lambda_k(r_k\theta + 1) = \lambda_j(r_j\theta + 1)$ is an affine equation in θ that is not an identity, so it has at most one root. Fix $\theta > 0$ outside these finitely many roots. Then the s_k are positive, because $r_k \ge 0$, and distinct. Independent dilations give $d_k = 0$ for every k, so $D = 0$. □

Theorem 7.4, parts (i)–(iii).

Proof. (i) Let $g(u) = \sum_k w_k \,\mathrm{clip}(s_k u) = 0$ for all u, with distinct $s_k > 0$, and let s_1 be the largest scale. At $u = 1/s_1$ every copy is inside its band or at its edge, so $g(1/s_1) = \sum_k w_k s_k/s_1 = 0$. If there is a second-largest scale s_2, then at $u = 1/s_2$ the first copy is saturated and the others are not, so $w_1 + \sum_{k \ne 1} w_k s_k/s_2 = 0$. The first equation gives $\sum_{k \ne 1} w_k s_k = -w_1 s_1$, so $w_1(1 - s_1/s_2) = 0$ and $w_1 = 0$. If there is no other scale, the first equation is $w_1 = 0$. Remove s_1 and repeat. The statement on probes inside the bands is Theorem 7.2(iv).

(ii) For every y, $1-\tanh y = 2e^{-2y}/(1+e^{-2y})$. Let $g(u)=\sum_k w_k \tanh(s_k u)=0$ for all u, with distinct $s_k>0$, and let s_1 be the smallest scale. Letting $u\to\infty$ gives $\sum_k w_k = 0$, so $\sum_k w_k(1-\tanh(s_k u))=0$ for all u. Multiply by $e^{2s_1 u}$ and let $u\to\infty$: the first term tends to $2w_1$, and every other term is at most $2|w_k|e^{-2(s_k-s_1)u}$, which tends to 0. So $w_1=0$. Remove s_1 and repeat.

Small probes. g is real-analytic, so if it vanishes near 0 it vanishes everywhere, and the argument above applies. A direct route: the odd Taylor coefficients of tanh are $\tau_{2n-1}=2^{2n}(2^{2n}-1)B_{2n}/(2n)!$, which are nonzero because the Bernoulli numbers B_{2n} are nonzero. If g vanishes near 0, then $\sum_k (w_k s_k)(s_k^2)^{n-1}=0$ for $n=1,\dots,K$, a Vandermonde system in the distinct numbers s_k^2, so every $w_k=0$. In Step 1 of the theorem, $|F_t|\le\sum_i c_i|\ell_i||m_{i,t}|$, so small target paths need only small probes.

(iii) $\sum_k w_k s_k = 5-8+3=0$ and $\sum_k w_k s_k^3 = 5-32+27=0$, so $\sum_k w_k\psi(s_k u) = u\sum_k w_k s_k - \gamma u^3\sum_k w_k s_k^3 = 0$. For a cubic shape the date-t flow map is $\int \lambda\, m_t(r)\, d\tilde\mu - \gamma\int \lambda^3 m_t(r)^3\, d\tilde\mu$, which depends on $\tilde\mu$ only through the measures $\lambda\,\tilde\mu$ and $\lambda^3\tilde\mu$ in r. Both are identified. Write A and B for these two measures on memories, which are finite and atomic. By Step 1 the flow map is identified; it is a polynomial in past prices, and the coefficient of x_j in F_t is $\int r^{t-1-j}\,dA$, while the coefficient of x_j^3 is $-\gamma\int r^{3(t-1-j)}\,dB$. So all moments of A and all moments of the image of B under $r\mapsto r^3$ are identified, and a finite atomic measure on $[0,1)$ is determined by its moments. □

Proposition 7.6.

Proof. (i) The proofs of these results use the dealer's rule only through causality and through its being the same map in both markets. (ii) The linear parts of the two expected flows agree because the kernels agree, so the merged weight at each memory is the same. For each trader, $|\mathbb{E}[c\,\mathrm{clip}(Y)] - c\,\mathbb{E}Y| \le c\,\mathbb{E}[(|Y|-1)_+]$ with $Y=\ell m^+\sim N(\ell\bar m,\sigma^2)$. Since $|Y|\le|\ell\bar m|+\sigma|Z|$, this is at most $c\,\mathbb{E}[(\sigma|Z|-\delta)_+] = c\big(2\sigma\phi(\delta/\sigma) - 2\delta(1-\Phi(\delta/\sigma))\big) \le c\sqrt{2/\pi}\,\sigma e^{-\delta^2/(2\sigma^2)}$. □

Numerical checks. The scripts are `research_extension/nonlinear_scoping/identification_and_figure.py`, `sim_wiener.py`, `sim_passive_clip.py` and `dealer_anticipation.py`. Two populations with the same $\tilde\mu$ (a trend follower of capacity 2 and a contrarian of capacity 1, against one trend follower of capacity 1, all at $r=0.7$ and $|\ell|=0.1$) give identical responses to 50 random probe sequences of standard deviation 10, with gap exactly 0, under both shapes. The constructive recovery in the main text locates the kinks at 3.702 and 14.813 (true 3.704 and 14.815) and returns net capacities 0.24989 and 0.99988.

S7 Which questions prices answer cheaply

The cost results of Section 5 concern composition. This section asks the same question of every linear question about the population, and finds a dichotomy: some questions are answered at the usual statistical rate whatever the number of groups and the spacing of their memories, some cannot be answered at all without a bound on memory, and the cost of each is the value of a problem of polynomial approximation. The results and proofs are adapted from the first author's thesis (in preparation).

Setting. Write $M=\sum_i w_i\delta_{r_i}$ for the merged measure, so that $k_m=\int r^m\,dM$, and $\mathcal{M}(\bar r,\bar W)$ for the positive measures on $[0,\bar r]$ of total mass at most $\bar W$, with $\bar r<1$ a memory bound. A *linear question* is $F_g(M)=\int g\,dM$ with g continuous on $[0,\bar r]$; examples are a single tap ($g(r)=r^m$), the

discounted feedback ($g(r) = (1-\beta r)^{-1}$, $\beta < 1$), and the total feedback weight $\Phi = \int dM/(1-r) = \sum_m k_m$, which with a and κ fixes the feedback budget $S = a+\kappa\Phi$. The observer runs N independent experiments, each returning the first L kernel taps with independent $N(0,\sigma^2)$ noise. One way to obtain them is a steering probe that holds the price at a unit impulse (as in the proof of Theorem 7.7), with aggregate feedback flow observed: the flow at date $t+1$ is then k_t. Price records are discussed at the end of the section. For accuracy $\epsilon > 0$ and level $1-\alpha$, $N_{\mathrm{cert}}(\epsilon; F_g)$ is the least N for which some interval built from the data has half-width at most ϵ and covers $F_g(M)$ with probability at least $1-\alpha$ for every $M \in \mathcal{M}(\bar r, \bar W)$.

For $g(r) = \sum_t a_t r^t$ write $\tau_g(T) = \bar W \sup_{r\le\bar r} |\sum_{t\ge T} a_t r^t|$, $A_g(T) = \sum_{t<T} a_t^2$, $T_g(\epsilon) = \min\{T \ge 1 : \tau_g(T) \le \epsilon/2\}$, and $E_T(g) = \min_{\deg p<T} \max_{[0,\bar r]} |g-p|$. For $c \in \mathbb{R}^L$ put $p_c(r) = \sum_{t<L} c_t r^t$, and for a function h on $[0,\bar r]$ put $\mathcal{R}(h) = (\sup h)_+ + (-\inf h)_+$, which is at least $\|h\|_\infty$. The modulus of the question is

$$\omega_L(\delta) = \sup\{F_g(M_1) - F_g(M_0) :\ M_0, M_1 \in \mathcal{M}(\bar r, \bar W),\ \|k_{0:L}(M_1) - k_{0:L}(M_0)\|_2 \le \delta\},$$

the largest difference in the answer between two populations whose observed taps differ by at most δ.

Theorem S7.1 (The cost of a linear question)**.** *Let z_β be the standard normal β-quantile and $\alpha \in (0, 1/2)$.*

(i) Upper bound. *If $L \ge T_g(\epsilon)$, then $N_{\mathrm{cert}}(\epsilon; F_g) \le \lceil 4 z_{1-\alpha/2}^2 \sigma^2 A_g(T_g(\epsilon)) \epsilon^{-2} \rceil$.*

(ii) Truncation is necessary. *For every T there are $M, M' \in \mathcal{M}(\bar r, \bar W)$ whose first T taps coincide and with $|F_g(M) - F_g(M')| \ge 2\bar W E_T(g)$; any certificate that uses only the first T taps has worst-case half-width at least $\bar W E_T(g)$.*

(iii) Long memory makes some questions impossible. *On the positive measures on $[0,1)$ of mass at most $\bar W$, with no memory bound, if $\limsup_{r\uparrow 1} |g(r)|\sqrt{1-r} = \infty$, then for $\alpha < 1/4$ no finite N certifies F_g to any finite accuracy.*

(iv) Exact cost. *For every depth L and $\delta > 0$,*

$$\omega_L(\delta) = \min_{c\in\mathbb{R}^L} \{\delta\|c\|_2 + \bar W\, \mathcal{R}(g - p_c)\},$$

and $\omega_L(0) = 2\bar W E_L(g)$. If N experiments certify F_g to half-width ϵ at level $1-\alpha$, then $\omega_L(2z_{1-\alpha}\sigma/\sqrt{N}) \le 2\epsilon$; conversely, if $\omega_L(2z_{1-\alpha/2}\sigma/\sqrt{N}) \le 2\epsilon$, an affine interval $\sum_t c_t \bar k_t + b \pm \epsilon$ certifies F_g with N experiments. With $N_\omega(\epsilon) = \min\{N : \omega_L(2z_{1-\alpha}\sigma/\sqrt{N}) \le 2\epsilon\}$ and $\lambda_\alpha = z_{1-\alpha/2}/z_{1-\alpha}$,

$$N_\omega(\epsilon) \le N_{\mathrm{cert}}(\epsilon; F_g) \le N_\omega(\epsilon/\lambda_\alpha).$$

Proof. (i) Let $\bar k_t$ be the average of the N replicates of tap t, and $T = T_g(\epsilon)$. The estimator $\hat F = \sum_{t<T} a_t \bar k_t$ has mean $\sum_{t<T} a_t k_t$ and variance $\sigma^2 A_g(T)/N$. Its bias is $|\sum_{t\ge T} a_t k_t| = |\int \sum_{t\ge T} a_t r^t\, dM| \le \tau_g(T) \le \epsilon/2$ for every M in the class. So $\hat F \pm (\epsilon/2 + z_{1-\alpha/2}\sigma\sqrt{A_g(T)/N})$ covers $F_g(M)$ with probability at least $1-\alpha$, and its half-width is at most ϵ once $z_{1-\alpha/2}\sigma\sqrt{A_g(T)/N} \le \epsilon/2$.

(ii) By duality for best uniform approximation, $E_T(g) = \max \int g\, d\nu$ over signed measures ν on $[0,\bar r]$ of total variation at most one that annihilate every polynomial of degree below T. Take a maximizer and write $\nu = \nu^+ - \nu^-$. Orthogonality to constants gives $\nu^+([0,\bar r]) = \nu^-([0,\bar r]) =: c \le 1/2$. The measures $M = (\bar W/c)\nu^+$ and $M' = (\bar W/c)\nu^-$ lie in the class, their first T moments, which

are their first T taps, coincide, and $F_g(M) - F_g(M') = (\bar{W}/c)E_T(g) \geq 2\bar{W}E_T(g)$. Two populations that produce identically distributed data cannot both be covered by an interval of half-width below half the difference of their answers.

(iii) Compare $M_0 = 0$ with $M_1 = c\,\delta_r$. Under M_1 the taps are $k_t = cr^t$, so the Kullback–Leibler divergence between the two data laws over any number of taps is at most $Nc^2/(2\sigma^2(1-r^2))$. Take $c = \sigma\sqrt{(1-r^2)/N}$, which is at most $\bar{W}$ for r near one; the divergence is at most $1/2$, so by Pinsker's inequality the total variation distance is at most $1/2$. The answers differ by $c|g(r)| \geq \sigma\sqrt{(1-r)/N}\,|g(r)|$, which is unbounded along a sequence $r \uparrow 1$. If an interval of half-width ϵ covered F_g with probability at least $1-\alpha$ under both, then once the answers differ by more than 2ϵ the event that the interval contains $F_g(M_0)$ has probability at least $1-\alpha$ under M_0 and at most α under M_1, so the total variation distance would be at least $1 - 2\alpha > 1/2$.

(iv) *Duality.* For any $c \in \mathbb{R}^L$ and any admissible pair, with $h = g - p_c$,

$$F_g(M_1) - F_g(M_0) = \int h\,dM_1 - \int h\,dM_0 + \langle c, k_{0:L}(M_1) - k_{0:L}(M_0)\rangle \leq \bar{W}\mathcal{R}(h) + \delta\|c\|_2,$$

because $\int h\,dM_1 \leq \bar{W}(\sup h)_+$ and $-\int h\,dM_0 \leq \bar{W}(-\inf h)_+$ for measures of mass at most $\bar{W}$. So $\omega_L(\delta)$ is at most the minimum. For the reverse inequality let $\mu = (M_0, M_1)$ and $\Lambda(\mu, c) = \int h\,d(M_1 - M_0) + \delta\|c\|_2$. Since $\inf_c\{\delta\|c\|_2 - \langle c, v\rangle\}$ is 0 if $\|v\|_2 \leq \delta$ and $-\infty$ otherwise, $\omega_L(\delta) = \sup_\mu \inf_c \Lambda(\mu, c)$. The set $\mathcal{M}(\bar{r}, \bar{W})^2$ is convex and weak-$*$ compact, Λ is affine and weak-$*$ continuous in μ (as h is continuous) and convex and continuous in c, so Sion's minimax theorem gives $\omega_L(\delta) = \inf_c \sup_\mu \Lambda(\mu, c)$. The inner supremum is $\bar{W}\mathcal{R}(g - p_c) + \delta\|c\|_2$, attained by mass $\bar{W}$ at a maximizer of h if $\sup h > 0$ and at a minimizer if $\inf h < 0$. For $\delta > 0$ the objective is convex, continuous and coercive in c, so the infimum is attained. At $\delta = 0$ the same argument gives $\omega_L(0) = \inf_c \bar{W}\mathcal{R}(g - p_c)$; adding a constant to p_c makes $g - p_c$ change sign, where $\mathcal{R}$ is the oscillation, so the infimum is $\bar{W}$ times the least oscillation of $g - p$ over polynomials of degree below L, which is $2\bar{W}E_L(g)$.

Lower bound. The averages $\bar{k} = (\bar{k}_0, \ldots, \bar{k}_{L-1})$ are sufficient and distributed as $N(k_{0:L}(M), (\sigma^2/N)I_L)$, so the total variation distance between the data laws of two populations is $2\Phi_{\mathcal{N}}(\sqrt{N}\|\Delta k\|_2/(2\sigma)) - 1$, with $\Phi_{\mathcal{N}}$ the standard normal distribution function. For an admissible pair with $\|\Delta k\|_2 < 2z_{1-\alpha}\sigma/\sqrt{N}$ this is below $1 - 2\alpha$. If their answers differed by more than 2ϵ, an interval of half-width ϵ covering each with probability $1-\alpha$ would separate the two laws with total variation at least $1 - 2\alpha$, as in (iii). Hence $\omega_L(t) \leq 2\epsilon$ for every $t < 2z_{1-\alpha}\sigma/\sqrt{N}$. The modulus is concave (the admissible pairs form a convex set and the constraint is a norm), finite and nondecreasing, hence continuous on $(0, \infty)$, so $\omega_L(2z_{1-\alpha}\sigma/\sqrt{N}) \leq 2\epsilon$.

Upper bound. Put $\delta = 2z_{1-\alpha/2}\sigma/\sqrt{N}$, let c attain the minimum in the dual form and $h = g - p_c$. The interval $\sum_t c_t\bar{k}_t + b \pm \epsilon$ with $b = \bar{W}[(\sup h)_+ - (-\inf h)_+]/2$ has bias at most $\bar{W}\mathcal{R}(h)/2$ for every population, by the bounds in the duality step, and normal noise with standard deviation $\sigma\|c\|_2/\sqrt{N}$. It covers F_g with probability at least $1-\alpha$ whenever $\epsilon \geq \bar{W}\mathcal{R}(h)/2 + z_{1-\alpha/2}\sigma\|c\|_2/\sqrt{N} = \omega_L(\delta)/2$.

Sandwich. The left inequality restates the lower bound. Concavity and $\omega_L(0) \geq 0$ give $\omega_L(\lambda t) \leq \lambda\omega_L(t)$ for $\lambda \geq 1$. If $\omega_L(2z_{1-\alpha}\sigma/\sqrt{N}) \leq 2\epsilon/\lambda_\alpha$, then $\omega_L(2z_{1-\alpha/2}\sigma/\sqrt{N}) \leq 2\epsilon$, and the upper bound applies. □

Part (iv) has the structure of the classical theory of estimating a linear functional over a convex class (Donoho, 1994): one modulus governs both the lower bound and an affine interval. What is specific to this market is the dual form. The cost of a question about the traders is the value of a problem of polynomial approximation of g on the memory range, in which the coefficients are penalized because each multiplies a noisy tap. The number of groups and the spacing of their

memories do not enter.

Proposition S7.2 (Which questions are parametric). *Let* $\omega_\infty(\delta) = \inf_L \omega_L(\delta)$. *Then* $\sup_{0<\delta\le 1} \omega_\infty(\delta)/\delta < \infty$, *so that* $N_{\mathrm{cert}}(\epsilon; F_g) = O(\sigma^2\epsilon^{-2})$ *with observation depth* L *of order* $\log(1/\epsilon)$, *if and only if* g *agrees on* $[0,\bar r]$ *with a power series* $\sum_t a_t r^t$ *whose coefficients are square summable. When* g *is a power series on* $[0,\bar r]$ *with* $\bar r > 0$, *this is the condition* $\sum_t a_t^2 < \infty$ *on its Taylor coefficients.*

Proof. If $a \in \ell^2$, take $c = (a_0, \dots, a_{L-1})$. By Cauchy–Schwarz, $\mathcal{R}(g - p_c) \le 2 \sup_{r\le\bar r} |\sum_{t\ge L} a_t r^t| \le 2\|a\|_2 \bar r^L/\sqrt{1-\bar r^2}$, so $\omega_L(\delta) \le \delta\|a\|_2 + 2\bar W\|a\|_2\bar r^L/\sqrt{1-\bar r^2}$, which is at most $C\delta$ once $L \ge \log(1/\delta)/\log(1/\bar r)$; Theorem S7.1(iv) then gives the parametric cost. Conversely, suppose $\omega_\infty(\delta) \le C\delta$ for $\delta \in (0,1]$. For $\delta_n = 1/n$ choose L_n and $c^{(n)} \in \mathbb{R}^{L_n}$ with $\delta_n\|c^{(n)}\|_2 + \bar W\mathcal{R}(g - p_{c^{(n)}}) \le 2C\delta_n$. Then $\|c^{(n)}\|_2 \le 2C$ and $\|g - p_{c^{(n)}}\|_\infty \le 2C\delta_n/\bar W \to 0$. Padded with zeros, the $c^{(n)}$ are bounded in ℓ^2, so a subsequence converges weakly to some $a \in \ell^2$. For each $r \in [0,\bar r]$ the sequence $(r^t)_t$ lies in ℓ^2, so $p_{c^{(n)}}(r) = \langle c^{(n)}, (r^t)\rangle \to \sum_t a_t r^t$ along the subsequence, while $p_{c^{(n)}}(r) \to g(r)$. Power series that agree on an interval are equal, which gives the last sentence. □

Where the questions of this paper sit. A single tap and the discounted feedback have square-summable coefficients and are parametric: they cost of order $\sigma^2\epsilon^{-2}$ experiments however many groups there are and however close their memories. Square-summable power series are the restrictions of the Hardy space H^2 of the unit disc, whose functions grow at most like $(1-r)^{-1/2}$ as $r \uparrow 1$; that is exactly the growth beyond which part (iii) rules out certification without a memory bound. The total feedback weight Φ, which fixes the feedback budget and with it stability and worst-case transient loss, has $a_t \equiv 1$ and sits on the boundary: Niroula et al. (2026) show that it costs $\sigma^2\epsilon^{-2}\log(1/\epsilon)$ experiments and that the logarithm cannot be removed. Composition sits at the far end. The weight of a group is not a linear functional of M at all, and by Theorem 5.1 its cost grows like $(\Delta\eta)^{-(4K-2)}$. Market-wide questions are cheap; questions about who trades are not.

Price records. The theorem is stated for observed kernel taps. With price records only, the first $T+1$ price taps are a function of the first T kernel taps ($h_0 = \kappa$ and $h_{t+1} = ah_t + \kappa\sum_{j\le t} k_j h_{t-j}$), so part (ii) holds with $T+1$ price taps in place of T kernel taps, and every lower bound for kernel taps that rests on equal taps holds for price taps. In the other direction, the map from the first $L+1$ price taps to the first L kernel taps is triangular, and its linearization at a stable economy is a lower-triangular Toeplitz operator whose symbol, the reciprocal of the square of the closed-loop transfer function up to a factor, is bounded on the unit circle uniformly in L. Estimated kernel taps therefore have ℓ^2 errors of order $\sigma N^{-1/2}$ to first order, and the parametric case of Proposition S7.2 carries over to price records to first order, on classes with a uniform stability margin. We do not state the higher-order terms.

Machine check. For finite populations, the weak-duality half of (iv), the affine estimate of (i), part (iii) and the parametric direction of Proposition S7.2 are machine-checked (`p1_question_weak_duality`, `p1_question_truncation_error`, `p1_long_memory_impossible`, `p1_parametric_modulus`, `Paper1QuestionsCost.lean`). The lower half of (iv), part (ii) and the converse of the proposition rest on Sion's minimax theorem, duality for best uniform approximation and weak compactness in ℓ^2, and are proved on paper.

Numerical check. With two groups at memories $0.5 \pm \Delta\eta$, the error of the truncated estimator of the discounted feedback ($\beta = 0.9$) from noisy taps does not depend on $\Delta\eta$, while

the error of the weights by Prony's method grows like $(\Delta\eta)^{-3}$: fitted slopes -0.02 and -3.02 over $\Delta\eta \in \{0.2, 0.1, 0.05\}$, 2,000 replications each (`research_extension/theorem_checks/mc_questions.py`).

S8 Surveillance over time and against strategic traders

The cost results treat the population as fixed. Real populations drift: algorithms enter, exit and are retuned, some of them in response to the supervision itself, and the groups being watched choose how to trade. This section gives three results: a speed limit on tracking a moving population, a floor on what probing can learn about traders who learn from prices, and an equilibrium in which surveillance without tags pushes algorithms together and never detects them.

S8.1 Tracking a moving population

Fix the setting of Theorem 5.1: $K \geq 2$ groups with rates $r_c + \Delta\eta\, s_i$, rescaled nodes s and weights ω in a rich class, persistence and impact known, and the market Schur stable on the class. Time runs in periods. In period t the population is $\theta_t = (s_t, \omega_t)$, and the supervisor runs n reset experiments. A path is *admissible at speed* v if every θ_t lies in the class and $\|\theta_{t+1} - \theta_t\|_\infty \leq v$. At date T the supervisor estimates ω_T from all data up to T.

Theorem S8.1 (Tracking a moving population)**.** *There are constants $c, C, \bar{v} > 0$ and $v_0^* > 0$, depending only on the class, such that, with $\Gamma = \sigma^2 v (\Delta\eta)^{-(4K-2)}/n$:*

(i) Lower bound. *For every estimator and every T large enough, some path admissible at speed $v \leq \bar{v}$ has* $\Pr\big(\max_i |\hat{\omega}_{T,i} - \omega_{T,i}| \geq c \min(1, \Gamma^{1/3})\big) \geq 1/4$.

(ii) Upper bound. *If $\Gamma \leq 1/C$, Prony's method applied to the taps averaged over the last $m \asymp (\sigma^2(\Delta\eta)^{-(4K-2)}/(nv^2))^{1/3}$ periods has* $\max_i |\hat{\omega}_{T,i} - \omega_{T,i}| = O_P(\Gamma^{1/3})$ *along every path admissible at speed v, for $T \geq m$.*

(iii) Speed limit. *If $v \geq v^* := v_0^*\, n(\Delta\eta)^{4K-2}/\sigma^2$, no estimator tracks the weights to accuracy below c, however long the record.*

(iv) Market-wide questions. *For a linear question F_g with square-summable coefficients whose answer moves by at most v_F per period along the path, the window average of the truncated estimator of Theorem S7.1(i) tracks it with error $O_P((\sigma^2 v_F/n)^{1/3})$, whatever K and $\Delta\eta$.*

Proof. (i) Use the family R_c of Lemma S5.3: for c near zero, the K-node rules whose nodes are the zeros of $\ell_K + c\,\ell_{K-1}$ match all moments of orders up to $2K-2$, so all members give populations that agree in the first $2K-1$ rescaled moments. Their nodes are simple zeros of a polynomial that is analytic in c, so nodes and weights are analytic in c; by Lemma S5.3(iii) the weights are not all constant, so some weight $\omega_{i_0}(c)$ has $|\omega'_{i_0}(c^*)| \geq 2b > 0$ at some c^* arbitrarily close to zero, and $|\omega'_{i_0}| \geq b$ on an interval $I = [c^*, c^* + \iota]$ on which the rules lie in the class (scaled to total weight W). Let $B = \sup_I \|\frac{d}{dc}(s,\omega)\|_\infty$. On I, the rescaled moments of orders $p \geq 2K-1$ are Lipschitz in c with constant at most $C_1(1+p)$, so, exactly as in the proof of Theorem 5.1, the price responses of the populations at c and c' differ in ℓ^2 by at most $C_2|c-c'|(\Delta\eta)^{2K-1}$, and one reset experiment carries divergence at most $C_3(c-c')^2(\Delta\eta)^{4K-2}/\sigma^2$.

Compare two paths. On the first, $c_t = c^*$ for all t. On the second, $c_t = c^*$ up to date $T-m$ and $c_t = c^* + (v/B)(t - T + m)$ afterwards; it is admissible at speed v as long as $vm/B \leq \iota$. The two data laws differ only in the last m periods, in the j-th of which the populations are vj/B apart

along the family, so the divergence between them is at most $\sum_{j=1}^{m} nC_3(vj/B)^2(\Delta\eta)^{4K-2}/\sigma^2 \le nC_3v^2m^3(\Delta\eta)^{4K-2}/(B^2\sigma^2)$, using $\sum_{j\le m} j^2 \le m^3$. At date T the weights ω_{i_0} differ by at least bvm/B. Choose $m = \min\{\lfloor (B^2\sigma^2/(2nC_3v^2(\Delta\eta)^{4K-2}))^{1/3}\rfloor, \lfloor \iota B/v\rfloor\}$, which is at least one for $v \le \bar{v}$ and small $\Delta\eta$. The divergence is then at most $1/2$, and Le Cam's two-point argument gives an error of at least half the separation, $bvm/(2B)$, with probability at least $1/4$ under one of the two paths. If the first term attains the minimum, the separation is of order $\Gamma^{1/3}$; if the second does, it is $b\iota/2$, a constant.

(ii) The rescaled moments of a population are linear in its merged measure, and on the class the rescaled moment of order p is Lipschitz in (s,ω) with a constant that does not depend on $\Delta\eta$, because $|s_i| \le 1$. So the average of the rescaled moments over the last m periods differs from those at date T by at most C_4vm. The averaged taps estimate the rescaled moments of that average with error $O_P(\sigma(\Delta\eta)^{-(2K-1)}(nm)^{-1/2})$, as in Theorem 5.2. The Prony map has a Lipschitz inverse near the class (Theorem 5.2), so the weight error is $O_P(vm + \sigma(\Delta\eta)^{-(2K-1)}(nm)^{-1/2})$, and the stated m balances the two terms at order $\Gamma^{1/3}$. The condition $\Gamma \le 1/C$ keeps the moment error within the radius of the inverse.

(iii) This is the second case of (i): if $v \ge v^*$ with v_0^* small enough, the window $\iota B/v$ is the binding one, the divergence is at most $1/2$, and the separation is a constant.

(iv) The bias of the window average is at most $v_F m$ plus the truncation error, and its noise is $O_P(\sigma\sqrt{A_g}(nm)^{-1/2})$; balancing gives the rate, with no dependence on the population beyond the bound on the weights. □

What this means. Theorem 5.1 says composition is expensive to learn; Theorem S8.1 says that when composition moves, the expense becomes a speed limit. The supervisor's error falls only like the cube root of the data rate, and populations that change faster than $v^* \propto n(\Delta\eta)^{4K-2}/\sigma^2$ cannot be tracked at all. In the proof, a population that drifts across the distance between two members of the hard family in a few identification times cannot be told from one that stays still. At illustrative parameters (memories with half-lives of 5 to 9 days), where telling such a pair apart takes thousands of years of daily prices, composition is untrackable unless it is stable over millennia. Market-wide questions are tracked at the cube-root rate whatever the number of groups; and along the hard directions their answers hardly move, because those directions preserve the low moments that market-wide answers mostly depend on.

Numerical check. With two groups moving along the family and the window chosen to minimize the median error, the weight error grows with the speed at fitted slope 0.33 (against 1/3) and with the spacing at slope -2.01 (against -2, the cube root of $(\Delta\eta)^{-6}$). The error for the discounted feedback grows with speed at slope 0.34; its dependence on the spacing, slope 0.98, comes only through the speed of its answer, which along the family is of order $(\Delta\eta)^3$ (`research_extension/tracking/tracking_check.py`).

S8.2 Traders who learn from prices: an observer effect

Suppose the traders adjust their rules in response to what they earn, and so in response to the supervisor's probes. A probe of size A moves a group's position by an amount proportional to A and the price by an amount proportional to A, so its profit and loss is proportional to A^2: with position $w_i m_t$ and price changes Δx_{t+1}, the profit over a probe's response is $w_i A^2 \Pi_i$, where $\Pi_i = \sum_t m_t^{(1)} \Delta x_{t+1}^{(1)}$ is computed on the unit response. A rule that moves its weight by η times realized profit therefore moves by at most $\eta A^2 \max_i |w_i \Pi_i|$ per probe. Call a class of learning rules

of sensitivity γ if, with n probes of size A per period, the population can move at speed up to $v_0 + \gamma n A^2$, and the supervisor knows only this bound.

Proposition S8.2 (An observer effect)**.** *In the setting of Theorem S8.1, with n probes of size A per period and learning rules of sensitivity γ, every probing policy leaves a minimax tracking error of at least*

$$c \min\Big(1, \big(\sigma^2 (\Delta\eta)^{-(4K-2)} (v_0/(nA^2) + \gamma)\big)^{1/3}\Big) \;\geq\; c \min\Big(1, \big(\sigma^2 \gamma (\Delta\eta)^{-(4K-2)}\big)^{1/3}\Big).$$

For a market-wide question whose answer moves at most γ_F per unit of probe energy, the floor is $c \min(1, (\sigma^2 \gamma_F)^{1/3})$.

Proof. A probe of size A is a unit probe with noise σ/A, so Theorem S8.1(i) applies with σ replaced by σ/A and v by $v_0 + \gamma n A^2$: then $\Gamma = \sigma^2 (v_0 + \gamma n A^2)(\Delta\eta)^{-(4K-2)}/(nA^2)$, which is at least $\sigma^2 \gamma (\Delta\eta)^{-(4K-2)}$. The lower bound of Theorem S8.1(i) needs only that some admissible path moves along the hard family at that speed, which a class of learning rules known only through its sensitivity allows. The market-wide case is the same computation with Theorem S8.1(iv). □

Probing harder buys information at rate A^2 and moves the target at rate A^2, so the two cancel: surveillance of traders who learn from prices has a floor that no probing design removes. Two caveats bound the result. It is a minimax statement over learning rules of a given sensitivity; a supervisor who knows the learning rule can anticipate the drift it causes. And the floor for composition carries the factor $(\Delta\eta)^{-(4K-2)}$, while the floor for market-wide questions does not.

S8.3 A surveillance and hiding game

A group that expects to be targeted once identified may choose to trade like its neighbours. Consider two untagged groups and let one of them choose the spacing $D \in (0, D_0]$ between its memory and its neighbour's, where D_0 is the spacing of its preferred strategy and moving away from it costs $k(D_0 - D)^2$. A supervisor with N observations runs a test with false-alarm probability α; by Theorems 5.1 and 5.4, its detection probability depends on the data and the spacing through ND^6 at leading order, $\pi(ND^6)$ with π continuous and nondecreasing and $\pi(0) = \alpha$. Detection costs the group $L > 0$. Tags, if the supervisor imposes them, make the group's weight identifiable at a cost that does not depend on D (Theorem 6.4).

Proposition S8.3 (Surveillance and hiding)**.** *Let $D^*(N)$ minimize $k(D_0 - D)^2 + L\pi(ND^6)$ over $(0, D_0]$.*

(i) Crowding. *If π is differentiable at ND_0^6 with positive derivative, then $D^*(N) < D_0$.*

(ii) Surveillance without tags is self-limiting. $\limsup_{N\to\infty} \pi(ND^*(N)^6) \leq \alpha + kD_0^2/L$. *When the stakes are high, $L \gg kD_0^2$, the group is detected about as often as a false alarm, however much data the supervisor has.*

(iii) Tags remove crowding. *If the detection probability does not depend on D, then $D^* = D_0$.*

(iv) Many groups. *Let $M \geq 2$ untagged groups choose memory positions p_g, group g paying $k_g(p_g - p_g^\circ)^2 + L_g \pi(ND_g^6)$, where p_g° is its preferred position and D_g its distance to the nearest other group. In every Nash equilibrium (existence is not established here) in which the distance Δ_g from p_g° to the nearest other group's position stays bounded as N grows, $\limsup_{N\to\infty} \pi(ND_g^6) \leq \alpha + k_g \limsup_N \Delta_g^2/L_g$ for every group.*

Proof. (i) The derivative of the objective at D_0 is $6LND_0^5\pi'(ND_0^6) > 0$, so a small move below D_0 lowers it. (ii) For any $c > 0$ and N large enough that $cN^{-1/6} \le D_0$, optimality against $D = cN^{-1/6}$ gives $L\pi(ND^{*6}) \le k(D_0 - cN^{-1/6})^2 + L\pi(c^6) \le kD_0^2 + L\pi(c^6)$. Letting $c \to 0$ and using continuity of π at zero gives the bound. (iii) The cost $k(D_0 - D)^2$ is minimized at D_0. (iv) Holding the others' positions fixed, group g can move to within $cN^{-1/6}$ of its nearest neighbour, at cost at most $k_g(\Delta_g + cN^{-1/6})^2$ and detection probability at most $\pi(c^6)$, because π is nondecreasing. Equilibrium gives $L_g\pi(ND_g^6) \le k_g(\Delta_g + cN^{-1/6})^2 + L_g\pi(c^6)$; letting $N \to \infty$ and then $c \to 0$ gives the bound. □

The proposition gives the paper's results on cost an equilibrium reading. More data without tags make high-stakes algorithms trade alike, and so make composition harder to learn, while the detection probability stays near the false-alarm level. Tags undo the crowding. Two predictions follow that algorithm-level data could test: where algorithm identification is introduced, the dispersion of memory rates across algorithm classes should rise; and in untagged markets, the memories of strategies with the most to lose from targeting should converge as surveillance improves. With the Gaussian test, the optimal spacing in the high-stakes case falls like $N^{-1/4}$ and detection stays at the 5% false-alarm level up to $N = 10^{16}$; in the low-stakes case the group hides for moderate N and stops hiding once data are plentiful (`research_extension/tracking/hiding_game_check.py`). With several groups, part (iv) bounds detection group by group in every equilibrium; whether an equilibrium exists, and what it looks like, is not studied here.

Machine check. The divergence of a moving window, Le Cam's two-point step, the bias–noise trade-off, the observer-effect inequality and parts (ii)–(iv) of Proposition S8.3 are machine-checked (`p1_track_kl_window`, `p1_track_two_point`, `p1_track_tradeoff`, `p1_observer_floor`, `p1_hiding_detection_bound`, `p1_hiding_tag_no_crowding`, `Paper1TrackingGame.lean`; `p1_hiding_nash_bound`, `Paper1NewsLawHiding.lean`). The analytic properties of the family R_c, the moment Lipschitz bounds and the closed-loop step are on paper.

S9 One running market, news of any law, and the value of tags in money

The upper bounds above are stated for reset experiments, and the passive bounds for Gaussian news. This section removes both restrictions for the composition cost and turns the cost into money. Probing a market that has been running from an unknown state needs the same order of data as reset experiments; Gaussian news is the least favourable news law, and any other law in a broad class changes only the constant; and the cost of learning composition from prices is a lower bound on the money value of class tags.

S9.1 One running market

The market of (1) has been running since an unknown date, so its state when probing begins is unknown and arbitrary. Its news ε_t may have any law with finite variance, and prices are recorded with independent observation noise ν_t. Starting ℓ periods before the first of T records, the observer trades $v_t = A\xi_t$, where the ξ_t are independent fair signs, independent of the market. The market is linear, so the recorded price is

$$\tilde{x}_t = \sum_{i \ge 0} h_i v_{t-i} + n_t,$$

where h is the response to a unit trade ($h_0 = \kappa$), the sum runs over the probes sent so far, and n_t collects the free response to the initial state, the response to news and the observation noise. The probes are independent of n. Write $\bar{s}^2 = T^{-1}\sum_{t\le T}\mathbb{E}n_t^2$, which is bounded under Schur stability whenever the initial state and the news have finite second moments.

Theorem S9.1 (Probing one running market). *(i)* Taps. *For every $j \le \ell$, the estimate $\hat{h}_j = (AT)^{-1}\sum_{t=1}^{T}\tilde{x}_t\xi_{t-j}$ satisfies*

$$\mathbb{E}(\hat{h}_j - h_j)^2 \le \frac{2\|h\|_2^2 + \bar{s}^2/A^2}{T}.$$

(ii) Upper bound. *In the setting of Theorem 5.2, with $\ell \ge 2K$, there are constants c_1, L, C depending only on the class such that, for $e \le 1$ with $2^{2K}Ce(\Delta\eta)^{-(2K-1)} \le c_1$, Prony's method applied to $\hat{h}_0, \dots, \hat{h}_{2K}$ has a solution and every solution recovers the weights to within $L\,2^{2K}Ce(\Delta\eta)^{-(2K-1)}$, with probability at least $1 - (2K+1)(2\|h\|_2^2 + \bar{s}^2/A^2)/(Te^2)$. So $T = O\big((\Delta\eta)^{-(4K-2)}\big)$ periods suffice for any fixed accuracy and confidence, whatever the state of the market when probing began and whatever the law of its news.*

(iii) Lower bound. *Suppose the news and the observation noise are Gaussian ($\sigma_\nu^2 > 0$), the market starts from rest any number of periods before the record, and the probes, of any law independent of the market, satisfy $\mathbb{E}\sum_t v_t^2 \le PT$. For the Gauss and Gauss–Radau pair of Theorem 5.1, a test with worst-case error below $1/4$ needs*

$$T \;\ge\; \frac{c\,(\Delta\eta)^{-(4K-2)}}{1 + P/\sigma_\nu^2},$$

with $c > 0$ depending only on the class.

Together, (ii) and (iii) say that one running market needs data of the same order as reset experiments: $(\Delta\eta)^{-(4K-2)}$ periods, no fewer and no more.

Proof. (i) Condition on the noise path n; the probes are independent of it, so a bound for every path is a bound after averaging. Since $\xi^2 = 1$,

$$\hat{h}_j - h_j = \frac{1}{T}\Big(\sum_{(t,i)} h_i\,\xi_{t-i}\xi_{t-j} + \sum_t \frac{n_t}{A}\xi_{t-j}\Big),$$

where the first sum runs over $t \le T$ and the lags $i \ne j$ of probes already sent. Products of distinct independent fair signs are orthonormal: $\mathbb{E}[\chi_S\chi_R] = 1$ if $S = R$ and 0 otherwise, where $\chi_S = \prod_{a\in S}\xi_a$ (flipping one sign in $S\triangle R$ reverses the product). Each term of the first sum is the product over a two-element set $\{t-i, t-j\}$, and such a set arises from at most two pairs (t,i): from (t,i) itself, and from the one pair whose j-th probe is the probe $t-i$. Each term of the second sum is a single sign, and distinct t give distinct signs. Pair products are orthogonal to single signs. Hence, using $2xy \le x^2 + y^2$ within each set of at most two coinciding pairs,

$$\mathbb{E}\Big(\sum h_i\xi_{t-i}\xi_{t-j} + \sum \tfrac{n_t}{A}\xi_{t-j}\Big)^2 \le 2\sum_{(t,i)} h_i^2 + \sum_t \frac{n_t^2}{A^2} \le 2T\|h\|_2^2 + \frac{T\bar{s}^2}{A^2}.$$

(ii) Chebyshev's inequality and a union bound over the $2K+1$ taps give the event $\max_{t\le 2K}|\hat{h}_t - h_t| \le e$ with the stated probability. On that event the deterministic part of the proof of Theorem 5.2 applies unchanged: the Lipschitz closed-loop deconvolution, the $(\Delta\eta)^{-m}$ rescaling and the uniform inverse of the moment map. The norm $\|h\|_2$ is bounded on the class by Schur stability.

(iii) Condition on the probes. Given v, the record is Gaussian with mean $h * v$ and covariance Σ (news through the closed loop, plus $\sigma_\nu^2 I$), and

$$\mathrm{KL}\big(N(\mu_A, \Sigma_A) \,\|\, N(\mu_B, \Sigma_B)\big) = \mathrm{KL}\big(N(0, \Sigma_A) \,\|\, N(0, \Sigma_B)\big) + \tfrac{1}{2}(\mu_A - \mu_B)^\top \Sigma_B^{-1}(\mu_A - \mu_B).$$

The first term is a passive divergence. Adding independent observation noise cannot increase it (data processing), and without that noise Theorem 5.4 bounds it by $C_1 T(\Delta\eta)^{4K-2}$, uniformly in the warm-up. For the second, $\Sigma_B \succeq \sigma_\nu^2 I$ and $\|(h^A - h^B) * v\|_2 \le \sup_\omega |\Delta\hat{H}(e^{i\omega})| \, \|v\|_2 \le C_2(\Delta\eta)^{2K-1}\|v\|_2$ (paragraph "What counts as an experiment"). The probes have the same law under both economies, so averaging gives $\mathrm{KL} \le T(\Delta\eta)^{4K-2}(C_1 + C_2^2 P/(2\sigma_\nu^2))$. By Pinsker's inequality, a worst-case error below $1/4$ needs this to exceed $1/2$. □

What this means. Reset experiments are a benchmark, not a requirement. A supervisor who trades random signs into a live market and correlates prices with its own trades learns the response as fast as one who could reset the market, and the composition barrier is the same. The proof of (i) uses nothing about the noise except its independence from the probes, so the market's history, its news law and any non-stationarity enter only through $\bar{s}^2$. In a simulation of a two-group market with unknown start and either Gaussian or Laplace news (400 runs, $T =$ 4,000), the mean squared error of every tap from h_0 to h_4 stays at most 0.71 of the bound in (i) (`research_extension/running/running_check.py`).

Machine check. Part (i) is machine-checked for every noise path, with the probes as independent fair signs on a finite probe space: orthonormality of sign products, the fibre count and the resulting bound (`p1_walsh_quadratic_bound`, `p1_running_tap_mse`). Part (ii) is machine-checked as stated: Chebyshev's inequality, the union bound and the deterministic Prony step of Theorem 5.2 (`p1_running_prony_upper`, `Paper1RunningMarket.lean`). Averaging over the noise path, the bound on $\|h\|_2$ over the class, and part (iii) are on paper.

S9.2 News of any law

Let the news be independent with a density f of mean zero and variance σ^2, and write $D_f(\Delta) = \int f(x) \log \frac{f(x)}{f(x-\Delta)}\, dx$ for its location divergence. Say that f has *quadratic location divergence* with constant I_f^* if $D_f(\Delta) \le \frac{1}{2} I_f^* \Delta^2$ for all Δ. Gaussian news has $I_f^* = 1/\sigma^2$, with equality. Laplace news with scale b has $D_f(\Delta) = |\Delta|/b - 1 + e^{-|\Delta|/b} \le \Delta^2/(2b^2)$, so $\sigma^2 I_f^* = 2$; Student-t news with five degrees of freedom has $\sigma^2 I_f^* = 1.25$, computed numerically.

Proposition S9.2 (News of any law)**.** *(i) Populations with the same kernel produce the same law of prices for every news law; populations with different kernels produce different laws for every news law with finite variance.*

(ii) If f has quadratic location divergence, then for any two economies and any record length T, the divergence between their price records from rest satisfies $\mathrm{KL}_f \le \sigma^2 I_f^* \cdot \mathrm{KL}_{\mathrm{Gauss}}$, where $\mathrm{KL}_{\mathrm{Gauss}}$ is the divergence under Gaussian news of the same variance. The passive lower bounds of Theorem 5.4 therefore hold with the number of observations divided by $\sigma^2 I_f^*$: the order $(\Delta\eta)^{-(4K-2)}$ is the same.

(iii) $\sigma^2 I_f^* \ge 1$, with equality for Gaussian news. Gaussian news is the least favourable law in this class.

Proof. (i) Prices are the closed-loop filter of the news, and the filter is determined by the kernel (Theorem 3.1); different kernels give different autocovariances whenever the news has finite variance. (ii) The filter from news to prices has leading coefficient one and a causal, stable inverse, $\varepsilon_t = x_t - ax_{t-1} - \kappa \sum_m k_m x_{t-1-m}$, so from rest the record determines the news exactly, and under economy θ the density of the record is $\prod_t f(x_t - m_t^\theta)$ with m_t^θ a linear function of past prices. The chain rule gives $\mathrm{KL}_f = \sum_t \mathbb{E}_A D_f(\Delta_t)$ with $\Delta_t = m_t^B - m_t^A$. Under economy A, Δ_t is a linear function of past news, so $\mathbb{E}_A \Delta_t^2$ depends on the news law only through σ^2, and the same identity under Gaussian news reads $\mathrm{KL}_{\mathrm{Gauss}} = \sum_t \mathbb{E}_A \Delta_t^2/(2\sigma^2)$. Hence $\mathrm{KL}_f \le \frac{1}{2} I_f^* \sum_t \mathbb{E}_A \Delta_t^2 = \sigma^2 I_f^* \, \mathrm{KL}_{\mathrm{Gauss}}$. (iii) Near zero, $D_f(\Delta) = \frac{1}{2} I(f) \Delta^2 + o(\Delta^2)$, with $I(f)$ the Fisher information for location, so $I_f^* \ge I(f) \ge 1/\sigma^2$ by the Cramér–Rao inequality for location, with equality only for the normal law. □

Laws outside the class, for example news with bounded support, whose shifted densities are mutually singular, can make prices far more informative; we do not analyse them. For the two-group pair of Theorem 5.1 and records of 200 periods from rest, the ratio $\mathrm{KL}_f/\mathrm{KL}_{\mathrm{Gauss}}$ is 1.999 under Laplace news ($\sigma^2 I_f^* = 2$) and 1.250 under Student-t news with five degrees of freedom (1.25): the pair is close, so the bound is attained to three digits (`research_extension/running/running_check.py`).

Machine check. The Laplace bound and its comparison with the Gaussian divergence (`p1_exp_neg_le`, `p1_laplace_location_kl`, `p1_laplace_vs_gaussian`) and the comparison step of (ii), as an inequality between integrals (`p1_newslaw_kl_compare`, `Paper1NewsLawHiding.lean`), are machine-checked. The chain-rule identity, the Laplace divergence formula, part (i) and part (iii) are on paper.

S9.3 The value of tags in money

Corollary S9.3 (A lower bound on the value of tags)**.** *In the setting of Theorem 5.1, let each experiment cost c_e. Any procedure that tells the Gauss and Gauss–Radau pair apart from prices with error probabilities $\alpha_0 + \beta_0 < 1$ costs at least $c\, c_e \sigma^2 (1 - \alpha_0 - \beta_0)^2 (\Delta\eta)^{-(4K-2)}$. With class tags, by Theorem 6.3(i), recovering the weights costs at most $C\, c_e$, independently of $\Delta\eta$. The money value of tags to a supervisor who needs composition, measured as the cost they save, is therefore at least $c_e\big(c\,\sigma^2 (1 - \alpha_0 - \beta_0)^2 (\Delta\eta)^{-(4K-2)} - C\big)$.*

The cost of an experiment is the execution cost of its probe, priced in Section S3. The corollary gives a floor, not an estimate: the value of tags to a supervisor who never needs composition is zero, and the corollary says nothing about the supervisor's loss from acting without it.

Machine check. The first statement is machine-checked for the moment-matched pair that Lean uses for Theorem 5.1 (binomial weights on equally spaced nodes, the same order), by composing the cost with the uniform lower bound (`p1_tags_value_money`, `Paper1NewsLawHiding.lean`); the tagged upper bound is that of Theorem 6.3.

S10 Questions left open in earlier versions

This section settles questions that earlier versions left open or answered only numerically: the exact size of the moment gap behind the constants of Theorem 5.1; the threshold for a cascade after a

breach, with mechanical impact and with forward-looking dealers, and the path the fall takes; a long-lived insider, alone and against a group that chooses its rule; an upper bound for a family of nonlinear rules; saturation of every order; rates for noisy stress identification with smooth shapes and with hard limits; learning and nonlinear traders; and the largest weights of the hard pair for every K. The rank condition of Theorem 6.5 and the finite-sample version of Proposition 5.5 are proved where those results are stated.

S10.1 The moment gap of the hard pair

Proposition S10.1 (The moment gap)**.** *Normalize the K-point Gauss and Gauss–Radau rules for the uniform probability on $[-1,1]$. Their moments agree up to order $2K-2$, and their moments of order $2K-1$ differ by exactly*

$$g_K = \frac{4^K}{K\binom{2K}{K}^2}, \qquad \frac{\pi}{4^K}\Big(1+\frac{1}{4K}\Big) < g_K < \frac{\pi}{4^K}\Big(1+\frac{1}{2K}\Big).$$

Proof. The Gauss rule integrates polynomials of degree $2K-1$ exactly, so its moment of order $2K-1$ is the true one. For the Gauss–Radau rule with a fixed endpoint and weight one on $[-1,1]$, the error for a function with continuous derivative of order $2K-1$ is $2^{2K-1}K[(K-1)!]^4 f^{(2K-1)}(\xi)/[(2K-1)!]^3$ for some ξ (Abramowitz and Stegun, 1964, 25.4.31). For $f(x)=x^{2K-1}$ the derivative is the constant $(2K-1)!$, so the error is exact, and halving it for the uniform probability gives $2^{2K-2}K[(K-1)!]^4/[(2K-1)!]^2 = 4^K/(K\binom{2K}{K}^2)$. The bounds follow from $4^K/\sqrt{\pi(K+1/2)} < \binom{2K}{K} < 4^K/\sqrt{\pi(K+1/4)}$. □

The gaps 0.222, 0.0533 and 0.0131 for $K=2,3,4$ reported with Theorem 5.1 are g_2, g_3 and g_4. The first tap at which the two price responses differ is h_{2K}, with $h^A_{2K} - h^B_{2K} = \pm\kappa^2 W g_K(\Delta\eta)^{2K-1}$. At leading order in $\Delta\eta$ the whole response difference is $g_K(\Delta\eta)^{2K-1}$ times a closed-loop response that depends on K but not on $\Delta\eta$. So the divergence in the proof of Theorem 5.1 carries the factor g_K^2, and the data requirement the factor $g_K^{-2} \approx 16^K/\pi^2$. This is one reason the constants grow so fast with K; the closed-loop factor adds more. The formula agrees with Gauss–Radau rules computed in 60-digit arithmetic to 3×10^{-46} for $K \le 30$, and the bounds hold for every $K \le 2{,}000$ (`research_extension/open_problems/open_checks.py`).

S10.2 The cascade after a breach

After a breach, traders whose positions hit their limits liquidate, and their sales move the price. Let the price fall by a shock $s \ge 0$ beyond what the linear market produces, and let $L(\delta) \ge 0$ be the quantity liquidated by traders whose limits lie within δ of the pre-shock price: nondecreasing, bounded by the total position Q, and, when positions are spread with density ℓ over distances to limits, $L(\delta) = \int_0^\delta \ell$. With price impact κ, liquidation rounds run $\delta_0 = s$, $\delta_{n+1} = s + \kappa L(\delta_n)$: each round's sales push further traders over their limits.

Theorem S10.2 (A threshold for cascades)**.** *(i) The rounds increase and, for continuous L, converge to the least fixed point $\delta^*(s)$ of $\delta = s + \kappa L(\delta)$; every fixed point at or above s lies above every round.*

(ii) Below the threshold. *If L is Lipschitz with constant $\bar\ell$ and $\kappa\bar\ell < 1$, the fixed point is unique and $|\delta^*(s) - \delta^*(s')| \le |s-s'|/(1-\kappa\bar\ell)$: the fall is continuous in the shock, amplified by at most $1/(1-\kappa\bar\ell)$.*

(iii) Above the threshold. *If $\kappa\ell(\delta_0) > 1$ at some δ_0 at which ℓ is continuous, the map $\psi(\delta) = \delta - \kappa L(\delta)$ decreases near δ_0, so its maximum on $[0, \delta_0]$ is attained at some $\delta_1 < \delta_0$ with $\bar{s} = \psi(\delta_1) > \psi(\delta_0)$. Then $\delta^*(\bar{s}) \le \delta_1$, while every fixed point for a shock $s > \bar{s}$ exceeds δ_0: an arbitrarily small additional shock produces a further fall of at least $\delta_0 - \delta_1$.*

Proof. (i) By induction the rounds increase, because L is nondecreasing, and stay below any fixed point $\delta' \ge s$, because $\delta_{n+1} = s + \kappa L(\delta_n) \le s + \kappa L(\delta') = \delta'$. They are bounded by $s + \kappa Q$, so they converge, and continuity of L makes the limit a fixed point. (ii) Subtracting the fixed-point equations for s and s' gives $|\delta - \delta'| \le |s - s'| + \kappa\bar{\ell}|\delta - \delta'|$. (iii) $\psi'(\delta_0) = 1 - \kappa\ell(\delta_0) < 0$, so $\psi(\delta) > \psi(\delta_0)$ for some $\delta < \delta_0$, and the maximum of ψ on the compact interval $[0, \delta_0]$ is attained before δ_0. A fixed point for s is a solution of $\psi(\delta) = s$; for $s > \bar{s} \ge \psi$ on $[0, \delta_0]$ there is none in that interval. For $\bar{s}$, δ_1 is a fixed point, so $\delta^*(\bar{s}) \le \delta_1$ by (i). □

What this means. The conjectured threshold holds. Cascades occur exactly where the liquidation triggered per unit of price move, $\kappa\ell$, exceeds one. Below that the market absorbs liquidation with bounded amplification. The density ℓ of positions near their limits is exactly what Theorem 7.1 says calm prices do not reveal. A supervisor can therefore learn from calm prices neither whether the market is above the threshold nor where it is. In a numerical example with a bump of positions half a unit from their limits, the largest jump in the fall is 0.002 on a grid of step 0.0005 when $\kappa \max \ell = 0.75$, and 0.45 when $\kappa \max \ell = 2$ (`research_extension/open_problems/open_checks.py`). The theorem uses mechanical impact; Propositions S10.5 and S10.10 add forward-looking dealers and the path of the fall.

Machine check. All three parts are machine-checked: the rounds stay below every fixed point and converge to the least one, the contraction bound, and the jump given a value of ψ above $\psi(\delta_0)$ before δ_0 (`p1_cascade_iterate_le_fixed`, `p1_cascade_rounds_converge`, `p1_cascade_contraction`, `p1_cascade_jump`, `Paper1OpenResolved.lean`). The derivative step of (iii) is on paper.

S10.3 A long-lived insider

An insider knows the value V and trades z_t over T periods; her trades enter (1) as κz_t, like a probe. She pays the pre-trade price plus $\theta\kappa z_t$, so $\theta = 1$ is the post-trade price and $\theta = 1/2$ the average price on a linear book. Write $\hat{H}(\omega) = \kappa/(1 - e^{-i\omega}(a + \kappa\hat{k}(\omega)))$ for the frequency response of the price to a trade, with $\hat{k}$ the transform of the kernel, and suppose the market is stable in the sense used throughout: $|a| + \kappa\sum_m |k_m| \le s < 1$.

Proposition S10.3 (A long-lived insider)**.** *(i)* Only the kernel matters. *For any causal strategy, the joint law of V, her trades and prices, and hence her profit, is the same for two populations with the same kernel. Her optimal strategies and value coincide, and no observer of prices and her trades learns more about composition than the kernel.*

(ii) No manipulation. *The expected execution cost of a deterministic schedule z beyond the value is $z^\top M_\theta z$, where M_θ is the symmetric Toeplitz matrix with diagonal $\theta\kappa$ and off-diagonal entries $h_{|t-u|}/2$, and $z^\top M_\theta z \ge \kappa\big(\frac{1}{1+s} - (1-\theta)\big)\|z\|^2 > 0$ for every $\theta \ge 1/2$. No schedule, round trips included, has negative expected cost, however strong the feedback, as long as the market is stable.*

(iii) The informed schedule. *With $\Delta = V - p^*$, the deterministic schedule that maximizes expected profit $\Delta \mathbf{1}^\top z - z^\top M_\theta z$ is $z^* = \frac{\Delta}{2} M_\theta^{-1}\mathbf{1}$, with value $\frac{\Delta^2}{4}\mathbf{1}^\top M_\theta^{-1}\mathbf{1} \le \Delta^2 T/[4\kappa(\frac{1}{1+s} - (1-\theta))]$. Both depend on the population only through the kernel.*

(iv) A rational market maker nets out feedback. *If instead prices are set by a market maker as $\mathbb{E}[V \mid$ order flow so far$]$, and the feedback flow at each date is a function of earlier prices, then total and net order flow (total minus feedback) determine each other date by date. Prices, the insider's optimal strategy and her profit are those of the market without feedback traders, and the feedback flow has no price impact. The market maker needs to know only the aggregate flow rule, that is the kernel.*

Proof. (i) Aggregate feedback flow is the kernel applied to past prices on every path (Theorem 3.1), so prices are the same function of news and trades in both markets, and her strategy is a function of what she observes. (ii) Expected prices are h convolved with her trades, so the expected cost is $\sum_t z_t \sum_{u \le t} h_{t-u} z_u - (1-\theta)\kappa\|z\|^2 = z^\top M_\theta z$. A symmetric Toeplitz form is $\frac{1}{2\pi}\int |\hat{z}(\omega)|^2 S(\omega)\, d\omega$ with symbol $S_\theta(\omega) = \theta\kappa + \sum_{d\ge 1} h_d \cos(d\omega) = \operatorname{Re}\hat{H}(\omega) - (1-\theta)\kappa$. With $w = e^{-i\omega}(a + \kappa\hat{k}(\omega))$, $|w| \le s$, and $\operatorname{Re}(1-w)^{-1} \ge (1+s)^{-1}$ for $|w| \le s < 1$, because $(1-\operatorname{Re} w)(1+s) - |1-w|^2 \ge (1-s)(s+\operatorname{Re} w) \ge 0$. Hence $S_\theta \ge \kappa(\frac{1}{1+s} - (1-\theta))$, which is positive for $\theta \ge \frac{1}{2}$. (iii) The objective is strictly concave by (ii), and the bound uses the smallest eigenvalue. (iv) Prices are functions of earlier order flow, so the feedback flow at t is a function of order flow before t. Then the order flow up to t and the net flow up to t are functions of each other, by induction on t, so the market maker's information, and the price, are the same. □

What this means. The open question was whether a long-lived insider could exploit or reveal the groups' memory. She cannot reveal it: everything she does depends on the kernel alone. And in a stable market she cannot manipulate it either, since the feedback never makes a round trip profitable. With a rational market maker, feedback flow is priced out entirely, so the paper's results on feedback concern markets where impact is mechanical, as in a limit order book, not markets where a dealer prices flow by its information. For 200 random stable kernels the smallest value of $\operatorname{Re}\hat{H}/\kappa$ is never below $1/(1+s)$, and $M_{1/2}$ is positive definite (`open_checks.py`). Proposition S10.9 lets a group choose its rule against her, and Proposition S10.5 settles the case of a rational market maker.

Machine check. The inequality $\operatorname{Re}(1-w)^{-1} \ge (1+s)^{-1}$, the resulting bound on the symbol and the netting step of (iv) are machine-checked (`p1_re_inv_one_sub_ge`, `p1_no_manipulation_symbol`, `p1_netting_prefix`, `Paper1OpenResolved.lean`). The Toeplitz representation, part (iii), the step from information to conditional expectations in (iv), and part (i) (an instance of Theorem 3.1) are on paper.

S10.4 Traders with a dead band

Theorem 7.7 gives lower bounds for any rules but no upper bound, and for some families composition is not identified at all. Here is a family for which it is. Let group i trade $w_i\psi_i(y_{i,t})$, where y_i is its memory state and $|\psi_i(u) - u| \le \delta_0$ for all u. This covers a dead band, $\psi(u) = \operatorname{sign}(u)(|u| - \delta_0)_+$, where the trader does nothing until its signal leaves the band and then trades linearly beyond it, and rounding to lots of size $2\delta_0$. It does not cover saturation, whose deviation from linearity is unbounded.

Proposition S10.4 (Dead bands are identifiable)**.** *In the setting of Theorem 5.2, with* $W_1 = \sum_i |w_i|$ *and* $|a| + \kappa \sum_m |k_m| \le s < 1$*, run reset experiments with a unit impulse scaled to size* A*.*

(i) The scaled price response differs from the linear response by at most $\kappa\delta_0 W_1/((1-s)A)$ *at every tap.*

(ii) Prony's method applied to the scaled, averaged taps recovers the weights to within $C(\kappa\delta_0 W_1/((1-s)A) + \sigma A^{-1} N^{-1/2}\sqrt{\log(4K/\beta)})(\Delta\eta)^{-(2K-1)}$ *with probability at least* $1-\beta$*, whenever this is below the radius of Theorem 5.2. Accuracy* ε *needs probes of size* $A \gtrsim \delta_0(\Delta\eta)^{-(2K-1)}/\varepsilon$ *and* $N \gtrsim \sigma^2/(A^2\varepsilon^2(\Delta\eta)^{4K-2})$ *experiments.*

Proof. (i) Let d_t be the difference between the price paths with and without the band. The linear part of every group's flow is the kernel applied to past prices, so $d_t = a d_{t-1} + \kappa \sum_m k_m d_{t-1-m} + e_t$, with $e_t = \kappa \sum_i w_i(\psi_i(y_{i,t-1}) - y_{i,t-1})$ and $|e_t| \le \kappa\delta_0 W_1$. By induction $|d_t| \le s \sup_{u<t} |d_u| + \kappa\delta_0 W_1$, so $|d_t| \le \kappa\delta_0 W_1/(1-s)$; divide by A. (ii) The scaled taps are the linear taps plus a bias bounded by (i) plus noise of standard deviation σ/A per experiment. The proof of Theorem 5.2 applies with the bias added to the radius of the event. □

Large probes identify dead-band traders because the band is a bounded distortion, which a probe of size A dilutes by $1/A$. In a simulation of two dead-band groups, the largest tap bias stays below the bound of (i) for probe sizes from 1 to 32 and falls with fitted slope -0.99 in A (`open_checks.py`).

Machine check. The bias bound of (i), for any bounded deviation fed through a stable closed loop, is machine-checked (`p1_deadband_bias`, `Paper1OpenResolved.lean`). The reduction of the group flows to that recursion and part (ii) are on paper; part (ii) is the deterministic Prony step of Theorem 5.2, which is machine-checked, with the bias added to the radius.

S10.5 Forward-looking dealers after a breach

Theorem S10.2 lets liquidation move the price round by round. Suppose instead that the price is set by competitive dealers who see the whole sequence coming. As in Grossman and Miller (1988), dealers absorb supply at a discount proportional to the inventory they must carry, κ per unit, and they know the liquidation rule L. A rational-expectations outcome is a total fall δ at which the liquidation the fall triggers, $L(\delta)$, is exactly what justifies the fall.

Proposition S10.5 (Cascades with forward-looking dealers)**.** *(i) The rational-expectations falls are the fixed points of* $\delta = s + \kappa L(\delta)$*. The price moves to one of them at once, without rounds.*

(ii) Below the threshold of Theorem S10.2(ii) there is exactly one, and it is the outcome of the rounds.

(iii) Above the threshold, with $\delta_1 < \delta_0$ *as in Theorem S10.2(iii), every shock* s *with* $\psi(\delta_0) < s < \psi(\delta_1)$ *and* $\psi(0) \le s$ *has at least three equilibria: a mild fall below* δ_1*, an intermediate one, and a crash beyond* δ_0*. Which occurs depends on what dealers expect, so a crash can be self-fulfilling.*

(iv) If dealers are risk-neutral toward liquidation flow that is predictable from public information, so that predictable flow carries no inventory discount, the fall is $\delta = s$*: no cascade at all.*

Proof. (i) Dealers clear the market at a fall that prices the inventory they end up holding, $\kappa L(\delta)$ beyond the shock, and with rational expectations that inventory is the liquidation the fall itself triggers. (ii) Theorem S10.2(ii). (iii) ψ is continuous, $\psi(0) \le s < \psi(\delta_1)$, $\psi(\delta_0) < s$, and $\psi(\delta) \ge \delta - \kappa Q \to \infty$, so the intermediate value theorem gives solutions of $\psi(\delta) = s$ in $[0, \delta_1)$, (δ_1, δ_0) and (δ_0, ∞). (iv) As in Proposition S10.3(iv), flow that is a function of public information is netted out at no discount. □

What this means. Anticipation does not remove the threshold; it changes what happens above it. Below the threshold, forward-looking dealers reproduce the round-by-round outcome at once. Above it, the market has several self-consistent prices, and a crash is one of them. Cascades in this model therefore need two things: positions dense enough near their limits, $\kappa\ell > 1$, and dealers who charge for absorbing even predictable liquidation. In the example of Theorem S10.2 with $\kappa \max \ell = 2$, the shock 0.249 has equilibria at falls of 0.253, 0.500 and 0.747 (`research_extension/open_problems/open_checks2.py`). The existence of three equilibria is machine-checked from the intermediate value theorem (`p1_cascade_three_equilibria`, `Paper1OpenResolved.lean`). Combined with Proposition S10.3(iv), a rational market maker also settles the insider's equilibrium with reacting groups. Whatever rules the groups follow, including rules that react to the insider through prices, their flow is a function of public information and is netted out, so the equilibrium is Kyle's (Kyle, 1985).

S10.6 Saturation of every order

Proposition 7.3 gives the first nonlinear order. Let ψ be odd and analytic near zero, $\psi(u) = \sum_{j\ge 0} a_j u^{2j+1}$ for $|u| < \varrho_\psi$, with $a_0 = 1$; for tanh, $a_1 = -\frac{1}{3}$, $a_2 = \frac{2}{15}$, $\varrho_\psi = \pi/2$, and no a_j vanishes. For a population, let $\mu_j = \sum_i c_i \ell_i^{2j+1} \delta_{r_i}$ be its odd leverage moment of order $2j+1$ at each memory, and $M_{2j+1} = \sum_i c_i \ell_i^{2j+1}$ its total. Since memories lie in $[0, 1)$, μ_0 is determined by the kernel.

Proposition S10.6 (Saturation of every order)**.** *In the setting of Proposition 7.3, let P and P' be stable and have $\mu_j(P) = \mu_j(P')$ for every $j < n$.*

(i) There are $A_0, C_n > 0$ with $\|x^P - x^{P'}\|_2 \le C_n |A|^{2n+1}$ for $|A| \le A_0$, so telling P from P' with both errors at most $\alpha < 1/2$ needs $N \ge 4\sigma^2(1-2\alpha)^2/(C_n^2 A^{2(2n+1)})$.

(ii) The first tap differs by $x_1^P - x_1^{P'} = \kappa^{2n+2} a_n \Delta M_{2n+1} A^{2n+1} + O(A^{2n+3})$. If $a_n \Delta M_{2n+1} \ne 0$, the averaged first tap tells them apart once $N \ge C\sigma^2/(a_n^2 \Delta M_{2n+1}^2 A^{2(2n+1)})$, so the order $A^{-2(2n+1)}$ is attained.

(iii) The coefficient of A^{2n+1} in the first tap is $\kappa^{2n+2} a_n M_{2n+1}$. Small probes therefore identify every odd leverage moment, the one of order $2n+1$ at information rate $A^{2(2n+1)}$.

Proof. (ii) and (iii): after the pulse every memory state is κA, so the first tap is $a\kappa A + \kappa \sum_i c_i \psi(\ell_i \kappa A) = a\kappa A + \kappa \sum_j a_j (\kappa A)^{2j+1} M_{2j+1}$; the totals of order below $2n+1$ agree. (i) Stability and $|\psi(u)| \le C|u|$ near zero give $|\ell_i m_{i,t}| \le C|A|\bar{r}^{\,t}$-type geometric bounds, so for $|A| \le A_0$ every state lies where the series converges. Write $\psi = \sum_{j<n} a_j u^{2j+1} + R_n$ with $|R_n(u)| \le K_n |u|^{2n+1}$ there. Since $m_{i,t}$ depends on trader i only through r_i, the order-$(2j+1)$ part of the aggregate flow is $a_j \int m_t(r)^{2j+1} \mu_j(dr)$, the same functional of the price path for P and P' when $j < n$. So the difference $d = x^P - x^{P'}$ solves the stable linear recursion of the kernel driven by two terms. The first is the order $j \ge 1$ terms evaluated on the two paths, bounded by $C' A^2$ times a geometric average of $|d|$. The second is the remainders, of ℓ^2 norm at most $C''|A|^{2n+1}$. For A small the first

term does not break the contraction, and $\|d\|_2 \le C_n|A|^{2n+1}$ follows as in Proposition 7.3. The test bound is Le Cam's, as there. □

This proves the expectation stated after Theorem 7.4: information about the order-n term scales with the probe size to the power $2(2n+1)$. Learning the leverage distribution moment by moment gets steeply harder; probes large enough to reveal high moments are the ones that push traders to their limits. For two populations with one memory, equal kernels and equal cubic measures but fifth moments 1 and 1.94, the response difference grows with fitted slope 5.00 in A, and the first tap's coefficient matches $\kappa^6 a_2 \Delta M_5$ to five digits (`open_checks2.py`). The sup-norm version of the contraction step is the bias bound `p1_deadband_bias` applied with the contraction factor $s + cA^2$ and $E = CA^{2n+1}$; the rest is on paper.

S10.7 A rate for noisy stress identification

Proposition S5.6 gives consistency without a rate. For smooth shapes, the rate is parametric. Call a population *irreducible* if every $c_i \neq 0$, every $\ell_i \neq 0$, and the pairs $(r_i, |\ell_i|)$ are distinct; Theorem 7.4 identifies populations only up to merging traders that violate this.

Proposition S10.7 (Parametric rate for smooth shapes)**.** *Let ψ be odd and analytic near zero with $a_j \neq 0$ for infinitely many j (as for* tanh*), and let θ_0 be irreducible.*

(i) The two-date flow map $F_\theta(u_1, u_2) = \sum_i c_i \psi(\ell_i(r_i u_1 + u_2))$ has an injective differential in θ at θ_0, as a map into functions on any neighbourhood of zero.

(ii) Hence there is a finite design of at most $3n$ probe pairs, as small as desired, at which the reset responses have an injective Jacobian at θ_0.

(iii) Least squares on that design, started from the consistent estimator of Proposition S5.6, satisfies $\|\hat\theta_N - \theta_0\| = O_P(N^{-1/2})$ up to relabelling and the sign flips $(c_i, \ell_i) \mapsto (-c_i, -\ell_i)$, which leave the flow unchanged, and so recovers $\tilde\mu$ at the parametric rate.

Proof. (i) A perturbation $(\delta c_i, \delta\ell_i, \delta r_i)$ changes F by $\sum_i [\alpha_i \psi(\ell_i z_i) + \beta_i z_i \psi'(\ell_i z_i) + \gamma_i u_1 \psi'(\ell_i z_i)]$, with $z_i = r_i u_1 + u_2$, $\alpha_i = \delta c_i$, $\beta_i = c_i \delta\ell_i$ and $\gamma_i = c_i \ell_i \delta r_i$. Suppose this vanishes near zero. Its homogeneous part of degree $2k+1$ is

$$a_k \sum_i \big[(\alpha_i \ell_i + (2k+1)\beta_i)\ell_i^{2k} z_i^{2k+1} + (2k+1)\gamma_i \ell_i^{2k} u_1 z_i^{2k}\big] = 0.$$

Take k with $a_k \neq 0$, set $u_1 = 1$ and $u_2 = y$, and group traders by memory. For distinct memories, the polynomials $(y+r)^{2k+1}$ and $(y+r)^{2k}$ are linearly independent once $2k+2$ exceeds twice the number of memories, by Hermite interpolation as in Theorem 6.5(iii). So for every memory class and every such k, $\sum_{i \in \text{class}} (\alpha_i \ell_i + (2k+1)\beta_i)\ell_i^{2k} = 0$ and $\sum_{i \in \text{class}} \gamma_i \ell_i^{2k} = 0$. Within a class the ℓ_i^2 are distinct. Sequences $k \mapsto (p + qk)\lambda^k$ with distinct λ are linearly independent on any infinite set of k, since the largest λ dominates. So $\alpha_i \ell_i = \beta_i = \gamma_i = 0$, and irreducibility gives $\delta\theta = 0$. (ii) Probes steer the first price to any u_1 and, with the second probe, the second price over an interval around any value. If a perturbation leaves all small two-date responses unchanged, the one-date response gives a vanishing perturbation of $u \mapsto \sum c_i \psi(\ell_i u)$, and the two-date response then gives a vanishing perturbation of F on a neighbourhood of zero. So the Jacobian over all small probe pairs is injective, and finitely many pairs, at most the dimension $3n$, already give full rank. (iii) With a nonsingular Jacobian on a finite design and Gaussian averaged observations, least squares in a neighbourhood of θ_0 is asymptotically linear with error $O_P(N^{-1/2})$ (van der Vaart, 1998). The consistent estimator puts $\hat\theta$ in that neighbourhood with probability tending to one. □

For 50 random irreducible tanh populations of up to three traders, the differential of the two-date map has full rank (`open_checks2.py`). The earlier Monte Carlo run, with errors falling close to $N^{-1/2}$, is consistent with (iii). The constant depends on the design: small probes give rates of order $(NA^{2(2n+1)})^{-1/2}$ for the order-n moments, by Proposition S10.6. The hard limit is treated in Proposition S10.11.

S10.8 Traders who learn

Proposition S8.2 covers learning rules with bounded sensitivity to probing. The general statement is about information, not dynamics. Let each trader's flow at t be any measurable function of the public price history, its own past trades (hence its own profit and loss) and a private randomization independent of news and of other traders. This covers gradient learning, reinforcement learning and Bayesian updating on public data. Call the law of aggregate flow at t given the price history and past aggregate flows the population's *aggregate flow kernel*.

Proposition S10.8 (Learning traders)**.** *(i) Two populations of learning traders with the same aggregate flow kernel produce the same joint law of prices and aggregate flows for every probe strategy, and the same equilibria under any concept defined from payoffs and the public history. Prices identify the aggregate flow kernel and nothing finer.*

(ii) Without private randomization the kernel is a deterministic map, and (i) is Theorem 7.7.

(iii) If the traders are linear with exponential memory and learning moves their memories and weights by at most v per period, Theorem S8.1 applies: composition is tracked with error of order $\Gamma^{1/3}$ below the speed limit v^* and not at all above it, and market-wide questions are tracked at the cube-root rate whatever K. If learning also responds to probing, Proposition S8.2 adds its floor.

Proof. (i) Induction on t. Given the same law of the history up to $t-1$, the price at t is the same function of that history, the probe, the news and aggregate flow, and the conditional law of aggregate flow is the same by assumption. So the law of the history up to t is the same. Payoffs that depend on public history and aggregate flow have the same law. (ii) is the deterministic case. (iii) restates Theorem S8.1 and Proposition S8.2 for this class. □

The deterministic pathwise step is the machine-checked invariance of any causal flow market (`p1_flow_market_invariance`); the randomized induction is on paper. Proposition S10.12 gives the cost of learning composition when the traders are nonlinear.

S10.9 An insider against a strategic group

Proposition S10.3 takes the groups' rules as given. Now let one group choose its rule. The group has exponential memory with rate r and chooses its weight w in $W = [-\bar{w}, \bar{w}]$, where $\bar{w}$ keeps the market stable with a margin: $|a| + \kappa(\bar{w}/(1-r) + \sum_m |k_m|) \le s < 1$, with k the kernel of the other traders. Its flow at t is $f_t = wm_t$, with $m_t = rm_{t-1} + x_{t-1}$, and it trades at the date-t price. Its payoff is its expected trading profit against the value, $\mathbb{E}\sum_t f_t(V - p_t)$. The insider of Proposition S10.3 trades $z_t = \Delta\zeta_t$ with $\Delta = V - p^*$, $\mathbb{E}\Delta^2 = \Sigma$ and a schedule $\zeta \in \mathbb{R}^T$ that she chooses. News has variance σ^2 per period. Both choose at the start, and an equilibrium is a Nash equilibrium in which the group may mix over W.

Write $H(w)$ and $G(w)$ for the lower-triangular Toeplitz matrices of the price response to a trade and to news, and R for the memory filter, $(Rx)_t = m_t$. Taking expectations over Δ and the news, the payoffs are

$$U_I(w,\zeta) = \Sigma\big(\mathbf{1}^\top\zeta - \zeta^\top M_\theta(w)\zeta\big), \qquad U_G(w,\zeta) = w\big(\Sigma\, I(w,\zeta) - \sigma^2 N(w)\big),$$

with $M_\theta(w)$ as in Proposition S10.3(ii), the informed gain $I(w,\zeta) = \mathbf{1}^\top RH\zeta - \zeta^\top H^\top R^\top H\zeta$ (the group buys after her purchases push the price up, before the value is revealed) and the noise cost $N(w) = \operatorname{tr}(RGG^\top)$ (it also buys after news pushes the price up).

Proposition S10.9 (An insider against a strategic group)**.** *(i)* Existence. *An equilibrium exists. In every equilibrium the insider's schedule is pure: against a mixed strategy μ of the group it is $\zeta^* = \frac{1}{2}\big(\mathbb{E}_\mu M_\theta(w)\big)^{-1}\mathbf{1}$.*

(ii) Only the kernel matters. *With any number of strategic groups, prices and the insider's trades depend on the groups' strategies only through the aggregate kernel they produce. Observers of prices and her trades learn the equilibrium kernel and nothing finer.*

(iii) Entry. *Let I_0 and N_0 be the informed gain and the noise cost at $w = 0$ against $\zeta^*(0) = \frac{1}{2}M_\theta(0)^{-1}\mathbf{1}$. If $\Sigma I_0 \neq \sigma^2 N_0$, no equilibrium has the group inactive: a small weight with the sign of $\Sigma I_0 - \sigma^2 N_0$ gives it a positive payoff.*

(iv) Pure equilibrium. *If $U_G(\cdot,\zeta)$ is quasi-concave on W for every ζ with $\|\zeta\| \le \sqrt{T}/(2c)$, $c = \kappa(\frac{1}{1+s} - (1-\theta))$, there is an equilibrium in which the group's weight is pure too.*

Proof. (i) By Proposition S10.3(ii), $M_\theta(w) \succeq cI$ with $c > 0$ for every $w \in W$, so $\mathbb{E}_\mu M_\theta \succeq cI$ for every mixture. Her payoff is strictly concave in ζ, and her unique best response ζ^* has norm at most $\sqrt{T}/(2c)$. Restrict her to that ball B, which is compact. For a fixed horizon the entries of $H(w)$ and $G(w)$ are polynomials in w, so both payoffs are continuous on $W \times B$, and Glicksberg's theorem (Glicksberg, 1952) gives an equilibrium in mixed strategies. Her equilibrium strategy is pure by strict concavity. It is an equilibrium of the unrestricted game because her best response in $\mathbb{R}^T$ lies in B. (ii) For every realization of the groups' weights, Proposition S10.3(i). (iii) $U_G(\cdot,\zeta)$ is differentiable with derivative $\Sigma I_0 - \sigma^2 N_0$ at $w = 0$ when $\zeta = \zeta^*(0)$. If the group were inactive, the insider would play $\zeta^*(0)$, and a small weight of that sign would be a profitable deviation. (iv) Her best response is a continuous function of w, and the group's best responses form a nonempty, convex-valued, upper hemicontinuous correspondence, so Kakutani's theorem applies on $W \times B$. □

What this means. A group that chooses its rule against an insider becomes a momentum trader that rides on her information. The losses that momentum suffers on news limit how far it goes. In an example with $a = 0.3$, $\kappa = 0.5$, $r = 0.6$, $\theta = 1$, $T = 10$ and $\Sigma = \sigma^2 = 1$, the informed gain at entry is 4.49 and the noise cost 3.49, so the group enters. Best-response iteration on a grid of 4001 weights converges to a pure equilibrium at $w = 0.032$, well inside the stability bound 0.515, and $U_G(\cdot,\zeta^*)$ is concave on a grid of W. The group earns 0.0165, and the insider's expected profit falls from 3.673 without the group to 3.531, about four percent (`research_extension/open_problems/open_checks3.py`). The group's choice is still summarized by the kernel, so the hiding results hold in equilibrium. Uniqueness, and existence of a pure equilibrium without the quasi-concavity condition, are not shown.

S10.10 How a cascade unfolds

Theorem S10.2 and Proposition S10.5 characterize where the fall ends. Two questions remain: how fast the rounds get there, and what the path looks like when dealers see it coming but capital is slow to arrive. For the second, follow Duffie (2010). Competitive dealers absorb the liquidation I at date 0 and hold it until outside capital arrives at date T, which takes the inventory at the price $p^* - s - \kappa I$, the mechanical price of Theorem S10.2. Holding inventory costs dealers γ per unit per period, an inventory-risk premium as in Grossman and Miller (1988). Liquidation is triggered by the date-0 price, so $I = L(\delta_0)$, with $\delta_0 = p^* - p_0$ the immediate fall.

Proposition S10.10 (How a cascade unfolds)**.** *(i)* Rate of the rounds. *If $\kappa\ell \le q < 1$ on $[s, \delta^*]$, the rounds satisfy $\delta^* - \delta_n \le q^n(\delta^* - s)$. If ℓ is continuous at δ^* with $q^* = \kappa\ell(\delta^*) < 1$, then $(\delta^* - \delta_{n+1})/(\delta^* - \delta_n) \to q^*$: the rounds converge geometrically at exactly the rate q^*.*

(ii) Critical slowing down. *If $\kappa\ell(\delta^*) = 1$ and ℓ is differentiable at δ^* with $\ell'(\delta^*) > 0$, as at the critical shock $\bar{s}$ of Theorem S10.2(iii) when $\delta^*(\bar{s}) = \delta_1$, then $n(\delta^* - \delta_n) \to 2/(\kappa\ell'(\delta^*))$: the rounds converge only like $1/n$.*

(iii) Slow-moving capital. *With forward-looking dealers and slow capital, the price falls at once by a δ_0 that solves $\delta = s + (\kappa + \gamma T)L(\delta)$, and then recovers by $\gamma L(\delta_0)$ per period to a fall of $s + \kappa L(\delta_0)$ at T. The fall overshoots its long-run level by $\gamma T L(\delta_0)$.*

(iv) A lower threshold. *Theorem S10.2 and Proposition S10.5 hold for the immediate fall with κ replaced by $\kappa + \gamma T$: cascades occur where $(\kappa + \gamma T)\ell > 1$. Patient capital, $\gamma T \to 0$, gives Proposition S10.5.*

Proof. (i) $\delta^* - \delta_{n+1} = \kappa(L(\delta^*) - L(\delta_n)) \le q(\delta^* - \delta_n)$ by the mean value inequality, since the rounds lie in $[s, \delta^*]$ by Theorem S10.2(i); iterate. For the exact rate, the ratio is $\kappa(L(\delta^*) - L(\delta_n))/(\delta^* - \delta_n)$, which tends to $\kappa\ell(\delta^*)$ because $\delta_n \to \delta^*$. (ii) With $e_n = \delta^* - \delta_n > 0$, a second-order expansion gives $e_{n+1} = e_n - \frac{1}{2}\kappa\ell'(\delta^*)e_n^2 + o(e_n^2)$. So $1/e_{n+1} - 1/e_n \to \frac{1}{2}\kappa\ell'(\delta^*)$, and the Cesàro mean gives $1/(ne_n) \to \frac{1}{2}\kappa\ell'(\delta^*)$. (iii) Competitive dealers holding I require the price to rise by γI per period, so $p_t = p_T - \gamma I(T - t)$ with $p_T = p^* - s - \kappa I$. Hence $\delta_0 = s + (\kappa + \gamma T)I$, and rational expectations require $I = L(\delta_0)$. (iv) The fixed-point equation of (iii) is that of Theorem S10.2 with impact $\kappa + \gamma T$. □

What this means. Away from the threshold a cascade settles in a few rounds, at a rate set by $\kappa\ell$ at the final fall. Near the threshold it slows down without bound, so a cascade that keeps going round after round signals a market close to its critical point. The decay rate also reveals $\kappa\ell(\delta^*)$, the local density of positions near their limits that calm prices hide (Theorem 7.1), but only after a breach. Slow capital makes things worse in two ways: the fall overshoots, and the cascade threshold falls. With a bump of positions as in the example of Theorem S10.2, peak density 1.6 and $\kappa = 0.5$, $\kappa \max \ell = 0.8 < 1$, so there is no cascade with mechanical impact. A shock of 0.42 gives rounds that converge at rate 0.606, against $\kappa\ell(\delta^*) = 0.603$, to a fall of 0.575. With $\gamma T = 0.4$, $(\kappa + \gamma T) \max \ell = 1.44 > 1$: the price falls at once by 0.780, overshooting by 0.16, and recovers to 0.62 (`open_checks3.py`). The bound in (i) and the price path of (iii) are machine-checked (`p1_cascade_rounds_geometric`, `p1_slow_capital_fall`, `Paper1OpenResolved2.lean`); the limits in (i) and (ii) are on paper.

S10.11 A rate under hard limits

Proposition S10.7 needs a smooth shape. Under hard limits, $\psi(u) = \max(-1, \min(1, u))$, the response has kinks, but the parametric rate survives if the design avoids them. Along a ray of probes $u = tv$, $t > 0$, in the two-date map of Proposition S10.7, trader i contributes $c_i \ell_i z_i(v)\, t$ until its kink $\tau_i(v) = 1/|\ell_i z_i(v)|$ and $c_i \operatorname{sign}(\ell_i z_i(v))$ after it, with $z_i(v) = r_i v_1 + v_2$.

Proposition S10.11 (A rate under hard limits)**.** *Let θ_0 be irreducible. Take two rays v, v' with $v_1 v_2' \neq v_2 v_1'$, $z_i(v) \neq 0 \neq z_i(v')$ for every i, distinct kinks along each ray, and probe sizes that range past every kink.*

(i) Along each ray the flow is piecewise linear in t, with $n+1$ pieces. On the piece where the set S of traders has saturated, the slope is $\sum_{i \notin S} c_i \ell_i z_i$ and the intercept $\sum_{i \in S} c_i \operatorname{sign}(\ell_i z_i)$.

(ii) Take a design with at least two probe sizes strictly inside each piece on both rays, $4(n+1)$ in all. For θ near θ_0 no design point crosses a kink, the responses are smooth in θ, and their Jacobian at θ_0 is injective.

(iii) Least squares on this design, started from the consistent estimator of Proposition S5.6, has $\|\hat{\theta}_N - \theta_0\| = O_P(N^{-1/2})$, up to relabelling and the sign flips $(c_i, \ell_i) \mapsto (-c_i, -\ell_i)$.

Proof. (i) Direct. (ii) The kinks are continuous in θ near θ_0 because $z_i \neq 0$, so a point strictly inside a piece stays inside it. On a piece the response is affine in t, so two points give its slope and intercept, which are smooth in θ. Consecutive pieces differ by one trader: the slope falls by $c_i \ell_i z_i$ and the intercept rises by $c_i \operatorname{sign}(\ell_i z_i)$. A perturbation that leaves the design's responses unchanged therefore has, for every i, $\delta c_i = 0$ and $c_i \delta(\ell_i r_i) v_1 + c_i \delta \ell_i \, v_2 = 0$, and the same with v'. Since v, v' are independent and $c_i \neq 0$, $\delta(\ell_i r_i) = \delta \ell_i = 0$, and $\ell_i \neq 0$ gives $\delta r_i = 0$. (iii) As in Proposition S10.7(iii). □

The conditions on the rays exclude finitely many directions for a given θ_0. A fixed design works for every θ_0 that leaves two design points inside each piece; otherwise a first stage locates the kinks and a second places the points. For two traders, two rays and 40 probe sizes per ray, the median error of least squares falls with fitted slope -0.49 in N (`open_checks3.py`). Hard limits cost care in where to probe, not in how many times. When kinks coincide on both rays the Jacobian can drop rank, and that case is not covered.

S10.12 The cost of composition when traders are nonlinear

Proposition S10.8 leaves open the cost of learning composition when traders are nonlinear. Let group i trade $c_i \psi(\ell_i m_{i,t})$, with ψ odd, analytic near zero with $\psi'(0) = 1 = \sup|\psi'|$ and bounded derivatives of every order on $\mathbb{R}$, as for tanh, memories $r_i \in [0, \bar{r}]$, $\bar{r} < 1$, and linear weights $w_i = c_i \ell_i$. Probes have size at most $\bar{A}$, records have length T, and the market is stable in the sense of Proposition 7.3.

Proposition S10.12 (Composition when traders are nonlinear)**.** *(i)* Equal leverage: nonlinearity does not help. *Let every group in P and P' have the same leverage ℓ, and let their weights form the hard pair of Theorem 5.1, whose moments agree below order $2K-1$. Then $\|x^P - x^{P'}\|_2 \leq C(\bar{A}, T)(\Delta\eta)^{2K-1}$ for every probe of size at most $\bar{A}$, so telling them apart needs $N \geq c\,\sigma^2 (\Delta\eta)^{-(4K-2)} / C(\bar{A}, T)^2$ experiments, as in the linear case.*

(ii) Distinct leverages: a cost that does not depend on $\Delta\eta$. *If ψ has infinitely many nonzero Taylor coefficients and the $|\ell_i|$ are distinct, the first taps after pulses of $2K$ suitable sizes have nonsingular Fisher information for $(w_i, |\ell_i|)_{i\le K}$, whatever the memories, $\Delta\eta = 0$ included, provided every $w_i \neq 0$. The weights are estimated with error $O_P(N^{-1/2})$, with a constant that depends on the leverages and the pulse sizes but not on the memories.*

(iii) Moving populations. *If the parameters drift by at most v per period, the lower bounds of Theorem S8.1(i) and (iii) hold in case (i), with constants that depend on $\bar{A}$ and T. In case (ii), averaging over a window of $m \asymp (\sigma^2/(\mathcal{I}v^2))^{1/3}$ periods tracks the weights with error $O_P((\sigma^2 v/\mathcal{I})^{1/3})$, where $\mathcal{I}$ is the per-period information of (ii).*

Proof. (i) With a common leverage, aggregate flow is $\int \psi(\ell m_t(r))\,\nu(dr)$ with $\nu = \sum_i c_i \delta_{r_i}$ and $m_t(r) = \sum_{j\ge 0} r^j x_{t-1-j}$. On a common price path, the flows of P and P' differ by $D_t = \int g_t\, d(\nu - \nu')$ with $g_t(r) = \psi(\ell m_t(r))$. The measure $\nu - \nu' = (\nu_w - \nu'_w)/\ell$ annihilates polynomials of degree below $2K-1$ and is supported on an interval of length proportional to $\Delta\eta$, so Taylor's theorem about its centre gives $|D_t| \le C \sup|g_t^{(2K-1)}|(\Delta\eta)^{2K-1}$. On $[0, \bar{r}]$ the derivatives of m_t are bounded by multiples of $\sup|x| \le C\bar{A}$, and those of ψ are bounded, so $\sup_t \sup|g_t^{(2K-1)}|$ is bounded by Faà di Bruno's formula. The two price paths differ by the response of the stable closed loop to κD, which is at most $\kappa \sup|D|/(1-\rho)$, with $\rho = \max(\rho_P, \rho_{P'})$, by the argument of Proposition S10.4(i), since $|\psi'| \le 1$ makes the flows Lipschitz in the path with the constants of the linear loop, and the ℓ^2 norm over T dates is at most $\sqrt{T}$ times that. Le Cam's two-point argument gives the bound on N. (ii) After a pulse of size A every memory state is κA, so the first tap is $a\kappa A + \kappa\sum_i (w_i/\ell_i)\psi(\ell_i \kappa A)$, which does not depend on the memories. Its derivatives in w_i and ℓ_i have Taylor coefficients in $u = \kappa A$ proportional to $a_j \ell_i^{2j}$ and $a_j w_i\, 2j\, \ell_i^{2j-1}$ at u^{2j+1}. A vanishing combination with coefficients α_i, β_i gives $\sum_i (\alpha_i + 2j\beta_i w_i/\ell_i)\ell_i^{2j} = 0$ for infinitely many j. Sequences $(p+qj)\lambda^j$ with distinct $\lambda = \ell_i^2$ are linearly independent on any infinite set of j, so $\alpha_i = \beta_i = 0$. The derivatives are therefore linearly independent functions of A, and $2K$ pulse sizes give a nonsingular Jacobian. Composition is also identified globally: the Taylor coefficients of the first tap give $\sum_i w_i \ell_i^{2j}$ for infinitely many j, which determine the pairs (w_i, ℓ_i^2). So a minimum-distance estimator over a growing set of sizes is consistent, as in Proposition S5.6, and the rate follows as in Proposition S10.7(iii). (iii) The proofs of Theorem S8.1(i) and (iii) use only a per-period bound on the divergence between the members of the hard pair, which (i) supplies. The upper bound balances a drift bias of order vm against a standard deviation of order $\sigma(\mathcal{I}m)^{-1/2}$. □

What this means. Nonlinearity helps a supervisor only if it separates the groups. If they share a leverage, it adds functionals of the same memory measure, and the moment problem, with its $(\Delta\eta)^{-(4K-2)}$ cost, remains. If their leverages differ, saturation tells them apart by how hard they push, not by how long they remember, and the cost no longer depends on how close their memories are, though large probes are still needed to see it. With two tanh groups of equal weight, four pulse sizes from 0.25 to 1.5 and memories $0.5 \pm \Delta\eta/2$, the asymptotic variance of the estimated share grows by factors 76, 67 and 65 per halving of $\Delta\eta$ when both leverages are one, against $2^6 = 64$ from (i), and the information is singular at $\Delta\eta = 0$. With leverages one and two the factors are 16.5, 6.2 and 3.1, and the variance is finite at $\Delta\eta = 0$ (`open_checks3.py`).

S10.13 The largest weights of the hard pair

Proposition S10.13 (Largest weights)**.** *For every $K \ge 2$, the largest weight of the normalized K-point Gauss–Radau rule exceeds the largest weight of the K-point Gauss rule. The difference is $\pi/(4K^2) + O(K^{-3})$.*

Proof. Work with weights summing to 2; normalization halves both. Write $\beta_m = \binom{2m}{m}4^{-m}$, so $P_{2m}(0)^2 = \beta_m^2$ and $P'_{2m+1}(0) = (2m+1)P_{2m}(0)$.

Step 1: central binomial bounds. Since $\beta_{m+1}/\beta_m = (2m+1)/(2m+2)$, the sequence $q_m = \beta_m^2(m+\frac{1}{4})$ increases, with $q_{m+1}/q_m - 1 = 1/(4(m+1)^2(4m+1))$, and $r_m = \beta_m^2(m+\frac{1}{4}+\frac{1}{32m})$ decreases, with $r_{m+1}/r_m - 1 = -(3m+4)/(4(m+1)^3(32m^2+8m+1))$. Both tend to $1/\pi$ by Wallis' formula, so $1/(\pi(m+\frac{1}{4}+\frac{1}{32m})) \le \beta_m^2 \le 1/(\pi(m+\frac{1}{4}))$ for $m \ge 1$.

Step 2: two Sonin functions. With $x = \cos\theta$, $u(\theta) = \sqrt{\sin\theta}\,P_K(\cos\theta)$ solves $u'' + Qu = 0$ with $Q(\theta) = (K+\frac{1}{2})^2 + 1/(4\sin^2\theta)$ (Szegő, 1975), which follows from Legendre's equation. Q is symmetric about $\pi/2$ and nonincreasing on $(0,\pi/2]$. Then $S_1 = u^2 + u'^2/Q$ has $S_1' = -Q'u'^2/Q^2 \ge 0$ there, and $S_2 = Qu^2 + u'^2$ has $S_2' = Q'u^2 \le 0$. Hence on $(0,\pi)$, $S_1 \le S_1(\pi/2) = P_K(0)^2 + P_K'(0)^2/Q_0$ and $S_2 \ge S_2(\pi/2) = Q_0 P_K(0)^2 + P_K'(0)^2$, with $Q_0 = K^2 + K + \frac{1}{2}$.

Step 3: the Gauss weights from above. At a Gauss node $u = 0$ and $u'^2 = \sin^3\theta\,P_K'(x)^2$, so the weight $2/((1-x^2)P_K'(x)^2) = 2\sin\theta/S_2(\theta) \le 2/S_2(\pi/2)$. By Step 1, the largest Gauss weight is at most πG_K, with $G_K = (K+\frac{1}{2}+\frac{1}{8K})/Q_0$ for even K and $G_K = (K-\frac{1}{2}+\frac{1}{8(K-1)})/K^2$ for odd K.

Step 4: a Radau weight from below. The interior Radau nodes are the zeros of $P_{K-1} + P_K$, with weights $(1-x)/(K^2P_{K-1}(x)^2)$ (Abramowitz and Stegun, 1964, 25.4.31). The zeros θ_ν of P_K satisfy $(\nu-\frac{1}{2})\pi/(K+\frac{1}{2}) < \theta_\nu < \nu\pi/(K+\frac{1}{2})$ (Szegő, 1975, Theorem 6.21.2), and P_{K-1} changes sign between consecutive zeros of P_K, so some Radau node has $|x| \le \sin(5\pi/(4K+2)) < d := 5/(K-\frac{1}{2})$. At that node $P_{K-1} = -P_K$, and $(1-x^2)P_K' = K(P_{K-1} - xP_K)$ gives $P_K'(x) = -KP_K(x)/(1-x)$. Substituting into S_1,

$$\frac{1-x}{K^2P_{K-1}(x)^2} = \frac{\sin\theta}{K^2S_1(\theta)}\Big[(1-x) + \frac{K^2(1+x) + Kx + x^2/(4(1+x))}{Q(\theta)}\Big].$$

With $\sin\theta \ge 1-d^2$, $Q_0 \le Q \le \Phi := (K+\frac{1}{2})^2 + 1/(4(1-d^2))$ and $a = K^2/\Phi$, the bracket times $\sin\theta$ is at least $T_K = (1-d^2) + a(1-d^2-d^3) - d\,|1-d^2-a| - dK/Q_0$. By Step 1, $S_1(\pi/2) \le F_K/\pi$, with $F_K = 2/(K+\frac{1}{2})$ for even K and $F_K = 2K^2/((K-\frac{1}{2})Q_0)$ for odd K. So the largest Radau weight is at least $\pi T_K/(K^2F_K)$.

Step 5: comparison. For each parity, $T_K/(K^2F_K) - G_K$ is a rational function of K; the sign of $1-d^2-a$ is constant for $K \ge 62$. Exact computation shows that its numerator and denominator have no real zero above 61.1 and that it is positive at $K = 62$. So the largest Radau weight exceeds the largest Gauss weight for every $K \ge 62$. For $2 \le K \le 61$ the difference of the largest weights, computed in 30-digit arithmetic, is positive, the smallest being 4.2×10^{-4} before normalization; a 60-digit computation confirms it up to $K = 200$.

The asymptotic difference. The interior Gauss–Radau nodes and weights are those of the $(K-1)$-point Gauss–Jacobi rule for the weight $1+x$, divided by $1+x$. Uniformly over nodes in any compact subinterval of $(-1,1)$, Gauss–Jacobi weights for the weight $(1-x)^\alpha(1+x)^\beta$ satisfy $w = \frac{\pi}{n+(\alpha+\beta+1)/2}(1-x)^{\alpha+1/2}(1+x)^{\beta+1/2}(1+O(n^{-2}))$ (Hale and Townsend, 2013). So the interior Radau weights are $\frac{\pi}{K}\sqrt{1-x^2}(1+O(K^{-2}))$, and the Gauss weights are $\frac{\pi}{K+1/2}\sqrt{1-x^2}(1+O(K^{-2}))$. The largest weights sit at the nodes nearest zero, where $\sqrt{1-x^2} = 1 - O(K^{-2})$, and the endpoint weight $2/K^2$ is smaller. Halving, the difference is $\frac{\pi}{2}(\frac{1}{K} - \frac{1}{K+1/2}) + O(K^{-3}) = \frac{\pi}{4K^2} + O(K^{-3})$. □

This explains the constant $0.79 \approx \pi/4$ reported with Theorem 5.1. The scaled gap $K^2\times$(difference) is 0.903, 0.851, 0.831, 0.820, 0.813 and 0.809 at $K = 10, 20, \ldots, 60$, approaching $\pi/4 = 0.785$ (`open_checks2.py`). The explicit bounds of Steps 3 and 4 also hold against the exact weights for every K from 8 to 60, and the Radau node nearest zero is within $0.79/(K-\frac{1}{2})$ of it for $K \le 300$, well inside d (`open_checks3.py`). The two polynomial inequalities behind Step 1 are machine-checked (`p1_centralbinom_q_step`, `p1_centralbinom_r_step`,

`Paper1OpenResolved2.lean`); the rest is on paper and in the exact computation. Proposition S5.4 does not need the comparison: it covers every K through the family R_c.

S11 General memory: the minimax rate is logarithmic

Exponential memory is a modelling choice. Suppose instead that the feedback kernel is completely monotone, $k_m = \int_{[0,\rho]} r^m \, d\mu(r)$ with $\rho < 1$, where μ is any nonnegative measure of total mass W: a finite or infinite mixture of exponential memories with merged weight W. Write $\mathcal{M}(\rho, W)$ for this class, assume the closed loop is uniformly stable, $s \equiv a + \kappa W/(1-\rho) < 1$, and measure accuracy by

$$d(\mu, \mu') = \sup\Big\{ \int f \, d(\mu - \mu') : \ f \text{ 1-Lipschitz on } [0,\rho], \ f(0) = 0 \Big\},$$

which equals the Wasserstein-1 distance between measures of equal mass and also compares estimates whose mass is not exactly W. Let $E_L(|x|)$ denote the error of the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree L, and let $c_* = \inf_{L \geq 1} L\, E_L(|x|)$. Bernstein (1914) showed that $L\, E_L(|x|)$ converges to a positive constant (0.2802; Varga and Carpenter, 1985), so $c_* > 0$; numerically $c_* = 1/4$, attained at $L = 2$ (Table S2).

Lemma S11.1 (Moment-matched group pairs)**.** *For every $n \geq 2$ there are two probability measures on $[0,\rho]$, each with exactly n atoms, whose moments of orders $0, \dots, 2n-2$ coincide and whose Wasserstein-1 distance is at least $\rho\, E_{2n-2}(|x|) \geq c_* \rho/(2n-2)$.*

The atoms are the points where the best approximation of $|r - \rho/2|$ by polynomials of degree $2n-2$ attains its maximal error with alternating signs, and the weights are those of the divided difference on these points, which annihilates every polynomial of degree $2n-2$ (Supplement S1). Each measure is an n-atom merged measure in the class; one atom may sit at rate 0 (a group with $\eta = 1$), which the class allows.

Theorem S11.2 (Minimax rate for general memory)**.** *Let N independent reset experiments observe the impulse response with Gaussian noise of variance σ^2 per tap, and let $B(\rho) = (1+\sqrt{2})(1+2/\rho)$ and $C_h = \kappa^2/(1-s)^2$.*

(i) Lower bound. *Let $\Lambda_N = \log(N/\sigma^2) + \log\big(2C_h^2 W^2/((1-\rho)\rho)^2\big)$. Whenever $\Lambda_N > 4\log(1/\rho)$, every estimator $\hat{\mu}$ satisfies*

$$\sup_{\mu \in \mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu}, \mu) \geq \frac{c_*\, \rho W \log(1/\rho)}{\Lambda_N} \Big) \ \geq \ \frac{1}{4e},$$

and the bound already holds over n-group economies, for a finite n that grows with N.

(ii) Upper bound. *With a and κ known, invert the averaged response into kernel values $\hat{k}_0, \dots, \hat{k}_L$, with $L + 1 = \lfloor \log(N/\sigma^2)/(4 \log B(\rho)) \rfloor$, and let $\hat{\mu}$ be any measure on $[0,\rho]$ minimizing $\max_{j \leq L} |\int r^j d\hat{\mu} - \hat{k}_j|$ (a linear program). For every $\beta \in (0,1)$ there is N_0 depending only on $(\rho, W, a, \kappa, \beta)$ such that for $N/\sigma^2 \geq N_0$,*

$$\sup_{\mu \in \mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu}, \mu) > \frac{3\pi \rho W \log B(\rho)}{\log(N/\sigma^2)} \Big) \ \leq \ \beta.$$

Hence the minimax accuracy is of exact order $1/\log(N/\sigma^2)$, and reaching accuracy w requires and suffices $N/\sigma^2 = \exp(\Theta(1/w))$.

The full proofs of Lemma eflem:altpair and Theorem efthm:general follow.

Machine check. Lean verifies part (i), for N reset experiments of any length, using only Lean's standard axioms. The formal proof replaces the pair of Lemma S11.1 by an elementary one: the binomial weights $W\binom{2L+1}{k}/2^{2L}$ on the rates $\rho k/(2L+1)$, odd k against even k, which have $n = L+1$ atoms each and equal moments of orders $0, \dots, 2n-2$. The 1-Lipschitz function $f(r) = (h/\pi)(1 - \cos(\pi r/h))$, $h = \rho/(2L+1)$, vanishes at 0 and at even nodes and equals $2h/\pi$ at odd nodes, so the pair is at distance at least $2\rho W/(\pi(2n-1))$, more than the $c_*\rho W/(2n-2)$ of the lemma, without Bernstein's theorem. Because the raw moments agree, the kernels agree below lag $2n-1$ and differ by at most $2W\rho^{2n-1}/(1-\rho)$ in ℓ_1 over all lags; the closed-loop ℓ_1 bounds, the Gaussian divergence and the testing lemma then show that, with $C = 2\kappa^2 W/((1-\rho)(1-s)^2)$, $\Lambda = \log(2NC^2/\sigma^2) > 0$ and $L = \lceil \Lambda/(4\log(1/\rho)) \rceil$, no estimator reaches accuracy $2\rho W \log(1/\rho)/(\pi(\Lambda + 6\log(1/\rho)))$ with probability at least $3/4$ under both economies. This is the order of part (i), with success probability $3/4$ in place of $1/(4e)$ and constants of its own. An estimator $\hat{\mu}$ enters through its integrals $\int f\, d\hat{\mu}$ of 1-Lipschitz functions with $f(0) = 0$, which is how $d(\hat{\mu}, \mu)$ is defined. Lean also verifies part (ii). Mathlib has no Jackson theorem, so the formal proof builds one: on $M = n+2$ equally spaced nodes the Fejér–Korovkin kernel $K(u) = |\sum_{k \le n} a_k e^{iku}|^2 / \sum_k a_k^2$, $a_k = \sin(\pi(k+1)/(n+2))$, is nonnegative, has node mean one and node mean of $K(u)(1-\cos u)$ equal to $1 - \cos(\pi/(n+2))$ (from the recurrence of the sines and the orthogonality of roots of unity), so the positive operator it defines maps an L-Lipschitz function on $[-1,1]$ to a polynomial of degree at most n within $L\pi/(n+2)$, by Cauchy–Schwarz on the nodes. Written in Chebyshev polynomials and rescaled to $[0, \rho]$, its coefficients sum in absolute value to at most $\rho(n+1)B(\rho)^n$. With an explicit bound for the deconvolution, $\alpha e(1+\beta')^t$ for kernel value t, and the sub-Gaussian union bound, any measure on $[0, \rho]$ that fits the first $L+1$ deconvolved kernel values at least as well as the truth, the linear-programming estimate among them, is within a finite-sample bound of μ with probability at least $1-\beta$; taking $L = \lfloor \log(N/\sigma^2)/(8\log D) \rfloor$, $D = B(\rho)(1+\beta')$, gives $16\pi\rho W \log D/\log(N/\sigma^2)$ for all N/σ^2 beyond a threshold depending only on $(a, \kappa, \rho, s, W, \beta)$. The formal constants are larger than the theorem's 3π and $B(\rho)$, because the formal deconvolution bound grows geometrically in L; the order is the same.

In words. If trader memory is a general mixture of exponentials rather than a few clean ones, accuracy improves only with the logarithm of the data: to halve the error, the data must be squared. The few-groups model of the previous results is therefore the optimistic case.

Theorem S11.2 replaces a polynomial barrier with a logarithmic one. With exponential memory and K known groups, halving the error multiplies the required number of experiments by a constant; with general memory, halving the error squares it. The lower bound holds already for finite mixtures, so knowing that the memory is a finite mixture of exponentials does not help unless the number of groups is also known (Theorem S12.1). The constants of the two bounds differ, by a factor that grows as $\rho \to 1$, so the dependence on how long memory lasts is not pinned down. The source of the gap is visible in the proof: as $\rho \to 1$ the moment-matched pairs separate their kernels more slowly (ρ^{2n-1} decays slowly), which weakens the lower bound through $\log(1/\rho)$, while the upper bound pays $\log B(\rho)$, which stays bounded away from zero, for the conditioning of polynomial approximation on $[0, \rho]$. Logarithmic rates of this kind are familiar from deconvolution with supersmooth errors (Fan, 1991) and from moment methods for mixing distributions (Kong and Valiant, 2017; Wu and Yang, 2020); here they arise from the closed loop of a market.

Table S2 checks the construction and the estimator. The pairs of Lemma S11.1 have exactly n atoms, agree in all moments below order $2n-1$ to 10^{-61}, and remain at Wasserstein distance about $0.4/n$, while the distance between their kernels shrinks by a factor of about 0.2 per atom at $\rho = 0.9$. The linear-programming estimator applied to a continuous memory distribution has error

Table S2: General memory: moment-matched group pairs and the moment estimator

A. Best approximation of $\lvert x\rvert$						
L	2	4	6	10	20	30
$L\,E_L(\lvert x\rvert)$	0.2500	0.2705	0.2756	0.2785	0.2797	0.2800
B. Pairs of Lemma S11.1						
Atoms per measure n	2	4	6	8	10	12
$W_1(\mu_A,\mu_B)$, $\rho=0.9$	0.2250	0.1037	0.0669	0.0494	0.0391	0.0324
$n\,W_1$, $\rho=0.9$	0.450	0.415	0.402	0.395	0.391	0.388
$\lVert k^A-k^B\rVert_2$, $\rho=0.9$	4.41×10^{-2}	1.03×10^{-3}	4.08×10^{-5}	1.96×10^{-6}	1.05×10^{-7}	5.95×10^{-9}
$\lVert k^A-k^B\rVert_2$, $\rho=0.99$	7.41×10^{-2}	5.84×10^{-3}	1.07×10^{-3}	2.76×10^{-4}	8.33×10^{-5}	2.76×10^{-5}
C. Linear-programming estimator, continuous memory distribution, $\rho=0.9$						
$\log_{10}N$	4	8	12	16	20	24
Moments used $(L+1)$	2	3	5	6	7	8
Error $d(\hat\mu,\mu)$	0.00878	0.00759	0.00432	0.00445	0.00382	0.00399
Error $\times\log(N/\sigma^2)$	0.202	0.245	0.179	0.226	0.229	0.276

A: Remez algorithm; the limit is Bernstein's constant 0.2802. B: moments of orders $0,\dots,2n-2$ agree to 10^{-61} (60-digit arithmetic); every W_1 exceeds $\rho E_{2n-2}(\lvert x\rvert)$. C: true merged measure Beta(2,3) on $[0,0.9]$ with mass 0.1; kernel values observed with noise $\sigma N^{-1/2}$, $\sigma=10^{-3}$; L from the rule in Theorem S11.2(ii); measure fitted on a 401-point grid; 20 replications.

times $\log(N/\sigma^2)$ between 0.18 and 0.28 for N from 10^4 to 10^{24}: a 10^{20}-fold increase in data halves the error. The exponential-memory model of the previous subsections, where the cost is polynomial in $1/\Delta\eta$, is therefore the optimistic case: relaxing the parametric form makes identification from prices much harder, not easier.

Full proofs.

Lemma S11.1. Let $L=2n-2$ and let p^* be the best uniform approximation of $\lvert x\rvert$ on $[-1,1]$ by polynomials of degree L, with error $E=E_L(\lvert x\rvert)$. By Chebyshev's equioscillation theorem (Rivlin, 1981) there are points $t_0<\cdots<t_{L+1}$ at which $\lvert x\rvert-p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i=1/\prod_{j\neq i}(t_i-t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i)=0$ for every polynomial p of degree at most L; multiplying by -1 if necessary, w_i has the sign of $\lvert t_i\rvert-p^*(t_i)$ for every i. Exactly n weights are positive and n negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x\mapsto\rho(x+1)/2$ gives n-atom probability measures π_A (positive part) and π_B (negative part) on $[0,\rho]$ with equal moments of orders $0,\dots,L$. For the 1-Lipschitz function $f(r)=\lvert r-\rho/2\rvert=(\rho/2)\lvert x\rvert$,

$$\int f\,d(\pi_A-\pi_B)=\frac{\rho}{2}\int\big(\lvert x\rvert-p^*(x)\big)\,d(\pi_A-\pi_B)=\frac{\rho}{2}(E+E)=\rho E,$$

because p^* integrates to the same value under both measures, $\lvert x\rvert-p^*=E$ on the support of π_A and $-E$ on that of π_B. By Kantorovich–Rubinstein duality $W_1(\pi_A,\pi_B)\geq\rho E\geq c_*\rho/L$.

Theorem S11.2. Write $K(z)=\sum_m k_m z^m$, $D=1-az-\kappa zK$ and $H=\kappa/D$, which is the generating function of the recursion $h_t=ah_{t-1}+\kappa\sum_m k_m h_{t-1-m}$ with $h_0=\kappa$. Since $k_m\geq0$ and $\sum_m k_m=\int(1-r)^{-1}d\mu\leq W/(1-\rho)$, the ℓ_1 norm of $az+\kappa zK$ is at most $s<1$, so $\lVert H\rVert_1\leq\kappa/(1-s)$ and $\lVert 1/H\rVert_1=\lVert D\rVert_1/\kappa\leq(1+s)/\kappa$.

(i) Let $\mu_A = W\pi_A$ and $\mu_B = W\pi_B$ with π_A, π_B from Lemma S11.1; both lie in $\mathcal{M}(\rho, W)$. With $\nu = \mu_A - \mu_B$, $\sum_m (k_m^A - k_m^B) z^m = \int (1 - rz)^{-1} d\nu(r) = z^{2n-1} \int r^{2n-1}(1 - rz)^{-1} d\nu(r)$, because the first $2n - 1$ terms of the geometric series integrate to zero. On $|z| = 1$ the modulus is at most $2W\rho^{2n-1}/(1 - \rho)$, so Parseval gives $\|k^A - k^B\|_2 \le 2W\rho^{2n-1}/(1 - \rho)$. Next, $H_A - H_B = z(K_A - K_B)H_A H_B$, so by Young's inequality $\|h^A - h^B\|_2 \le \|H_A\|_1 \|H_B\|_1 \|k^A - k^B\|_2 \le C_h \|k^A - k^B\|_2$. The Kullback–Leibler divergence between the N-sample Gaussian experiments is $N\|h^A - h^B\|_2^2/(2\sigma^2) \le (N/\sigma^2)\, 2C_h^2 W^2 \rho^{2(2n-1)}/(1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho) \ge \Lambda_N - 2\log(1/\rho)$. Take n the smallest integer with this property and $n \ge 2$; then $2n-2 < \Lambda_N/(2\log(1/\rho))$ when $\Lambda_N > 4\log(1/\rho)$. Since d is a norm on signed measures and $d(\mu_A, \mu_B) = W\, W_1(\pi_A, \pi_B) \ge W c_* \rho/(2n - 2) > 2c_* \rho W \log(1/\rho)/\Lambda_N$, the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma 2.6 and Theorem 2.2) gives $\max_{j \in \{A,B\}} P_j(d(\hat{\mu}, \mu_j) \ge c_* \rho W \log(1/\rho)/\Lambda_N) \ge \frac{1}{4} e^{-1}$.

(ii) *Approximation.* Let f be 1-Lipschitz on $[0, \rho]$ with $f(0) = 0$, so $|f| \le \rho$. The function $g(x) = f(\rho(x + 1)/2)$ is $(\rho/2)$-Lipschitz on $[-1, 1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial p of degree L with $\sup|g - p| \le (\pi/2)(\rho/2)/(L + 1) \le \rho$, hence $\sup|p| \le 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \le 2\sup|p| \le 4\rho$. The polynomial $q(r) = p(2r/\rho - 1)$ approximates f with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \le \sum_{j \le L} |b_j|\, \|T_j(2r/\rho - 1)\|_{\mathrm{coef}} \le 4\rho \sum_{j \le L} B(\rho)^j \le \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\mathrm{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\mathrm{coef}} \le (1 + \sqrt{2})^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of y^i by at most $(1 + \alpha)^i$, and $B(\rho) - 1 \ge 4$. On the event $\max_{j \le L} |\hat{k}_j - k_j| \le \delta$ the true μ attains objective at most δ, so the minimizer $\hat{\mu}$ satisfies $\max_{j \le L} |\int r^j d\hat{\mu} - k_j| \le 2\delta$ by the triangle inequality. *Moment comparison.* If $\max_{j \le L} |\int r^j d\hat{\mu} - k_j| \le 2\delta$, then $\hat{\mu}$ has mass at most $W + 2\delta$ and

$$\int f\, d(\hat{\mu} - \mu) = \int (f - q)\, d(\hat{\mu} - \mu) + \sum_{j \le L} q_j \Big(\int r^j d\hat{\mu} - k_j \Big) \le \frac{\pi\rho(W + \delta)}{2(L + 1)} + 2\delta\rho B(\rho)^{L+1}.$$

Kernel recovery. From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0, \ldots, k_L$ are the coefficients of $(1 - az - \kappa/H)/(\kappa z)$, which depend on $h_0, \ldots, h_{L+1}$ only. Work in the algebra of power series modulo z^{L+2} with the ℓ_1 norm, which is submultiplicative. If the averaged response is $\hat{H} = H + E$ and $\|1/H\|_1 \|E\|_1 \le \frac{1}{2}$, then $1/\hat{H} = (1/H)(1 + E/H)^{-1}$ and $\|1/\hat{H} - 1/H\|_1 \le 2\|1/H\|_1^2 \|E\|_1$, so $\max_j |\hat{k}_j - k_j| \le \delta \equiv 2((1 + s)/\kappa)^2 (L + 2) \max_{t \le L+1} |E_t|$. Each E_t is Gaussian with variance σ^2/N, so with probability at least $1 - \beta$, $\max_t |E_t| \le \sigma\sqrt{2\log(2(L + 2)/\beta)/N}$. The true μ is feasible for the linear program with value at most δ, so the minimizer satisfies the moment condition above. *Choice of L.* With $L + 1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho)) \rfloor$, $B(\rho)^{L+1} \le (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4} L\sqrt{\log(L/\beta)})$ and $\delta/(L + 1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L + 1)) \le 2\pi\rho W \log B(\rho)/(\log(N/\sigma^2) - 4\log B(\rho))$. For N/σ^2 larger than a threshold depending on $(\rho, W, a, \kappa, \beta)$ the side condition $\|1/H\|_1 \|E\|_1 \le \frac{1}{2}$ holds on the event and the total is below $3\pi\rho W \log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho, W)$. The final statement combines (i) and (ii).

S12 Near-colliding rates: a uniform rate for the merged measure

Theorem 5.2 needs the rates to stay γ-separated and every weight to stay away from zero. Neither restriction can be dropped for weight recovery: two atoms of weights w and $W - w$ at rates r and $r + \Delta$ and two of weights $W - w$ and w at the same rates have merged measures that converge to the same measure as $\Delta \to 0$, so no estimator recovers the individual weights uniformly over

near-colliding economies. What can be recovered uniformly is the merged measure itself, in the permutation-invariant Wasserstein distance.

Theorem S12.1 (Uniform rate without separation)**.** *Fix $K \geq 1$, a memory bound $\bar{r} < 1$ and $0 < m_{\min} \leq B$. Let $\mathcal{M}_K$ be the merged measures with at most K atoms in $[0, \bar{r}]$ and total mass $k_0 \in [m_{\min}, B]$, with no separation condition and no lower bound on individual weights, and write $\bar{M} = M/k_0$ for the normalized memory distribution. From N reset experiments as in Theorem 5.2, let $\hat{M}$ be any element of $\mathcal{M}_K$ whose first $2K$ kernel values are closest in the sup norm to the estimated ones. There is C depending only on $(K, \bar{r}, a, \kappa, B)$ such that, with probability at least $1 - \beta$,*

$$W_1(\hat{\bar{M}}, \bar{M}) \;\leq\; C\Big(\frac{\sigma\sqrt{\log(4K/\beta)}}{m_{\min}\sqrt{N}}\Big)^{1/(2K-1)} \qquad \text{uniformly over } \mathcal{M}_K.$$

Hence $N = O(\sigma^2 m_{\min}^{-2} \epsilon^{-(4K-2)})$ experiments give accuracy ϵ, and with the Wasserstein statement of Theorem 5.1 the minimax rate for the merged memory distribution is $N^{-1/(4K-2)}$, uniformly over near-colliding economies.

In words. Even with no lower limit on how close the groups' memories are or how small a group is, the combined memory distribution can still be estimated, and no faster rate is possible. What is lost when groups nearly coincide is the ability to tell them apart, not the ability to describe the combined picture.

Proof. As in the proof of Theorem 5.2, the kernel values $k_0, \ldots, k_{2K-1}$ are a Lipschitz function of the first $2K+1$ responses on the bounded set of responses the class produces, so a union bound gives $\max_{t<2K} |\bar{k}_t - k_t| \leq e = C_1\sigma\sqrt{\log(4K/\beta)/N}$ with probability at least $1-\beta$ (the Lipschitz constant is for a bounded neighbourhood of the class; when the noise is too large for that, the right-hand side of the theorem exceeds $\bar{r} \geq W_1$ once C is enlarged, so the bound holds trivially). The class is compact and the moment map continuous, so a closest element $\hat{M}$ exists, and $\max_t |k_t(\hat{M}) - k_t(M)| \leq 2e$ by the triangle inequality. Since all atoms lie in $[0,1]$, $|k_t| \leq k_0$, and the normalized moments $m_t = k_t/k_0$ satisfy $|m_t(\hat{M}) - m_t(M)| \leq 2e/k_0 + (|\hat{k}_t|/\hat{k}_0)(2e/k_0) \leq 4e/m_{\min}$. The affine map $r \mapsto 2r/\bar{r} - 1$ carries both normalized measures to distributions on $[-1,1]$ with at most K atoms, multiplies W_1 by $2/\bar{r}$ and the differences of moments of order below $2K$ by at most $(1 + 2/\bar{r})^{2K-1}$. The moment comparison inequality for K-atomic distributions on $[-1,1]$ (Wu and Yang, 2020), $W_1 \leq c_K \delta^{1/(2K-1)}$ when the first $2K-1$ moments differ by at most δ, then gives the bound with $C = c_K(\bar{r}/2)(1 + 2/\bar{r})(4C_1)^{1/(2K-1)}$. □

The theorem separates two questions that Theorem 5.2 answers together. Which memory distribution do the participants have? That has the same polynomial cost whether or not their rates collide. Which participant has which weight? That becomes arbitrarily expensive as rates approach each other (Theorem 5.1) and impossible when they coincide (Theorem 3.3). A supervisor who needs the first can have it at the rate of Theorem 5.1; one who needs the second has to be told the number of groups and a separation, or has to obtain tagged data (Section 6).

S13 Local information: the Cramér–Rao benchmark

Proposition S13.1 (Cramér–Rao benchmark)**.** *Let $\theta \in \Theta \subset \mathbb{R}^{2K}$ collect the weights $w_1, \ldots, w_K$ and rates $\eta_1, \ldots, \eta_K$, with Θ open. Each of N reset probes of size V_{probe} records the taps $h_0, \ldots, h_T$ with independent $N(0, \sigma_p^2)$ errors, so the averaged, rescaled record is $Y = h(\theta) + \sigma_h Z$ with $Z \sim$*

Table S3: Cramér–Rao sample-size benchmark N_{CR} of (23) for multi-group weight recovery ($T = 120$ taps)

Scenario	K	N_{CR} probes	σ_{crit}	ML success ($f = 1$)	RMSE/CRB
$\eta = 0.30$	1	0.1	6.05×10^{-3}	94%	1.02
$\Delta\eta = 0.30$	2	47.8	2.31×10^{-4}	94%	1.01
$\Delta\eta = 0.20$	2	597	6.55×10^{-5}	96%	0.98
$\Delta\eta = 0.10$ ($\delta = 0.05$ pair)	2	40,374	7.96×10^{-6}	93%	1.03
$\Delta\eta = 0.05$	2	2,620,320	9.88×10^{-7}	95%	1.03
spacing 0.10	3	17,349,791	3.84×10^{-7}	98%	1.22
spacing 0.20	3	27,038	9.73×10^{-6}	98%	0.88

Tolerance ± 0.01 at 95%. σ_{crit} is the single-probe noise at which one probe suffices; operational single-probe noise is 1.6×10^{-3}. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at σ_{crit}; success means every weight within ± 0.01. Fitted log–log slope of N_{CR} in $\Delta\eta$: -6.04. N_{CR} is a benchmark for an efficient, approximately normal estimator, not a lower bound for every procedure (Proposition S13.1(ii)).

$N(0, I_{T+1})$ *and* $\sigma_h = \sigma_p/(V_{\mathrm{probe}}\sqrt{N})$. *Let* $J_{tj} = \partial h_t/\partial\theta_j$ *and* $F_{\mathrm{unit}} = J^\top J$, *and assume* $T \geq 2K$ *and that* J *has full column rank at* θ, *so that* F_{unit} *is invertible.*

(i) The Fisher information of Y *is* $I(\theta) = \sigma_h^{-2} F_{\mathrm{unit}}$. *Every estimator* $\hat{w}_k(Y)$ *with finite variance at* θ *that is unbiased for* w_k *on a neighbourhood of* θ *satisfies* $\mathrm{Var}_\theta(\hat{w}_k) \geq \sigma_h^2 [F_{\mathrm{unit}}^{-1}]_{kk}$. *No further regularity is needed: the bound follows from the Hammersley–Chapman–Robbins inequality by letting the alternative approach* θ *along a line, without differentiating under the integral.*

(ii) Benchmark interpretation. *For an estimator that attains this variance and is approximately normal, a 95% interval of half-width* tol *needs*

$$N \geq N_{\mathrm{CR}} \equiv \Big(\frac{1.96}{\mathrm{tol}}\Big)^2 \Big(\frac{\sigma_p}{V_{\mathrm{probe}}}\Big)^2 [F_{\mathrm{unit}}^{-1}]_{kk}. \tag{23}$$

This is a sample-size benchmark, not a lower bound on the interval width of every procedure.

(iii) For $K = 2$, *fix the horizon* $T \geq 4$, *known* a *and* κ *with* $\kappa_{\min} \leq |\kappa| \leq \kappa_{\max}$, *and weights* $0 < w_i \leq w_{\max}$. *For all rates with* $0 \leq \eta_2 < \eta_1 \leq 1$, *writing* $\Delta\eta = \eta_1 - \eta_2$, $c\,(\Delta\eta)^{-6} \leq [F_{\mathrm{unit}}^{-1}]_{kk} \leq C\,(\Delta\eta)^{-6}$ *for both weight coordinates* k, *where* $c, C > 0$ *depend only on* T, $|a|$, $\kappa_{\min}$, $\kappa_{\max}$ *and* $w_{\max}$ *(not on* $\Delta\eta$ *or on where the rates lie). Hence* $N_{\mathrm{CR}} = \Theta((\Delta\eta)^{-6})$. *The upper constant* C *does not depend on* T. *The lower constant* c *does in general. Under a small-gain margin it can be taken free of* T: *if* $1 - \eta_2 \leq \rho < \theta < 1$ *and* $|a| + 2\kappa_{\max} w_{\max}\theta/(\theta - \rho) \leq \theta$, *then the bound holds for every* $T \geq 4$ *with* $c = (\theta - \rho)^8(1 - \theta)^3(3 + \theta)/(256\,\kappa_{\max}^4(1 + \theta)^2)$ *(`p1_prop_S7_1_iii_uniform`, `p1_prop_S7_1_iii_two_uniform`). The margin is sufficient, not sharp. That an estimator attains this order is a separate statement, given by Theorem 5.2; the minimax lower bound of the same order is Theorem S4.1.*

In words. Part (i) bounds the variance of every estimator that is unbiased near the truth; part (ii) turns it into the benchmark number of test trades an efficient estimator needs to pin down one group's trading weight to a stated precision. Each extra group with a nearby memory raises the number sharply, as Table S3 shows.

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table S3), so the critical noise levels, not estimator failure, are what make close groups expensive. Under ordinary surveillance budgets ($N \le 50$) unbiased estimators cannot resolve close weights.

Proof of Proposition S13.1. Part (i): for every alternative θ' the Gaussian likelihood ratio has second moment $\exp(\|h(\theta') - h(\theta)\|^2/\sigma_h^2)$, so the Hammersley–Chapman–Robbins inequality gives $(\mathbb{E}_{\theta'}\hat{w}_k - \mathbb{E}_\theta \hat{w}_k)^2 \le \mathrm{Var}_\theta(\hat{w}_k)\,(e^{\|h(\theta')-h(\theta)\|^2/\sigma_h^2} - 1)$; put $\theta' = \theta + ta$, use local unbiasedness, divide by t^2 and let $t \to 0$ to get $a_k^2 \le \mathrm{Var}_\theta(\hat{w}_k)\|Ja\|^2/\sigma_h^2$, and choose $a = F_{\mathrm{unit}}^{-1} e_k$. Part (iii): for $K = 2$ the columns of J for the two rates become collinear as $\Delta\eta \to 0$; expanding to third order, the smallest singular value of J is of order $(\Delta\eta)^3$, which gives the upper bound $[F^{-1}]_{kk} \le \sigma_{\min}(J)^{-2} = O((\Delta\eta)^{-6})$. The matching lower bound for the weight coordinate does not follow from the singular value alone; it uses the exact inverse row of the 4×4 confluent Vandermonde matrix in moment coordinates together with the bounded conditioning of the map from moments to the observed taps on the fixed-centre family.

S14 Finite-sample surveillance bounds

S14.1 A finite-window price-surveillance bound

Theorem S14.1 (Finite-window abstention)**.** *Observe $x_t = sx_{t-1} + w_t$, $w_t \sim N(0, \sigma^2)$, $x_0 = 0$, $t \le T$, with $s \in (0, 2)$. Let a possibly randomized rule Ψ output Stable, Unstable or Abstain, with $\sup_{s<1} P_s(\Psi = \text{Unstable}) \le \alpha$ and $\sup_{s\ge 1} P_s(\Psi = \text{Stable}) \le \alpha$, $\alpha < 1/2$. Choose $0 < c_0 < 1$ with $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. On $B_T = (1 - c_0/T, 1 + c_0/T)$,*

$$\inf_{s \in B_T} P_s(\Psi = \text{Abstain}) \;\ge\; 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

and for every finite T the minimax worst-case error of binary rules over $s \in (0, 2)$ is at least $1/2$.

Proof. Pair $s_0 \in (1 - c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\mathrm{KL}}(P_{s_0} \| P_{s_1}) = \frac{(s_1 - s_0)^2}{2\sigma^2} \sum_t \mathbb{E} x_{t-1}^2 \le c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \le \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \le \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval B_T is an abstention layer that any error-controlled monitor must respect in the scalar Gaussian case; richer disclosure can remove it.

S14.2 Passive distinction of a known pair in finite samples

Proposition S14.2 (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations $w_t \in [-b, b]$ of variance $s_w^2 = b^2/3$ at each price update, observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma^2)$, use independent three-step reset blocks and no probes, with $P_* > 3b$. Let $b_1 = a + \kappa q\eta$, $b_{2,0} = b_1^2 + \kappa q\eta(1 - \eta)$ and $b_{2,\delta} = b_{2,0} - \kappa q\delta^2$. In each block the covariance of the first and third observations differs by $\Delta = \kappa q s_w^2 \delta^2$. With*

$$R = \max_{r \in \{0,\delta\}} \Big\{ (s_w^2 + \sigma^2)\big[s_w^2(1 + b_1^2 + b_{2,r}^2) + \sigma^2\big] + s_w^4 b_{2,r}^2 \Big\},$$

Table S4: Sample requirement with real Bitcoin price noise: fitted slope of $\log N^*$ in $\log \Delta\eta$

	Theory	Real noise	Gaussian control	Pre-specified criterion	Result
Price only, $K = 2$ (H1)	-6	-6.01	-6.22	slope in $[-7.5, -4.5]$	pass
Tagged, $K = 2$ (H2)	bounded	256–724	256–1,024	max/min ≤ 8	pass (2.8)
Price only, $K = 3$ (H3)	-10	-9.81	-9.96	slope in $[-12.5, -7.5]$	holds

Synthetic groups at rates $r_c + \Delta\eta\, s$ (s the K-point Gauss nodes; $r_c = 0.6$ and $\Delta\eta$ from 0.5 to 0.125 for $K = 2$, $r_c = 0.5$ and $\Delta\eta$ from 0.6 to 0.25 for $K = 3$), $a = 0.5$, $\kappa = 0.8$, total weight 0.08, $\sigma = 10^{-3}$, $\sigma_f = 0.02$. The test passes if H1 and H2 hold; H3 is reported either way. The tagged row gives the range of N^* over the spacing grid. Requirements above 4,096 use the central-limit draw described in the text. Requirements at each spacing are in the full working paper. Test opened once.

the midpoint test of the average product has worst-case error at most α whenever $n \geq 4R/(\alpha\Delta^2)$ blocks are observed: zero controlled quantity and order δ^{-4} blocks.

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for 3.9×10^7 blocks at $\alpha = 0.1$; in simulation the midpoint test still errs 32–38% of the time with 2×10^5 blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

Proof. In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \leq 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2(\mathbb{E}w^4 - 3s_w^4) \leq 0$ for uniform innovations, so it is at most R. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$. □

S15 Do the rates survive real market noise?

Theorems 5.1 and 5.2 are proved for observation errors that are independent and Gaussian, and the full working paper checks them with Gaussian noise. Real price changes are different: one-minute returns have heavy tails, although they are almost uncorrelated from one minute to the next. To see whether the rates depend on the Gaussian assumption, we ran a pre-specified test (opened once) in which the trader groups are synthetic, with known memory spacing, and only the noise is real. Each reset experiment returns the price response to a unit impulse plus a window of consecutive one-minute log-price changes of Binance BTCUSDT, demeaned and scaled to unit variance. The changes come from 21 days, the 6th of each month from January 2025 to September 2026, that are used nowhere else in this project; their kurtosis is 19.4, against 3 for Gaussian noise. With class tagging, the experiment also returns each group's flow response, observed through independent windows of the same noise. A control arm uses i.i.d. Gaussian noise of the same variance. For each spacing, N^* is the smallest number of experiments at which every recovered weight lies within 25% of the smallest true weight in at least 75% of 200 replications, using Prony's method on prices alone or the per-group regression of Proposition 6.2 with tagging. The noise level and the spacing grids were fixed on Gaussian noise alone, before any real noise was downloaded.

Both pre-specified hypotheses hold, so the test passes (Table S4). With real noise, the price-only requirement for two groups rises from 64 experiments at $\Delta\eta = 0.5$ to 370,728 at $\Delta\eta = 0.125$, a factor of about 5,800, and the fitted slope is -6.01; for three groups it is -9.81. Over the same grid the tagged requirement stays between 256 and 724 experiments, and it falls slightly rather

than rises as the spacing shrinks. At every spacing the real-noise requirement is within one step of the grid (a factor $\sqrt{2}$) of the Gaussian control. The slopes are least-squares fits of $\log N^*$ on $\log \Delta\eta$ over the stated grids (seven spacings from 0.5 to 0.125 for two groups, five from 0.6 to 0.25 for three). No uncertainty interval was pre-specified; as a post hoc description, the ordinary least-squares standard errors are 0.17 and 0.47 with real noise (0.12 and 0.31 for the Gaussian control, slopes -6.22 and -9.96), and each N^* is itself read off a grid with steps of $\sqrt{2}$. For two groups, the last four price-only points lie above 4,096 and use the normal draw, so the heavy tails enter the fit only through the first three.

The test has a limit, stated in its specification. Averages of N real windows are resampled exactly up to $N = 4{,}096$; beyond that they are drawn as Gaussian vectors with the real windows' covariance, as the central limit theorem suggests. Real one-minute changes are almost uncorrelated (lag-one autocorrelation between -0.007 and 0.009 within the windows), so beyond 4,096 the real and Gaussian arms differ little by construction. The direct evidence about heavy tails therefore comes from the points in the exact range: every tagged requirement, and the price-only requirements for two groups at the three widest spacings (64, 362 and 1,448 experiments, against 64, 362 and 1,024 with Gaussian noise). The other four price-only points for two groups, and all points for three groups, check the rate with the real noise's variance and covariance but not its tails. Averaging should make heavy tails matter less as N grows, but the test does not show this.

In words. Where real Bitcoin price noise could be used directly, it needs about as many experiments as Gaussian noise, and over the whole grid halving the spacing between two groups still multiplies the number of experiments by about 64, while class-tagged records keep the requirement flat. At the largest sample sizes the test relies on a normal approximation to the real noise.

S16 What prices hide, in economic terms

The worked example states the barrier in signal-to-noise ratios. That is the general picture. Write $s = \sigma_\zeta^2/\sigma_\nu^2$ for a desk's signal-to-noise ratio, so that by (7) its memory rate is $1 - \eta(s)$ with

$$\eta(s) = \tfrac{1}{2}\big(\sqrt{s^2 + 4s} - s\big), \qquad \eta'(s) = \tfrac{1}{2}\Big(\frac{s+2}{\sqrt{s^2+4s}} - 1\Big) > 0,$$

and η is increasing and concave. With no inventory penalty, the first-order condition of the Remark on exponential memory gives the trading intensity $q_i = \beta_i/\lambda_{\mathrm{tc},i}$: what the desk expects to earn per unit of belief, over what it pays to trade.

Proposition S16.1 (Identification in economic primitives)**.** *Consider K Kalman desks (Lemma 2.1(i)) with signal-to-noise ratios $s_i \in [\underline{s}, \bar{s}] \subset (0, \infty)$ and intensities $q_i = \beta_i/\lambda_{\mathrm{tc},i}$, and write $c_- = \eta'(\bar{s})$ and $c_+ = \eta'(\underline{s})$.*

(i) Prices and aggregate order flow depend on the desks only through the measure

$$\sum_i q_i\, \eta(s_i)\, \delta_{1-\eta(s_i)}.$$

Desks are identified through their signal-to-noise ratios and intensities only; expected returns and trading costs enter only through their ratio, and two populations with the same intensity-weighted distribution of signal-to-noise ratios are observationally equivalent.

(ii) Fix a centre s_c and let the ratios lie in $s_c + \Delta s\,[-1,1] \subset [\underline{s}, \bar{s}]$. Recovering the intensity-weighted distribution of signal-to-noise ratios to Wasserstein accuracy of order Δs, or its weights $q_i \eta(s_i)$, from N noisy reset experiments with worst-case error probability below $1/4$ requires $N \geq c\,\sigma^2 (\Delta s)^{-(4K-2)}$; on separated configurations, Prony's method of Theorem 5.2 composed with η^{-1} achieves this order. The constants are those of Theorems 5.1 and 5.2 multiplied by powers of c_- and c_+.

In words. What prices reveal about Kalman-filter desks is the mix of how good their information is (their signal-to-noise ratios), weighted by how hard they trade. Expected returns and trading costs matter only through their ratio. Pinning down that mix carries the same steep cost as separating memory rates.

Proof. (i) By Lemma 2.1(i) desk i is a group with memory rate $1 - \eta(s_i)$ and weight q_i, so Theorem 3.3 applies, and $\beta_i, \lambda_{\mathrm{tc},i}$ enter the economy only through q_i. (ii) On $[\underline{s}, \bar{s}]$, η is a C^1 bijection onto its image with $c_- \leq \eta' \leq c_+$. Lower bound: take the two economies of Theorem 5.1 with centre $r_c = 1 - \eta(s_c)$ and rate spacing $\Delta\eta = c_- \Delta s$. By the mean value theorem their gains are images of ratios in $s_c + \Delta s\,[-1,1]$, so they are two populations of Kalman desks in the class, and the bound $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)} = c\,c_-^{-(4K-2)} \sigma^2 (\Delta s)^{-(4K-2)}$ applies to recovering their weights. Because η is c_+-Lipschitz, the Wasserstein distance between the two distributions of ratios is at least $1/c_+$ times that between the two distributions of rates, which Theorem 5.1 bounds below by a constant times $\Delta\eta = c_- \Delta s$. Upper bound: ratios in $s_c + \Delta s\,[-1,1]$ with spacing at least $\gamma \Delta s$ have gains in $\eta(s_c) + c_+ \Delta s\,[-1,1]$ with spacing at least $c_- \gamma \Delta s$, a class of the form required by Theorem 5.2 with rescaled separation $\gamma c_- / c_+$; Prony's method recovers the rates and weights at that theorem's rate, and η^{-1}, which is $1/c_-$-Lipschitz, maps rate errors to ratio errors. □

The private information a supervisor would most like to know, how good each desk's information is, is exactly what prices hide. In the worked example the two desks' ratios differ by a factor of almost two, and separating them already costs hundreds of thousands of dollars of probing capital. The proposition takes intensities to vary independently of information quality, because they also reflect costs and capital. The next proposition asks what happens when the economics ties them together.

Proposition S16.2 (Intensity tied to information quality)**.** *Let each desk's intensity be a known function of its signal-to-noise ratio, $q = \varphi(s)$ with φ positive and C^2 on $[\underline{s}, \bar{s}]$ (for a mean–variance desk, $\varphi(s)$ increases with s), and let $n_i \geq 0$ denote the mass of desks with ratio s_i (a divisible population, not an integer desk count). Write $s(\eta)$ for the inverse gain map and $w(\eta) = \varphi(s(\eta))\,\eta$ for the weight per desk as a function of its gain.*

(i) Unknown numbers of desks. *The merged measure is $\sum_i n_i\, w(\eta(s_i))\, \delta_{1-\eta(s_i)}$. Every positive measure with finitely many atoms in $[1 - \eta(\bar{s}), 1 - \eta(\underline{s})]$ is of this form, with $n_i = m_i / w(\eta_i)$ for an atom of mass m_i at $1 - \eta_i$. So when the numbers of desks of each type are not known, tying intensity to information quality removes no economy, and every lower bound of Proposition S16.1(ii) holds unchanged.*

(ii) Known numbers of desks. *If the n_i are also known, the set of economies is smaller, but the single-versus-pair problem keeps its order: one type at gain $\bar{\eta}$ with $2n$ desks and two types at $\bar{\eta} \pm \delta$ with n desks each have taps that differ by $n\,\delta^2\,\partial_\eta^2 [w(\eta)(1-\eta)^m]_{\bar{\eta}} + o(\delta^2)$, so separating them needs $N \geq c\,\sigma^2 \delta^{-4}$ reset experiments, the order the symmetric pair has when intensities vary freely.*

(iii) Self-confirming configurations. *Suppose the ratio each desk perceives is the one implied by the closed-loop dynamics it trades in, through a map T from configurations $(s_1, \dots, s_K)$ to perceived ratios that is continuous on a box $\prod_i [\underline{s}, \bar{s}]$ and maps it into itself. Then a configuration with $T(s) = s$ exists, and parts (i)–(ii) and Proposition S16.1 apply at it.*

Proof. (i) Desk type i is a group with gain $\eta_i = \eta(s_i)$, memory rate $1 - \eta_i$ and total weight $n_i \varphi(s_i)$, so it contributes $n_i \varphi(s_i)\eta_i = n_i w(\eta_i)$ to the atom at $1 - \eta_i$. Since $w > 0$, any atom mass $m_i \geq 0$ is attained with $n_i = m_i / w(\eta_i)$. (ii) Write $g_m(\eta) = w(\eta)(1-\eta)^m$; the tap difference is $n[g_m(\bar{\eta} + \delta) + g_m(\bar{\eta} - \delta) - 2g_m(\bar{\eta})] = n\delta^2 g_m''(\bar{\eta}) + o(\delta^2)$, with $\sum_m \sup |g_m''|^2 < \infty$ on gains bounded away from zero, so the Euclidean distance between the two tap vectors is $O(n\delta^2)$ and the Kullback–Leibler divergence over N experiments is $O(Nn^2\delta^4/\sigma^2)$; Le Cam's two-point argument gives the bound. (iii) is Brouwer's fixed-point theorem. □

So the barrier does not depend on intensities varying freely: it survives any known link between intensity and information quality as long as the supervisor does not also know how many desks of each kind there are, and even then the simplest separation problem keeps its δ^{-4} order. Whether the full $(\Delta\eta)^{-(4K-2)}$ exponent survives with known numbers of desks is open. Part (iii) shows only that self-confirming configurations exist under continuity; it does not derive the map T from a model of the desks' learning.

S17 Natural excitation and the cost of ambiguity

Natural price innovations can also separate the pair with zero controlled quantity: an exact finite-sample test on independent three-step blocks needs order δ^{-4} blocks (Online Supplement, Proposition S14.2), the same order as the passive Whittle bound of Theorem 5.4. Non-identification also has a price on the quote side: when the relevant set of economies includes costlier ones, a robust dealer widens its quote, which the companion paper analyses (Niroula et al., 2026); group non-identification alone does not establish that the worst relevant cost exceeds the informed one.

S18 A worked example: two Kalman desks

To make the orders concrete, consider two desks that track the same trend with Kalman filters (Lemma 2.1(i)) and differ only in their signal-to-noise ratios: $\sigma_\zeta^2/\sigma_\nu^2 = 0.09$ and 0.16. Their gains are $\eta_1 = 0.2584$ and $\eta_2 = 0.3279$, so $\bar{\eta} = 0.2931$ and $\delta = 0.0348$. With equal weights, the economy is observationally the split pair of Theorem S3.1 around $\bar{\eta}$, and the response difference first appears at lag two, $d_2 = -\kappa^2 q \delta^2 = -1.55 \times 10^{-4}$ (verified).

A supervisor who wants to tell this market from a single desk with gain $\bar{\eta}$ at 5% error using a single pulse and the third observation needs $A = 2\sigma\Phi^{-1}(0.95)/(\kappa^2 q \delta^2) = 212$ units of executed quantity, and, under a 0.5% margin and 4 basis points of fees, single-pulse capital of 18,166 par units. The pulse is very inefficient: the account-level spectral lower bound of the full working paper, evaluated at this pair ($\bar{\eta} = 0.2931$, $\delta = 0.0348$), gives about 77 par units for the optimal flat probe (rescaling the $\bar{\eta} = 0.4$ constant by δ^{-4} alone would give 262, because the constant also depends on $\bar{\eta}$). Converting with the measured Roll noise (factor $(0.40/0.01)^2$), the optimal probe costs roughly 1.2×10^5 USDT and the naive pulse about 3×10^7 USDT. Two desks whose signal-to-noise ratios differ by almost a factor of two are therefore expensive, but not impossible, to tell apart by probing; three or more desks are much harder (Theorem 5.1).

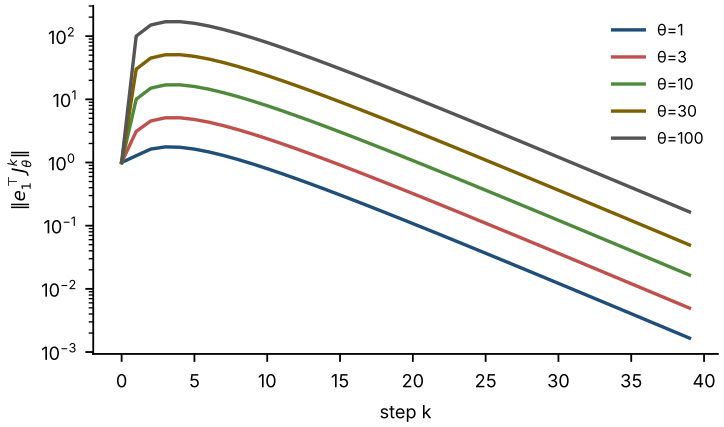


Figure S2: Price-row transient gain of the spectral-blindness family J_θ: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every θ.

S19 A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

Theorem S19.1 (Spectral blindness)**.** *For any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance $s_1 > 0$ and stationary price variance V large enough, there is a family of two-state markets $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ with shock covariance $\mathrm{diag}(s_1, s_2(\theta))$ such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain $\sup_k \|e_1^\top J_\theta^k\|$ grows without bound in θ. Here s_1 is the variance of the fresh price shock given the full prior state; a forecaster who sees prices only can face additional uncertainty about the hidden state, and the family is the one with positive coupling $\theta > 0$.*

In words. Two markets can look identical by the usual risk measures, the same eigenvalues and the same price variance, and yet one of them can produce far larger losses in the short run. Monitoring eigenvalues or volatility alone therefore cannot certify that a market is safe. The gain is measured in the state coordinates of J_θ; rescaling a hidden coordinate changes its numerical value while leaving the price dynamics unchanged, so translating it into market risk needs a specified shock set, units and loss.

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure S2).

Two clarifications prevent over-reading. First, the group economy of Section 2 is itself non-normal, yet its price output does not amplify (Lemma S1.1). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (full working paper). Second, in a network of M identical group economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma S1.1; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator. A stable normal contraction cannot exhibit Euclidean state-norm transient growth. This does not exclude scalar input-output overshoot relative to initial

impact, which also depends on the input and output maps. The positive-group no-overshoot result is model-specific; the companion paper studies a depth-depletion mechanism.

Proof of Theorem S19.1. With J_θ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and P_{11} is affine in s_2 with positive slope, so for each θ a unique $s_2(\theta) \geq 0$ gives $P_{11} = V$ when V exceeds the value at $s_2 = 0$. The one-step price variance is s_1 throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \geq |\theta|$.

S20 Pricing, evasion and the choice of an audit trail

Proposition S20.1 (Conditional endogenous-pricing invariance)**.** *Let $\kappa = \mathcal{K}(\Gamma_M(\kappa))$ on a common admissible interval, with identical innovation laws, initial-state conventions and other primitives across economies. Here $\Gamma_M(\kappa)$ is the specified covariance record at a fixed impact and $\mathcal{K}$ is a maintained dealer rule, not a derived general equilibrium.*

(i) If equal merged measures imply the same map Γ_M and the fixed point is unique (or a common selection rule depends only on that map), equal merged measures give equal selected impacts.

(ii) Suppose $\mathcal{K}$ is L_K-Lipschitz, each covariance map is L_Γ-Lipschitz in impact, $L_K L_\Gamma < 1$, and $\sup_\kappa \|\Gamma_A(\kappa) - \Gamma_B(\kappa)\| \leq \epsilon$. For existing fixed points,

$$|\kappa_A - \kappa_B| \leq \frac{L_K \epsilon}{1 - L_K L_\Gamma}. \tag{24}$$

Thus a covariance discrepancy of order $(\Delta\eta)^{2K-1}$ preserves that order only under these uniform conditions.

Limits of the pricing extension. Anonymous flow alone implies neither a covariance-only dealer objective nor uniqueness or continuity of its equilibrium. Multiple equilibria, threshold quote changes, private information and inventory-dependent rules are outside this proposition; for the question of invariance, Theorem 3.1 below covers them. Preserving a price-record testing bound additionally requires a uniformly Lipschitz observation law with the same nondegenerate noise; adding anonymous flow is a different statistical experiment and requires its own divergence bound. We therefore make no unconditional claim that endogenous pricing or additional aggregate-flow observations preserve the price-only lower bound.

Proof of Proposition S20.1. Equal maps have the same fixed-point set; uniqueness or the declared common selection yields (i). For (ii), put $d = |\kappa_A - \kappa_B|$. The fixed-point equations and the two Lipschitz bounds give $d \leq L_K(L_\Gamma d + \epsilon)$. Since $1 - L_K L_\Gamma > 0$, rearrangement proves the displayed bound. The scalar inequality is machine checked in `p1_endogenous_contraction_sensitivity`; the economic derivation and uniformity of the covariance map are assumptions, not consequences of that algebra. A discontinuous threshold pricing rule need not preserve a small discrepancy. □

Proposition S20.2 (Conditional limits of three evasion arguments)**.** *Maintain the exogenous reporting-noise and valid-instrument assumptions of Proposition 6.2; fix the finite number of linked class labels and a uniformly stable model family.*

(i) For a fixed split $u_{i,m} = \alpha_{i,m} u_i$ with known common ownership, independent mean-zero reporting errors and $\alpha_{i,m} \geq \underline{\alpha} > 0$, only the lagged-flow column of the population IV matrix scales.

Thus $\Sigma_{i,m} = \Sigma_i \operatorname{diag}(\alpha_{i,m}, 1)$ *and* $\det \Sigma_{i,m} = \alpha_{i,m} \det \Sigma_i$. *Subject to the remaining uniform moment and excitation conditions, fixed finitely many consistent sub-account estimates can be summed: the true identity is* $w_i = \sum_m w_{i,m}$, *and the estimated sum has sampling error. The share restriction implies* $M_i \underline{\alpha} \le 1$. *This is not robustness to arbitrarily many vanishing accounts; a common estimated rate does not identify ownership.*

(ii) Assume separately an execution cost lower bound $\mathbb{E}C_\zeta \ge c_\zeta \sigma_\zeta^2$ *with* $c_\zeta > 0$ *and a gross profit upper bound* $\mathbb{E}\Pi \le \overline{\Pi}$ *uniform in the spacing. Camouflage is unprofitable when* $c_\zeta \sigma_\zeta^2 > \overline{\Pi}$. *A variance requirement of order* $(\Delta\eta)^{-(4K-2)}$ *requires an additional additive exogenous-noise estimation model with uniformly controlled instruments. Neither this variance requirement nor the execution bound is established for arbitrary strategic orders.*

(iii) If intensity is scaled to $\gamma_i w_i$ *while* $1 - r_i \ge \eta_{\min} > 0$, *then* $q_i \le \gamma_i w_i / \eta_{\min} \to 0$ *as* $\gamma_i \to 0$. *The other groups remain in the aggregate budget; reducing one contribution does not itself establish that the ceiling is met.*

Economic scope. These are conditional calculations about fixed linked accounts, bounded execution costs and a persistence class. They do not solve an equilibrium game of evasion. In particular, endogenous orders can affect prices and invalidate the reporting-error instrument, and a persistence change can offset a reduction in weight. Adversarial account creation and response adaptation require further modelling.

Proof of Proposition S20.2. (i) The current-price regressor column is unchanged, whereas the lagged true flow column scales by $\alpha_{i,m}$; valid independent reporting noise has zero population covariance with the instruments. Multiplicativity of determinants gives the stated factor. Summing shares at least $\underline{\alpha}$ gives $M_i \underline{\alpha} \le 1$, checked in `p1_sybil_share_count_bound`. For a fixed finite collection the summed estimation errors remain of order $T^{-1/2}$ under the maintained joint central limit assumptions; this asymptotic step is not newly formalized here. (ii) Subtract the cost lower bound from the uniform gross-profit upper bound. Once the former exceeds the latter, expected net profit is negative, as checked in `p1_camouflage_unprofitable`. The cost coercivity and variance-to-difficulty link are explicit hypotheses, not consequences of fairness alone. (iii) Divide the scaled weight by $1 - r_i \ge \eta_{\min}$; without that bound, taking $1 - r_i = \gamma_i$ can keep $q_i = w_i$ constant. □

Consider M markets, a supervisory cycle of $\bar{T}$ periods, and in each market one algorithm whose conduct does harm $\ell > 0$ per period until it is attributed. With prices alone, attribution at the required error level takes $T_p(\Delta\eta) = c_p(\Delta\eta)^{-(4K-2)}$ periods; with class-tagged records it takes $T_t(\Delta\eta) = c_t(\Delta\eta)^{-2}$ periods, the orders of Theorems 5.4 and 6.3 with constants $c_p, c_t > 0$. We take the attribution times to be of exactly these orders, which is a modelling assumption: Theorem 5.4 gives only a lower bound for prices, and if price-only attribution is slower, G below is larger and disclosure is chosen at least on the band the proposition describes. $T_t \propto (\Delta\eta)^{-2}$ is the rate target; for weights at fixed accuracy T_t does not grow as $\Delta\eta$ shrinks, and the lower edge of the band disappears. Attribution that would take longer than the cycle does not happen within it. An audit trail costs F per cycle to operate and c per market to comply with, including any information leakage. The supervisor's expected loss is therefore $\mathcal{L}_{\text{price}} = M\ell \min\{\bar{T}, T_p\}$ without disclosure and $\mathcal{L}_{\text{disc}} = F + Mc + M\ell \min\{\bar{T}, T_t\}$ with it. Write $\Delta\eta_t = (c_t/\bar{T})^{1/2}$ and $\Delta\eta_p = (c_p/\bar{T})^{1/(4K-2)}$ for the spacings at which tagged and price-only attribution just fit within the cycle.

Proposition S20.3 (When an audit trail is chosen)**.** *Let* $K \ge 2$ *and* $\Delta\eta_t < \Delta\eta_p$, *and let* $G(\Delta\eta) = \min\{\bar{T}, T_p(\Delta\eta)\} - \min\{\bar{T}, T_t(\Delta\eta)\}$ *be the attribution delay that disclosure saves.*

(i) The supervisor strictly prefers disclosure if and only if $F/M + c < \ell\, G(\Delta\eta)$.

(ii) G *vanishes for* $\Delta\eta \le \Delta\eta_t$*, increases on* $(\Delta\eta_t, \Delta\eta_p]$ *to its maximum* $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$*, decreases wherever it is positive beyond* $\Delta\eta_p$*, and is negative for* $\Delta\eta > (c_p/c_t)^{1/(4K-4)}$*, where it returns toward zero from below. Hence the spacings at which disclosure is chosen form an interval around* $\Delta\eta_p$*, possibly empty: disclosure is never chosen for algorithms so similar that even tagged records cannot attribute them within the cycle, nor for algorithms so different that prices attribute them quickly.*

(iii) The interval is non-empty if and only if the audit trail covers more than $M^* = F/(\ell G^* - c)$ *markets (with* $M^* = \infty$ *if* $\ell G^* \le c$*), and it widens as* M *grows.*

Part (i) restates the comparison of losses. Part (ii) says that disclosure is valuable in a middle band, where algorithms are close enough that prices cannot attribute them within the cycle but far enough apart that tagged records can; the polynomial gap between the two delays, $(\Delta\eta)^{-(4K-2)}$ against $(\Delta\eta)^{-2}$, is what makes the band wide. Part (iii) is an economies-of-scale result: an audit trail with a fixed operating cost pays only when it spans enough markets. This illustrates a possible economy of scale for market-wide reporting; the cost function is stylised, not estimated from CAT, and the result neither explains CAT's legal architecture nor shows that it is optimal. A supervisor that can trade has a second option, active probing, whose cost is the frontier computed in the full working paper; because that cost grows at the same polynomial order, it shortens price-only attribution only for well-separated algorithms and leaves the comparison for close ones unchanged. The model takes the harm ℓ as given and holds the algorithms' behaviour fixed across regimes; neither is modelled here.

Proof of Proposition S20.3. (i) $\mathcal{L}_{\text{disc}} < \mathcal{L}_{\text{price}}$ is $F + Mc < M\ell G$. (ii) T_p and T_t are decreasing in $\Delta\eta$, with $T_t \ge \bar{T}$ exactly when $\Delta\eta \le \Delta\eta_t$ and $T_p \ge \bar{T}$ exactly when $\Delta\eta \le \Delta\eta_p$. On $\Delta\eta \le \Delta\eta_t$ both minima equal $\bar{T}$ and $G = 0$. On $(\Delta\eta_t, \Delta\eta_p]$, $G = \bar{T} - c_t(\Delta\eta)^{-2}$, which increases to $G^* = \bar{T} - T_t(\Delta\eta_p) > 0$. On $\Delta\eta > \Delta\eta_p$, $G = c_p(\Delta\eta)^{-(4K-2)} - c_t(\Delta\eta)^{-2} = (\Delta\eta)^{-2}\big[c_p(\Delta\eta)^{-(4K-4)} - c_t\big]$; the bracket is positive exactly when $\Delta\eta < \Delta\eta_0 = (c_p/c_t)^{1/(4K-4)}$, and $G'(\Delta\eta) = (\Delta\eta)^{-3}\big[2c_t - (4K-2)c_p(\Delta\eta)^{-(4K-4)}\big] < 0$ whenever $c_p(\Delta\eta)^{-(4K-4)} > c_t$, because $4K - 2 > 2$. So G decreases wherever it is positive beyond $\Delta\eta_p$, and is non-positive beyond $\Delta\eta_0$. Every superlevel set of G at a positive level is therefore an interval (the negative tail, where G rises back toward zero from below, never reaches a positive level), and the level $(F/M + c)/\ell > 0$ lies above $G = 0$, so the set where (i) holds is an interval contained in $(\Delta\eta_t, \Delta\eta_0)$ that contains $\Delta\eta_p$ whenever it is non-empty. (iii) The interval is non-empty if and only if the level is below the maximum, $F/M + c < \ell G^*$, that is $M > M^*$; the level decreases in M, so the superlevel set grows. □

S21 Robustness to nonlinear and adaptive trading rules

The theorems of Sections 3–5 use the linear exponential-memory recursion (2). Real desks cap positions, trade only beyond thresholds, adapt their gains and learn online. A small probe of size A moves the state by $O(A)$, so any rule whose order is a twice-differentiable function of an exponential-memory state is linear to first order and differs from it by $O(A^2)$; the theorems should then hold for small probes and fail for large ones. We check this, and locate where it fails, in simulations of five stylized rules at $a = 0.5$, $\kappa = 0.8$, total weight $q = 0.2$ and mean rate $\bar{\eta} = 0.4$, over 240 periods after a pulse probe.

The rules are: (1) *momentum with a position limit*, the group's order a smooth saturating (tanh) function of its memory state $y_{i,t}$, normalized to unit slope at the operating point; (2) a *threshold rule* that trades only when $|y_{i,t}| > \tau = 0.05$, in two versions, an exact dead zone and one averaged over background state variation of standard deviation 0.04; (3) an *adaptive-gain rule* in the form of recursive least squares with forgetting factor $1 - \eta_i$, whose gain falls when the regressor is large; (4) an *online neural readout*, a linear readout of a fixed two-layer tanh feature map with eight hidden units, trained by regularized stochastic gradient steps of size η_i; and (5) a *tabular value rule* that updates a 21-bin table at rate η_i, in two versions, kernel-smoothed across bins and a single-bin lookup. These are stylized: each is built so that its small-signal dynamics are an exponential filter with rate η_i, and they are not calibrated to any production system.

Two checks follow the theory. *Test A (merged measure):* one group with weight q and position limit 1 against two groups with weights $q/2$ at the same rate and limits 0.55 and 1.65, so the two economies have the same merged measure; the raw response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2$ should be $O(A^2)$ (slope 2 in $\log A$) and the probe-normalised gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ should be $O(A)$ (slope 1 in $\log A$) over $A \in [0.002, 0.064]$. *Test B (δ^{-4} law):* one group at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$, $\delta \in [0.015, 0.12]$, at probe size $A = 0.005$; the normalized response difference $\|\Delta h\|_2$ should scale as δ^2 and the account-level capital lower bound of the full working paper as δ^{-4}. The saturation threshold A_{sat} is the smallest probe at which $\|\Delta h\|_2$ at $\delta = 0.05$ departs by more than 10% from its small-probe value.

To check that these conclusions do not depend on hand-built rules, we pre-specified a second experiment in which online learners are trained from scratch inside the closed-loop market for $T_{\text{burn}} = 5{,}000$ periods of flow and noise innovations; their learned parameters are saved and reloaded into the probing simulator without rescaling. Four learning rules are used: (i) recursive least squares with forgetting factor $\lambda = 0.995$, learning its return-forecast slope; (ii) a two-layer tanh network with eight hidden units whose readout is trained by regularized online stochastic gradient descent on trading profit; (iii) a scalar policy-gradient rule trained on trading profit; and (iv) a tabular Q-learner with 21 states and 21 actions, TD(0) Bellman backups ($\gamma = 0.90$) and Boltzmann action choice. Each learner acts on an exponential-memory state $y_{i,t}$ with rate η_i: training sets its response gain and curvature, not its memory rate, so the experiment asks whether learned policies preserve the merged-measure and δ^{-4} predictions, not whether learning produces exponential memory. The specification fixed four predictions: (a) a single exponential fits each trained learner's small-probe impulse response with $R^2 \geq 0.90$; (b) two trained populations with the same merged measure differ by at most 1% in relative L_2 response at $A = 0.005$; (c) the capital lower bound to separate trained populations whose memory rates differ by 2δ scales with log–log slope in $[-4.15, -3.85]$; and (d) we report where equivalence and the δ^{-4} law fail when learning continues during the probe.

Table S5 and Figure S3 show both where the linear reduction holds and where it breaks down. In Panel A, for small probes every smooth or dithered rule behaves like the linear model: economies with the same merged measure have raw response gap $O(A^2)$ (slopes 2.011 to 2.064) and probe-normalised gap $O(A)$ (slopes 1.011 to 1.064), while the response difference and capital lower bound scale as δ^2 (slope 2.007) and δ^{-4} (slope -4.001), with capital at $\delta = 0.05$ within 0.1% of the linear 61.42. In Panel B, predictions (a)–(c) pass for all four trained learners: a single exponential explains $R^2 \geq 0.993$ of each small-probe impulse response, populations with the same merged measure differ by at most 10^{-5} in relative response at $A = 0.005$, and capital lower bound scales with slopes -4.014 to -3.973, inside the registered band. The level of capital differs across learners because training sets the response gain.

Panel C shows where this changes. When the two populations are trained on independent innovation streams, prediction (c) fails: the capital slopes are -1.75, -0.97, -1.62 and -0.38.

Table S5: Merged-measure equivalence and the δ^{-4} law under stylized nonlinear rules and closed-loop trained online learners

Rule	Test A raw ($O(A^2)$)	Test A norm. ($O(A)$)	Test B slope ($\|\Delta h\|_2$)	Test B slope (capital)	Capital at $\delta = 0.05$	A_{sat} / R^2_{exp}
Panel A. Stylized nonlinear and adaptive rules (fixed operating point)						
Linear benchmark	—	—	2.007	−4.001	61.42	—
1. Momentum (tanh limit)	2.015	1.015	2.007	−4.001	61.47	0.834
2. Threshold rule, averaged	2.043	1.043	2.007	−4.001	61.42	0.049
3. Adaptive gain (RLS)	2.064	1.064	2.007	−4.001	61.46	0.751
4. Online neural readout	2.011	1.011	2.007	−4.001	61.48	0.751
5. Tabular value, smoothed	2.039	1.039	2.007	−4.001	61.44	1.271
Panel B. Learners trained in the closed loop, saved and reloaded (frozen during the probe; shared training stream)						
6. Recursive least squares	3.000	2.000	2.008	−4.014	54.58	$R^2 = 0.993$
7. Two-layer network (SGD)	1.878	0.878	2.007	−4.012	65.19	$R^2 = 0.994$
8. Policy gradient on profit	3.000	2.000	2.004	−4.006	58.05	$R^2 = 0.995$
9. Tabular Q-learner (TD(0))	2.108	1.108	1.994	−3.973	186,821	$R^2 = 1.000$
Panel C. Where the trained-learner results change (range over the four learners)						
Independent training streams	rel. gap $\leq 10^{-5}$		0.13 to 0.92	−1.75 to −0.38	—	(c) fails
Learning during the probe	rel. gap $\leq 1.1 \times 10^{-3}$		1.99 to 2.01	−4.02 to −3.98	—	(c) holds
Exact dead zone ($\tau = 0.05$): $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe moves the state past τ only for $A > 0.139$.						
Single-bin value lookup: $\|\Delta h\|_2 = 0$ at $A = 0.005$; the probe leaves the central bin only for $A > 0.149$.						

Panel A: deterministic simulations around the operating point; Test A reports the raw response-gap slope in $\log A$ ($O(A^2)$) and the probe-normalised gap slope $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ ($O(A)$, plotted in Figure S3). Panel B: learners trained for 5,000 periods, parameters saved and reloaded without rescaling and frozen during the probe; the two compared populations are trained on the same innovation stream; relative equivalence gaps at $A = 0.005$ are at most 10^{-5}. A raw Test A slope above 2 means the gap is of higher order than A^2. Panel C: the same learners trained on independent innovation streams (first row) or updated at learning rate 0.35 during the probe (second row). Capital in par units, account-level lower bound of the full working paper (61.42 for the linear model); the Q-learner settles on a small response gain (0.022), so separating it costs far more capital.

Independently trained populations end with learned gains that differ by sampling error of order $T_{\text{burn}}^{-1/2}$, so they differ in merged weight as well as in memory rate, and a difference in weight is cheaper to detect than an $O(\delta^2)$ difference in rates. The δ^{-4} law therefore prices the separation of memory rates between populations whose feedback weights match; it is not a claim about telling apart any two trained populations. When learning continues during the probe at rate 0.35, the law survives (slopes -4.02 to -3.98) and the merged-measure gap rises only to 1.1×10^{-3}, so the breakdown we anticipated under prediction (d) did not occur in this design.

The linear theory fails in two further places. First, saturation, adaptive gains and the neural readout leave the small-signal regime once probes exceed $A_{\text{sat}} \in [0.751, 1.271]$ par units, and the dithered threshold rule leaves it already at $A_{\text{sat}} = 0.049$, the threshold itself. Second, rules with hard discontinuities are different in kind: an exact dead zone ($\tau = 0.05$) or a single-bin lookup does not respond at all to probes smaller than 0.139 and 0.149 par units ($\|\Delta h\|_2 = 0$), concealing those groups from small probes altogether. Neither smooth nonlinearity nor hard thresholds offers a cheaper route to separating memory rates than the linear benchmark; populations whose learned

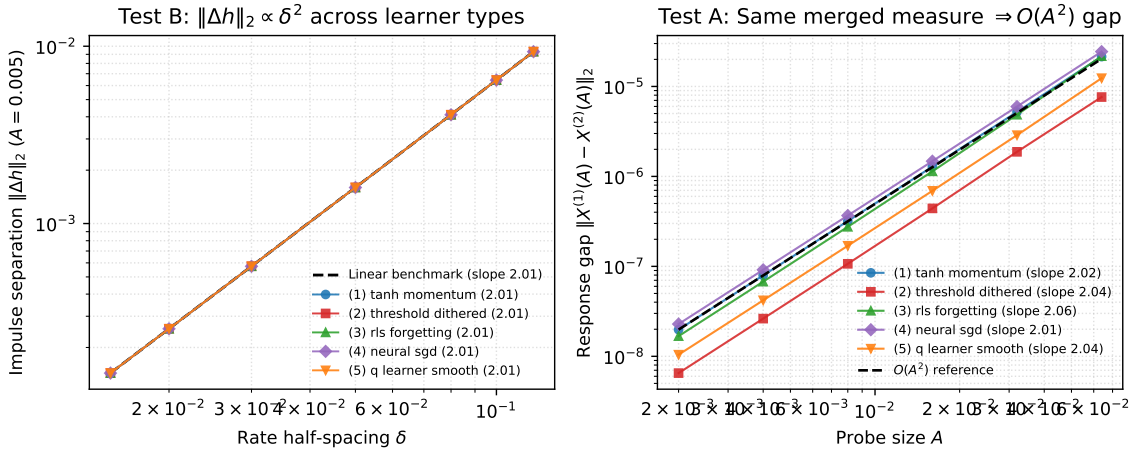


Figure S3: Left: Test B, probe-normalised response difference $\|\Delta h\|_2$ against the rate half-spacing δ at probe size $A = 0.005$, five stylized rules and the linear benchmark. Right: Test A, probe-normalised response gap $\|X^{(1)}(A) - X^{(2)}(A)\|_2/A$ between two economies with the same merged measure against the probe size A, with an $O(A)$ reference line (corresponding to an $O(A^2)$ raw gap).

weights differ are told apart by their weights, not their rates.

S22 Machine checks in detail

Theorem˜4.1. `Paper1InventoryEquilibrium.lean` checks the algebraic core of part (i):

- the dealers' first-order condition holds in every state;
- the next price loads on the next flow shock with coefficient κ/D;
- the no-feedback existence condition.

Part (iii) is proved on paper. For part (iv), `Paper1InventoryIFT.lean` applies Mathlib's inverse function theorem along a one-parameter family of feedback $G = \epsilon g$, with g strictly differentiable at $D = 1$. For all small ϵ it gives a zero near $(\kappa_-, 1)$, its uniqueness in a fixed neighbourhood, and the two first-order derivatives in the statement (`p1_inventory_small_feedback`). The version that is uniform over all small group measures, in the norm of the statement, is proved on paper.

Theorem 5.1. Lean proves the bound for every K and every horizon T, with a constant that does not depend on T. It uses the theorem's stability hypothesis $|a| + |\kappa|W/(1 - r_c - \Delta\eta) \le \varsigma < 1$, and holds for all error probabilities with $\alpha_0 + \beta_0 \le 1$ (`p1_thm_kcohort_lower_bound_uniform`): $4\sigma^2(1-\alpha_0-\beta_0)^2 \le N\, C_U^2 (\Delta\eta)^{4K-2}$. The checked pair is a different moment-matched pair with the same order: binomial weights on $2K$ equally spaced nodes, split into odd and even halves. For the Gauss and Gauss–Radau rules themselves, Lean checks the case $K = 3$ (positive weights, matched moments and the fifth-moment gap). Lean also checks both estimator statements for this pair, for every horizon (`p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1`). The first covers an estimator of the weights, listed in order of memory rate, that is accurate to $LW/4^L$ in each coordinate with probability at least 3/4 under both economies. The second covers an estimator of the merged measure that integrates every 1-Lipschitz function vanishing at zero to

within $2\Delta\eta W/(\pi(2K-1))$. For measures of equal mass, a Wasserstein error below that accuracy implies this. Both force $\sigma^2 \le N\,C_U^2(\Delta\eta)^{4K-2}$, here with $L = K-1$. The matching lower order $\|h^A - h^B\|_2 \gtrsim (\Delta\eta)^{2K-1}$ is proved on paper.

Theorem 5.2. Lean verifies this theorem for every $K \ge 1$, using only Lean's standard axioms. The formal proof follows the one above: a Lipschitz bound for the closed-loop deconvolution that does not depend on $\Delta\eta$, the $(\Delta\eta)^{-m}$ rescaling, a sub-Gaussian tail bound over the $2K$ taps, injectivity of the Jacobian of the moment map (proved by Hermite interpolation, without the determinant formula), and a uniform inverse function theorem on the compact class. The formal estimator is the representation with increasing nodes and nonzero weights that matches the $2K$ estimated moments; Lean shows that it is unique and that Prony's linear system has a unique solution whose polynomial has exactly its nodes as roots, so the roots are real. The formal statement allows sub-Gaussian noise and assumes $|r_c| + \Delta\eta \le 1$, so that every memory rate has modulus at most one.

Theorem 5.3. For every K, Lean checks part (i) and part (ii) starting from equal spectra at every frequency (`p1_thm_minphase_zeros`, `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged`). The check uses H in the rational form $\kappa z D(z)/((z-a)D(z)-\kappa N(z))$ of Theorem 3.3, and *stable* in the sense that the poles of H lie in the open unit disk. Both are derived from the market recursion. The eigenvalues of the transition of the state $(x_t, y_{1,t}, \dots, y_{K,t})$ are exactly the zeros of the denominator of H, so Schur stability of the recursion is the pole condition (`p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`). The generating function of the recursion's price response h_t to a unit trade satisfies $\left(\sum_t h_t x^t\right) x^{K+1}B(1/x) = x^{K+1}\kappa x^{-1}D(1/x)$ as formal power series, where B is the denominator of H; with $x = 1/z$ this is $\sum_t h_t z^{-t} = H(z)$ (`p1_impulse_generating`, `p1_tfBottomRev_eval`, `p1_tfTopRev_eval`).

Theorem 5.4. Lean checks the finite-sample consequence of parts (i) and (iii), not the divergence-rate statements. In the checked record the market starts from rest and runs for any number M of warm-up periods before the T recorded prices. The bound does not depend on M, but the passage to the stationary law ($M \to \infty$) is not checked. Part (iii) is checked for every pair of populations with nonnegative merged weights of total at most W, normalized rates in $[-1, 1]$ and equal moments below order $2K-1$ (`p1_thm_passive_cost_matched`). This covers the K-group Gauss and Gauss–Radau economies of Theorem 5.1: both rules are positive with nodes in $[-1, 1]$ and integrate polynomials of degree $2K-2$ exactly. Lean proves these properties of the two rules for $K = 2$ and $K = 3$ (`p1_thm_passive_cost_gauss_radau_two`, `p1_thm_passive_cost_gauss_radau_three`), so for these K the check covers the paper's economies with no extra hypothesis. For general K, the exactness of the two quadrature rules, a classical property, enters the check as the moment hypothesis. It also covers the binomial pair used in the numerical checks (`p1_thm_passive_cost_every_K`).

S23 Machine-checked declarations

The table lists, for each main result, the Lean 4 declarations that check it and what is proved only on paper. The project builds against a pinned Mathlib and has 1,235 named declarations in 144 modules, with no `sorry`. A full build on 13 November 2026 (Lean 4.34.1, Mathlib commit `d13f23b`) passed, together with the axiom audit below; the receipt, which records the SHA-256 hash of every source file, is `research_extension/current_manuscript_lean_20261008/formal_verification_receipt.json`. An axiom audit of every named declaration finds only Lean's three

standard axioms (`propext`, `Classical.choice` and `Quot.sound`). A machine check confirms the formal statement, not that it matches the prose. The prose statements above say where the two differ.

Result	Declarations (module)	On paper only
Lemma 2.1	`tracking_gain_explicit_root`, `tracking_prior_riccati_equation` (`Microfoundations.lean`); the third example's recursion, `p1_proportional_trade_recursion` (`Paper1StressIdentification.lean`)	Convergence of sample covariances. The third example's first-order condition is assumed in the checked statement, not derived
Theorem 3.1 (sufficiency)	`p1_eqAggFlow_kernel`, `p1_eqAggFlow_of_kernel`, `p1_kernel_of_eqAggFlow`, `p1_eq_market_invariance`, `p1_eq_insider_profit_invariance`, `p1_eq_group_level_differs` (`Paper1EquilibriumInvariance.lean`)	The unbounded-horizon moment statement; the passage from pathwise equality to equal equilibria
Proposition 3.2	`p1_anylinear_flow_kernel`, `p1_anylinear_iff`, `p1_anylinear_invariance` (`Paper1CoreExtensions.lean`)	The passage from pathwise equality to equal equilibria
Theorem 4.1	`p1_inventory_foc`, `p1_inventory_innovation`, `p1_inventory_benchmark` (`Paper1InventoryEquilibrium.lean`); `p1_inventory_small_feedback` (`Paper1InventoryIFT.lean`)	Part (iii); the version of part (iv) uniform over small group measures
Theorem 4.2	`p1_eq_realize_D`, `p1_eq_realize_dealer`, `p1_eq_scaled_cohort` (`Paper1EquilibriumReducedForm.lean`)	Parts (ii)–(iii), and part (iv) beyond the realization
Theorem 5.1	`p1_thm_kcohort_lower_bound_uniform` (`Paper1KCohortUniform.lean`); `p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1` (`Paper1KCohortEstimators.lean`)	The matching lower order of the response gap; the checked pair is a binomial moment-matched pair of the same order
Theorem 5.2	`p1_thm_prony_upper_bound` (`Paper1PronyUpper.lean`)	—

Result	Declarations (module)	On paper only
Theorem 5.3	`p1_thm_minphase_zeros` (`Paper1MinPhase.lean`); `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged` (`Paper1SpectralH.lean`); model link `p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`, `p1_impulse_generating` (`Paper1ModelLink.lean`)	—
Theorem 5.4	`p1_thm_passive_cost_matched`, `p1_thm_passive_cost_gauss_radau_two` (`Paper1PassiveMatched.lean`); `p1_thm_passive_cost_gauss_radau_three` (`Paper1GaussRadau3.lean`); `p1_thm_passive_cost_every_K` (`Paper1PassiveKCohort.lean`)	The divergence-rate statements; the stationary limit; Gauss–Radau exactness for general K enters as a hypothesis
Theorem S2.3	`p1_insider_reduced_form`, `p1_insider_best_response`, `p1_insider_kyle_limit`, `p1_insider_posterior_variance` (`Paper1EquilibriumInsider.lean`)	The fixed-point characterization in part (i), part (iii), and the numerical comparative statics
Proposition 6.1	`p1_prop_target_two_cohorts`, `p1_prop_target_every_K` (`Paper1SupervisorySeparation.lean`)	The checked two-group pair is a different moment-matched pair from the Gauss and Gauss–Radau pair, with the same orders; the every-K version assumes a dominant-group margin; for the quadrature pairs the largest-weight comparison is checked numerically for $K \le 200$, not proved
Proposition S5.4 and Lemma S5.3	The counting step of Lemma S5.3(iii) only: `p1_largest_value_not_constant`, `p1_largest_weight_not_constant` (`Paper1EveryKHalt.lean`)	Lemma S5.3(i)–(ii), the finiteness of the level sets, and all of Proposition S5.4
Proposition 5.5	Steps 2 and 3: the lower bound $\lvert D_A\rvert \ge c_3(\epsilon + \lvert\omega\rvert)$ (`p1_nearcrit_lower`), the uniform ratio bound (`p1_ratio_bound`), $\lvert u-1\rvert \le 4\rho$ and $u - 1 - \log u \le 2(u-1)^2$, and the rate bound $16\rho^2$ from Whittle's formula (`p1_nearcrit_kl_bound`) (`Paper1NearCritical.lean`)	Whittle's formula; Step 1 (the kernel-gap bound); the local properties of D^* from primitivity and their uniformity in $\Delta\eta$

Result	Declarations (module)	On paper only
Theorem 6.4	Part (i) in full on joint price and tagged-flow records: `p1_partial_lower_bound_joint`, `p1_partial_lower_bound_matched` (`Paper1PartialDisclosureLower.lean`); algebraic steps of (ii), (iii): `Paper1PartialDisclosure.lean`; the deterministic core of (ii), subtract-then-Prony with existence, uniqueness and the weight bound $L\delta$, $\delta = 2^{2K'}\eta(\Delta\eta)^{-(2K'-1)}$: `p1_partial_upper_deterministic` (`Paper1PartialDisclosureUpper.lean`)	The $O_P(N^{-1/2})$ rates of the tagged estimates that feed η (Monte Carlo checks in the text)
Theorem 6.5	Lower bound on joint records of all markets for any matched pair: `p1_pool_lower_bound_joint`, `p1_pool_lower_bound_matched`; explicit pair for $K = D = 2$: `p1_pool_explicit_pair_K2D2` (`Paper1PoolingLower.lean`); counting: `Paper1PoolingMarkets.lean`; upper bounds of (ii) and (iv) under condition (R): uniform local injectivity, bi-Lipschitz on compact classes and the minimum-distance error $(2/c)2^l e(\Delta\eta)^{-l}$ (`p1_pool_upper_local`, `p1_pool_upper_global`); injectivity and immersion at order $2K-1$, so $l_g \le 2K-1$ (`p1_pool_injective_top`, `p1_pool_fderiv_injective_top`) (`Paper1PoolingUpper.lean`) ; condition (R) at an explicit configuration for every K and D, with the groups i mod D and Hermite interpolation with mixed multiplicities (`p1_hermite_partial`, `p1_rank_condition_witness`, `p1_rank_condition_every_K`, `Paper1RankCondition.lean`)	Existence of matched pairs near generic configurations (constant rank); the step from one configuration to almost every configuration (vanishing of polynomial minors); the injectivity hypothesis of (iv) below order $2K-1$ (Monte Carlo checks in the text)
Proposition 6.6	none; it rests on Theorem 3.1	All of it
Proposition 6.2	`p1_tagged_iv_determinant_bound` (`Paper1TaggedIV.lean`); `p1_iv_boundary_decomposition` (`Paper1IVBoundary.lean`)	The stationary IV central limit theorem

Result	Declarations (module)	On paper only
Theorem 7.1	`p1_sat_sufficiency`, `p1_flow_market_invariance` (`Paper1SaturatingSufficiency.lean`)	The passage from pathwise equality to equal equilibria
Theorem 7.2	`p1_sat_stress_invariance`, `p1_sat_follows_linear`, `p1_satAggFlow_linear_of_band`, `p1_sat_capacity_differs` (`Paper1SaturatingSufficiency.lean`); coupling, union bound over dates and traders, and the test cost $T \geq (1-2\alpha)e^{1/(2\vartheta^2)}/(n+n')$ (`p1_coupling_bound`, `p1_exit_union_bound`, `p1_stress_test_cost_exp`, `Paper1SaturatingBounds.lean`)	The Gaussian tail of the memory states and the bound on their standard deviation; part (iv)
Proposition 7.3	For a cubic shape $u - \gamma u^3$ only: `p1_cubic_pulse_tap1_kernel_cancel`, `p1_cubic_pulse_tap1_price_gap` (`Paper1StressIdentification.lean`) ; the cost steps of (i) and (ii) (`p1_smooth_lower_cost`, `p1_smooth_upper_cost`, `Paper1SaturatingBounds.lean`)	For tanh: the bound $\|x^P - x^{P'}\|_2 \leq C_D\|A\|^3$ and the expansion of the first tap. The checked statements show the linear cancellation and the exact A^3 gap for the cubic shape
Theorem 7.4	General statement, any number of traders: `p1_thm_stress_identification`; hard limit: `p1_clip_indep_dilations`, `p1_thm_stress_identification_hard_limit`; cubic counterexample: `p1_cubic_not_indep_dilations` (`Paper1StressIdentificationGeneral.lean`); tanh: `p1_tanh_indep_dilations`, `p1_thm_stress_identification_tanh` (`Paper1StressIdentificationTanh.lean`); steps: `p1_probe_steer_target`, `p1_probe_flow_recovery_eq`, `p1_sat_signed_capacity_sufficiency` (`Paper1StressIdentification.lean`)	The small-probe statement in part (ii); the last sentence of part (iii); the statement in part (i) on probes inside the bands, which is Theorem 7.2(iv)

Result	Declarations (module)	On paper only
Theorem 7.7	Parts (i)–(ii) for any flow map: `p1_flow_market_invariance` (`Paper1SaturatingSufficiency.lean`); `p1_general_identification`, `p1_general_identification_iff`, `p1_responses_of_flow_eq`, `p1_identified_functional` (`Paper1GeneralIdentification.lean`); band-linear embedding: `p1_sat_follows_linear`	The transfer of the lower bounds in part (iii); the passage to equilibria in part (i)
Proposition S5.6	The consistency argument on any compact parameter set, in probability (`p1_identification_margin`, `p1_extremum_consistency_prob`) and the uniform bound for the truncated criteria (`p1_criterion_uniform_close`) (`Paper1NoisyStress.lean`); identification is Theorem 7.4	Continuity of the responses and of $\tilde{\mu}$ in the weak topology; the expectation bound $\sigma\sqrt{T_j/N}$
Theorem S8.1, Propositions S8.2, S8.3	Divergence of a moving window, Le Cam's two-point step, the bias–noise trade-off (`p1_track_kl_window`, `p1_track_two_point`, `p1_track_tradeoff`); the observer-effect inequality (`p1_observer_floor`); parts (ii) and (iii) of the game (`p1_hiding_detection_bound`, `p1_hiding_tag_no_crowding`) (`Paper1TrackingGame.lean`)	Analyticity and Lipschitz properties of the family R_c; the closed-loop and Prony steps, as in Theorems 5.1 and 5.2; part (i) of the game
Theorem S9.1	Parts (i) and (ii): orthonormal sign products, the tap bound for every noise path, Chebyshev, the union bound and the Prony step (`p1_walsh_quadratic_bound`, `p1_running_tap_mse`, `p1_running_prony_upper`, `Paper1RunningMarket.lean`)	Averaging over the noise path, the bound on $\|h\|_2$, part (iii)
Proposition S9.2, Corollary S9.3, Proposition S8.3(iv)	Laplace bound, comparison of divergences, money bound composed with the lower bound of Theorem 5.1, group-by-group equilibrium bound (`p1_laplace_vs_gaussian`, `p1_newslaw_kl_compare`, `p1_tags_value_money`, `p1_hiding_nash_bound`, `Paper1NewsLawHiding.lean`)	Chain-rule identity, Laplace divergence formula, Proposition S9.2(i) and (iii)

Result	Declarations (module)	On paper only
Proposition S5.5	The eigenvalue step: n eigenvalues within 4ρ of one give divergence at most $16n\rho^2$ (`p1_nearcrit_finite_sample`, `Paper1OpenResolved.lean`)	The Toeplitz sandwich
Theorem S10.2	Rounds below every fixed point and convergence to the least one, the contraction bound, the jump (`p1_cascade_iterate_le_fixed`, `p1_cascade_rounds_converge`, `p1_cascade_contraction`, `p1_cascade_jump`, `Paper1OpenResolved.lean`)	The derivative step of (iii)
Proposition S10.3	$\mathrm{Re}\,(1-w)^{-1} \geq (1+s)^{-1}$, the symbol bound, netting of public flow (`p1_re_inv_one_sub_ge`, `p1_no_manipulation_symbol`, `p1_netting_prefix`, `Paper1OpenResolved.lean`)	The Toeplitz representation, (i), (iii), and the step to conditional expectations in (iv)
Proposition S10.4	The bias of a bounded deviation through a stable closed loop (`p1_deadband_bias`, `Paper1OpenResolved.lean`); the Prony step of Theorem 5.2	The reduction of group flows to the recursion
Proposition S10.1	none	All of it (the Gauss–Radau error formula is classical); exact check in 60-digit arithmetic for $K \leq 30$
Proposition S10.5	Three equilibria from the intermediate value theorem (`p1_cascade_three_equilibria`, `Paper1OpenResolved.lean`)	The equilibrium condition, (ii), (iv)
Proposition S10.6	The sup-norm contraction step, through `p1_deadband_bias`	The expansion, the ℓ^2 bound and the test bound
Proposition S10.7	none	All of it
Proposition S10.8	The deterministic pathwise step (`p1_flow_market_invariance`)	The randomized induction, (iii)
Proposition S10.9	The symbol bound behind (i), through `p1_no_manipulation_symbol`	Existence, (ii)–(iv)
Proposition S10.10	The geometric bound of (i) and the price path of (iii) (`p1_cascade_rounds_geometric`, `p1_slow_capital_fall`, `Paper1OpenResolved2.lean`)	The limits in (i) and (ii), the equilibrium condition in (iii), (iv)
Proposition S10.11	none	All of it
Proposition S10.12	The closed-loop bias bound in (i), through `p1_deadband_bias`	The Taylor bound, (ii), (iii)

Result	Declarations (module)	On paper only
Proposition S10.13	The two polynomial inequalities behind the central binomial bounds (`p1_centralbinom_q_step`, `p1_centralbinom_r_step`, `Paper1OpenResolved2.lean`)	The Sonin-function bounds and the asymptotics; exact rational check for $K \geq 62$, 30-digit check for $K \leq 61$
Theorem S7.1, Proposition S7.2	For finite populations: the tap identity and weak duality, the upper half of (iv) (`p1_question_weak_duality`); the affine estimate of (i) (`p1_question_truncation_error`); part (iii) (`p1_long_memory_impossible`); the parametric direction of the proposition, modulus at most $\delta A + 2\bar{W}A\bar{r}^{L}/\sqrt{1-\bar{r}^{2}}$ (`p1_parametric_modulus`) (`Paper1QuestionsCost.lean`)	The lower half of (iv) (Sion's minimax theorem and the total-variation lower bound); (ii) (duality for best uniform approximation); the converse of the proposition (weak compactness in ℓ^2); measures that are not finitely supported
Theorem S11.2 (general memory)	Lower bound: `p1_thm_general_memory_lower` (`Paper1GeneralMemoryLower.lean`); upper bound: `p1_thm_general_memory_upper` (`Paper1GeneralMemoryUpper.lean`), with a Jackson theorem built in `Paper1Jackson.lean`; rate: `p1_thm_general_memory_rate` (`Paper1GeneralMemoryRate.lean`)	The constants of the paper statement (the formal constants are larger; the order is the same) and Lemma S11.1 via Bernstein's theorem (the formal proof uses an elementary pair)
Theorem S12.1	The arithmetic step of the lower bound: `p1_thm_uniformw1_minimax_lower_bound` (`Paper1FullFrontiers.lean`)	The probabilistic content of the lower bound, and the upper bound
Proposition S13.1	`p1_prop_S7_1_iii_uniform`, `p1_prop_S7_1_iii_two_uniform` (`Paper1CRUniform.lean`, `Paper1CRUniformW2.lean`)	All but the two-sided Fisher-information bound for $K = 2$, uniform in the horizon
Proposition S20.1	The displayed bound of part (ii), as an arithmetic step: `p1_endogenous_contraction_sensitivity` (`Paper1ClaimRepairs.lean`)	Part (i), and the fixed-point argument of part (ii)
Proposition S20.2	Two arithmetic steps: `p1_camouflage_unprofitable`, `p1_sybil_share_count_bound` (`Paper1ClaimRepairs.lean`)	The determinant factor of part (i) and the economic arguments
Theorem S19.1, Propositions S16.1, S16.2, S20.3, S14.2 and Theorem S14.1	none in this table	All of them (proofs in the supplement)
Proposition 7.5	`p1_delever_stress_invariance` (`Paper1CoreExtensions.lean`)	Parts (ii)–(iv), as for Theorem 7.2

Result	Declarations (module)	On paper only
Proposition 7.6	none	All of it

S24 Notation

Symbol	Meaning
x_t	pricing error (price minus fundamental)
a	price persistence (anchor), $0 \le a < 1$
κ	price impact per unit of executed quantity
$q_i,\ \eta_i,\ r_i$	group weight, adaptation rate, memory rate $r_i = 1 - \eta_i$
$w_i = q_i\eta_i$	merged weight
$M = \sum_i w_i \delta_{r_i}$	merged rate–weight measure
$k_m,\ G(z)$	feedback kernel and its transform
$H(z),\ h_j$	closed-loop transfer function and impulse response, $h_0 = \kappa$
$J,\ S$	state transition matrix and $S = a + \kappa \sum q_i$
$v_t,\ V$	executed probe quantity and total quantity $\sum \lvert v_t \rvert$
d_T	impulse-response difference between hypotheses, first T taps
σ	observation-noise standard deviation
$\delta,\ \Delta\eta$	half-separation of a known pair; spacing of unknown groups
K_*	transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$
T_h	Toeplitz matrix $(h_{\lvert i-j \rvert})$ of execution cost
$g_i(\omega)$	dissipation kernel $\mathrm{Re}\,\hat{H}_i(e^{i\omega}) - \kappa/2$
$B,\ \mu,\ c,\ \phi$	capital, maintenance-margin fraction, proportional cost, taker fee
V_{eff}	per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$
$\hat{Q},\ \hat{S}$	reduced-form loop gain and persistence indices